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14. ABSTRACT
This Joint Ordnance Test Procedure (JOTP) shall serve as the US national implementing document for NATO STANAG 4764 Edition 1 and AAS3P-23 Edition A. The categories of ammunition covered by this JOTP are high explosive, smoke, illumination, less-than-lethal, inert and training cartridges.

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DEPARTMENT OF DEFENSE

JOINT ORDNANCE TEST PROCEDURE (JOTP)-023

SAFETY AND SUITABILITY FOR SERVICE ASSESSMENT TESTING FOR MORTAR CARTRIDGES

Joint Services Munition Safety Test Working Group

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Joint Ordnance Test Procedure (JOTP)-023
Safety and Suitability for Service Assessment Testing for Mortar Cartridges

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DISTRIBUTION STATEMENT:

Distribution Statement A. Approved for public release; distribution is unlimited.

ABSTRACT:

Joint Ordnance Test Procedure (JOTP)-023 is intended to act as a munition type specific document dealing specifically with the necessary safety testing and assessments for Mortar Cartridges to enter service within the North Atlantic Treaty Organization (NATO) community. Two Safety and Suitability for Service (S3) test approaches, analytical and empirical, are presented in this JOTP with the intent that the manager of the test program shall select the more appropriate approach for the munition under test. The munitions covered by this JOTP are high explosive, smoke, illumination, less-than-lethal, inert and training cartridges.

COORDINATION DRAFT REVIEWED BY:

This document was coordinated with the following Standardization Offices: AR, AS, EA, MC, MI, NM, OS, TE, AF-11, and AF-99. In addition, the document was also coordinated with the Joint Weapon Safety Working Group, DODESB, OUSD LW&M, and select subject matter experts within the field of missile and rocket munitions safety testing.

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IMPLEMENTATION PLAN:

This Joint Ordnance Test Procedure (JOTP) shall serve as the US national implementing document for NATO STANAG 4764 Edition 1 and AAS3P-23 Edition A. The categories of ammunition covered by this JOTP are high explosive, smoke, illumination, less-than-lethal, inert and training cartridges.

APPROVING AUTHORITY:



THIERRY L. CHIAPELLO
Executive Director

8 July 2020
Date

DoD Explosives Safety Board
Office of the Under Secretary of Defense for
Acquisition and Sustainment

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DEPARTMENT OF DEFENSE
JOINT ORDNANCE TEST PROCEDURE

*Joint Ordnance Test Procedure (JOTP)-023 16 March 2020
DTIC AD No.

SAFETY AND SUITABILITY FOR SERVICE ASSESSMENT TESTING FOR MORTAR CARTRIDGES

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Cross-Reference Table	Table I3-1	APPENDIX I ANNEX 3

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1. INTRODUCTION.

This Joint Ordnance Test Procedure (JOTP) is aimed at the Safety and Suitability for Service (S3) Assessment Testing for Mortar Cartridges. JOTP-001 provides general discussion of Safety and Suitability for Service Assessment Testing. JOTP-023 is intended to act as an ammunition type specific document dealing specifically with the necessary safety testing and assessments for Mortar Cartridges to enter service within the North Atlantic Treaty Organization (NATO) community.

2. SCOPE.

2.1. Purpose.

The purpose of this JOTP is to guide personnel involved in the planning and implementation of S3 assessment testing of ammunition to enable appropriate evidence to be collected covering the entire life cycle. The objective of the safety test program defined by this JOTP is to provide data to demonstrate that the ammunition will be "safe for use", as defined in JOTP-001, throughout the potential deployment possibilities in NATO service.

2.2. Application.

The guidance provided in this JOTP is applicable to NATO, multi-national collaborative and national acquisition of Mortar Cartridges. The munitions covered by this JOTP include mortar cartridges as defined in Section 3 of this document. The categories of mortar cartridges covered by this JOTP are high explosive, smoke, illumination, less-than-lethal, inert and training cartridges. Cartridge unique hazards such as white phosphorous shall be considered in the detailed tests plans. This document excludes safety testing of weapons or fuzes.

This JOTP assumes that there has been considerable engineering design and development conducted prior to S3 qualification, and that there is a high expectation of successful safety qualification. This body of evidence should be used during the Preliminary Design Assessment and Hazard Analysis to determine if tailoring of the testing recommended within this standard is necessary. The tailoring shall be carried out as described in paragraph 6.5.

2.3. Limitations.

This JOTP is not intended to be used in the assessment of effectiveness, reliability or performance of a munition unless failure to be reliable or to perform effectively is deemed to represent a direct and immediate safety hazard to the user or other personnel. However, the data may be used in support of effectiveness, reliability, or performance assessment. This document does not define the In-Service Surveillance or Stockpile Reliability test requirements; however, the data may be used in the support of planning for these requirements. Refer to AR 702-6, AR 702-12, AR 740-1, STANAG 4675, AOP-62, AOP-63, and AOP-64 for further guidance. This document is not intended to address nuclear munitions.

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3. TERMINOLOGY.

Terminology in this JOTP are taken from the NATO Terminology Management System (NTMS) where possible. Terminology specific to this document and additional notes are added for clarity. Refer to JOTP-001 for terminology related to Safety and Suitability for Service test procedures.

3.1. Mortar.

The mortar includes the cannon and those components required for its operation, comprising the aggregate of the mortar, the associated platform (e.g., tracked or wheeled vehicle), and the ancillary equipment necessary to test, aim, launch, and guide the munition, as applicable (e.g., bipod, recoil unit, and fire control components). The vehicle-mounted (see Figure 3-1) and ground-mounted (see Figure 3-2) are the two (2) broad categories of mortars.

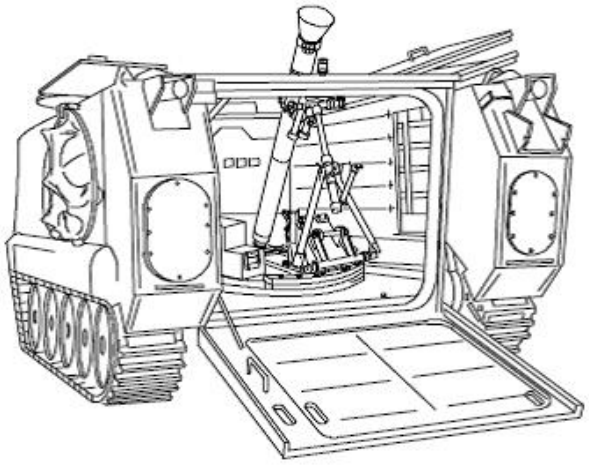


Figure 3-1: Typical Vehicle-Mounted Mortar.

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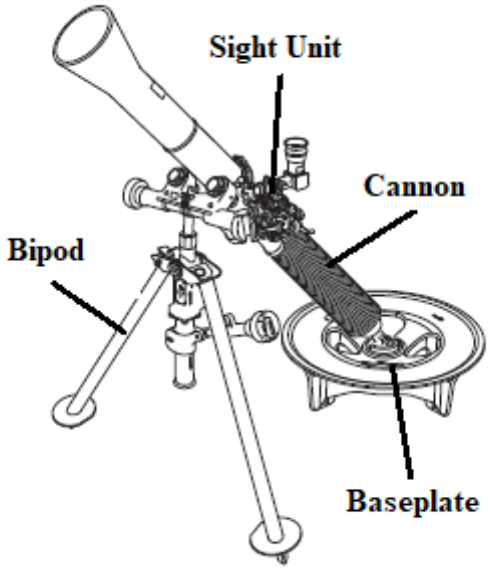


Figure 3-2: Typical Ground-Mounted Mortar.

3.2. Cannon.

The cannon is comprised of the breech, tube and firing assembly.

3.3. Mortar Cartridge.

The complete munition, comprising projectile and propellant system, to be fired from a mortar. The projectile normally comprises of the fuze, body filled with HE or other filling, obturator, and tail assembly. The propellant system normally includes an ignition cartridge and propelling charge(s).

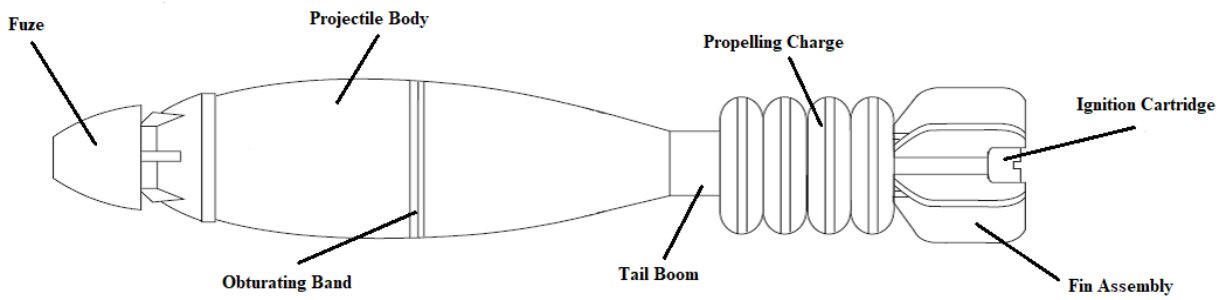


Figure 3-3: Typical Mortar Cartridge.

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3.4. Mortar Munitions.

Mortar munitions include all calibers/types of mortar cartridges. Common mortar system calibers are 60mm, 81mm, and 120mm.

3.5. Ignition Cartridge.

A component of the munition that contains energetic material that when ignited produces high pressure gasses, ignites additional propelling charges, and propels the mortar cartridge out of the cannon.

3.6. Propelling Charge.

A component of the munition that contains the energetic material that when ignited, produces higher pressure gasses than those produced by the ignition cartridge and propels the mortar cartridge out of the cannon.

3.7. Mortar Launched Guided Munitions.

Those cartridges capable of deviating from their initial ballistic trajectory upon firing to react to the target's maneuvers or to improve accuracy to the programmed target or target coordinates. When the cartridges are in the cannon they behave as conventional munitions and once launched they possess a guidance capability enabling directional control.

3.8. Temperature Conditioning.

Exposure of a packaged or bare munition to a thermal environment in preparation for a test event at a specified test temperature.

3.9. Pre-Stress.

Exposure of a munition to a sequence of one (1) or more environmental stresses (e.g., temperature, humidity, shock, vibration, etc.) prior to conducting a particular test event.

3.10. Solar Radiation Equivalent (SRE) Temperature.

In environmental testing, the SRE is defined as the maximum temperature achieved by energetic materials (e.g., propellant, cartridge, fuze) when exposed to cycles of high temperatures in combination with solar radiation during a laboratory test.

Note: The SRE is defined as the maximum temperature value experienced by the energetic material during the solar test. Determination of this value will require exposure of an inert, internally instrumented munition, with similar thermal characteristics to the complete round, to the full solar test requirement defined in MIL-STD-810. The SRE temperature should be determined for the

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packaged and unpackaged state. In the absence of this data, a value of +71 °Celsius (C) should be used for the SRE temperature.

3.11. Temperature Stabilization.

Temperature stabilization is achieved when the part of the munition considered to have the longest thermal lag is changing no more than 2 °C per hour. The stabilization time will typically be required for the cartridge in both the unpackaged and the packaged configurations and at the hot and cold temperature extremes. For packaged configurations, stabilization times are dependent upon the dimensions of the container, container dunnage, and the air gap between the munition and container.

Note: Since it may not be practical to monitor the part of a live munition with the longest thermal lag during test without damaging seals, the stabilization time may be determined prior to live munition testing using an inert, internally instrumented munition, with similar thermal characteristics to the complete round. The stabilization time will typically be required for the munition in both the unpackaged and the transport configurations and at the hot and cold temperature extremes. For packaged configurations, stabilization times are dependent upon the dimensions of the container, container dunnage, and the air gap between the munition and container.

3.12. Upper Firing Temperature (UFT).

The temperature to which the test munition is stabilized for hot test firing. This temperature is based on the climatic region predicted to be the worst case hot firing environment that the test munition will encounter during operations. In some cases, the UFT may be as high as the upper conditioning temperature (UCT). If no further information is available, a default of SRE temperature should be used. Although this value may be more severe than the stated upper firing temperature for the munition, the extreme value should be used to assess safety aspects of the mortar firing under worst-case service conditions. Appropriate precautions should be taken if exceeding the stated limit.

3.13. Upper Conditioning Temperature (UCT).

The temperature to which the test munition is stabilized for hot non-firing tests. This temperature is based on the climatic region predicted to be the worst case hot environment that the test munition will encounter during storage and transportation. If no further information is available, the SRE temperature should be used.

3.14. Lower Firing Temperature (LFT).

The temperature to which the test munition is stabilized for cold test firing. This temperature is based on the climatic region predicted to be the worst case cold firing environment that the test munition will encounter during operations. If no further information is available, a default of -46

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°C should be used. Although this value may be more severe than the stated lower firing temperature for the munition, the extreme value should be used to assess safety aspects of the mortar firing under worst-case service conditions. Appropriate precautions should be taken if exceeding the stated limit.

3.15. Lower Conditioning Temperature (LCT).

The temperature to which the test munition is stabilized for cold non-firing tests. This temperature is based on the climatic region predicted to be the worst case cold environment that the test munition will encounter during storage and transportation.

3.16. Mortar Safe Maximum Pressure (SMP) Curve.

The mortar SMP curve is a pressure against location curve which specifies, as a result of design, the particular value of pressure at each point along the tube which, if exceeded, could result in the occurrence of permanent deformation.

3.17. Mortar Design Pressure (DP) Curve.

The mortar DP curve is a pressure against location curve which specifies the particular value of pressure at each point along the tube which statistically should not be exceeded by more than one (1) round in 1,000,000 rounds under Extreme Service Conditions (ESC).

3.18. Margin of Safety.

The margin of safety is the difference between the mortar SMP curve and the mortar DP curve at any point along the tube.

3.19. Mortar Permissible Maximum Pressure (PMP) Curve.

The mortar PMP curve is the pressure against location curve which specifies the pressure at each point along the tube which, for reasons of safety, should not be exceeded statistically by more than 13 rounds in 10,000 under ESC.

3.20. Maximum Operating Pressure (MOP) Curve.

This is the curve derived from the pressures generated at each point in a specified mortar tube by a specified charge under the ESC which will not be exceeded statistically by more than 13 rounds in 10,000. The extreme conditions normally occur at the UFT in a new tube with the heaviest permissible weight of projectile. An allowance is made for occasion to occasion, tube to tube, lot to lot, and round to round variations. The MOP, which is derived by firing, is compared with Mortar PMP (or Projectile PMP if this is lower) in order to determine whether a particular charge design is likely to produce unsafe pressures in relation to the strength of design of the mortar or mortar cartridge.

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3.21. Mortar Proof Pressure (PP).

The mortar PP is that pressure, within specified tolerances, at which a mortar is proofed. The maximum mortar PP is to be the mortar DP: the minimum mortar PP should be the mortar PMP.

4. FACILITIES AND INSTRUMENTATION.

4.1. Facilities.

All test facilities utilized must suit specific test requirements and provide adequate protection for personnel and equipment in accordance with local and national regulations for testing of hazardous material. Note that although it is not necessary for all the facilities to be co-located, consideration should be given to the safe transport of potentially degraded test articles between test facilities. In addition to the requirements provided in Appendix F, Table F-1, test facilities shall be prepared for the handling and possible disposal of explosive items.

4.2. Instrumentation, Accuracy, and Calibration.

The accuracy and calibration requirements for instrumentation and test equipment used for control or monitoring are typically specified in the referenced test methods/procedures. Where such requirements are not specified in the referenced test methods/procedures, the accuracy shall be equal to at least 1/3 of the tolerance of the variable being measured. Recommended default tolerances for common test parameters are provided in Appendix F, Table F-2. Where there is conflict between these recommended values and any requirements stated in the referenced test methods/procedures, the requirements of the referenced methods/procedures take precedence. All instrumentation and test equipment shall be calibrated periodically to laboratory standards whose calibration is traceable to national laboratory standards. The test facility shall maintain the calibration records.

5. LIFE CYCLE ENVIRONMENTAL PROFILE.

5.1. Life Cycle Environmental Profile (LCEP).

Mortar cartridges are likely to encounter the life cycle environments shown in Figure 5-1. Detailed tests flows are provided in Appendix B of this document as sequential testing flowcharts and munition allocation tables. Test rationale are provided in Appendix A and test procedures in Appendices C, D and H. For each LCEP there are two (2) primary sequential streams, one (1) hot stream and one (1) cold stream and is considered as the standard for the JOTP.

The representative LCEP is based upon the applicable environmental factors for storage, transportation, and deployment selected from Appendix A of Allied Environmental Conditions Testing Publication (AECTP) 100, along with the generic usage profiles from AECTP 100 for mortars. Testing in accordance with this life cycle sequence and combining environments (i.e., vibration with temperature) is required to determine if the interaction (synergistic effect) and/or

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the sequence in which environments are experienced may result in a safety hazard. If the munition specific LCEP identifies environments or usage profiles significantly in excess of those provided, the test specifications should be adjusted accordingly.

5.2. LCEP Deviations.

All deviations from the representative LCEP contained in this document shall be approved by National S3 Authority(ies) or other appropriate authorities prior to the start of testing. The rationale used in tailoring shall be documented and retained as part of the Munition Safety Data Package in accordance with MIL-STD-882, MIL-HDBK-764, and ITOP 05-1-060.

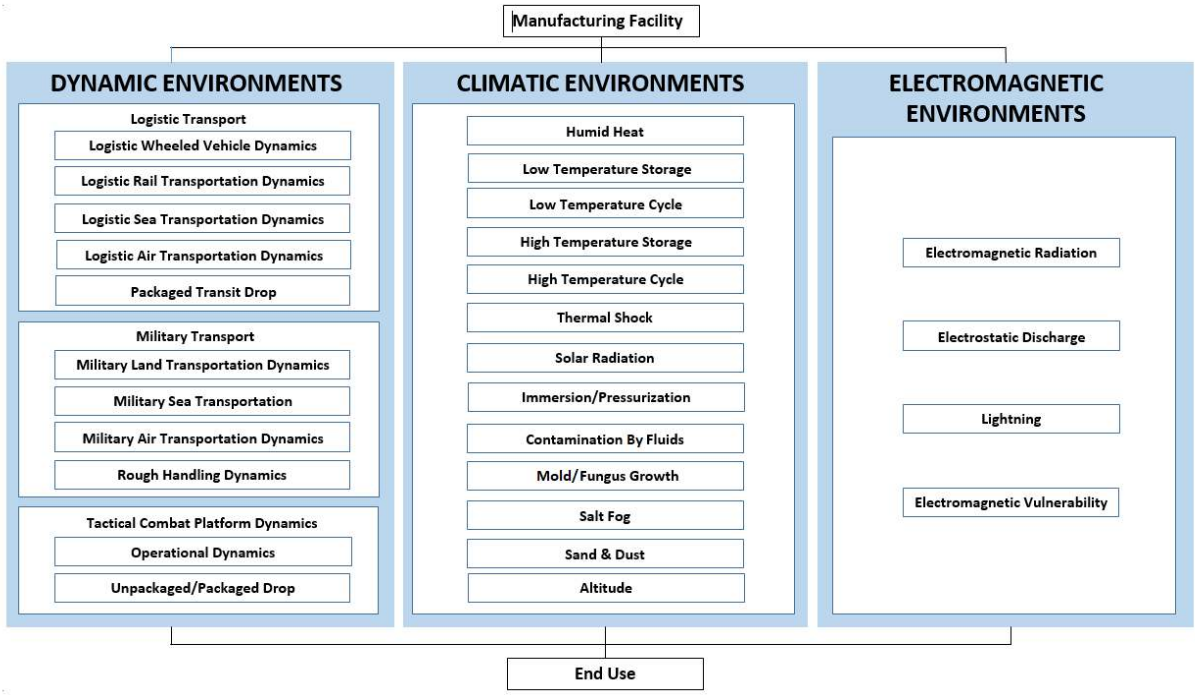


Figure 5-1: Expected Environments for Mortar Cartridges.

6. SAFETY TEST PLANNING.

6.1. Overall Test Objectives.

The objectives of the safety tests are to provide data to demonstrate that risks associated with the munition are sufficiently low that they meet the criteria of "safe for use" as defined in JOTP-001. To achieve this, the safety tests must provide data to determine the following:

a. Existence and nature of actual and potential munition hazards to personnel and equipment.

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b. Safety of the munitions throughout the planned LCEP including storage, transport, maintenance, training, operations, firing, and disposal.

6.2. Safety Data Sources.

Safety assessment of munitions is an evolutionary process, which begins in the early design phase of the munition and continues after deployment of the munition. The data gathered during the S3 tests described in this document should not be considered the exclusive source of data to support the safety assessment. Evidence shall be provided by the developing agency that the energetic materials used in the munition have been assessed and qualified. Other sources of safety data such as the ones described below shall be considered.

6.3. Preliminary Design Assessment, Test Data Review and Hazard Analysis.

Review of existing safety, design, and test data is strongly advised prior to development of the safety test plan. Review documentation concerning munition requirements, design, safety and prior tests in order to identify potential hazards and their causes. The degree to which this JOTP is followed and the degree to which other data are accepted in place of these JOTP tests depend on the characteristics of the munition and on the credibility and completeness of existing safety data. These reviews and this JOTP must be used to develop the detailed safety test plan and shall be in accordance with national health and safety standards and regulations. Thoroughly review all data related to the munition under test. This should also include any data that can be used by analogy from previously qualified designs. This review should include test data from component and munition level performance and safety testing (engineering-design or component-development tests). If this review shows that the munition conforms to a proven design, and that its performance (or that of similar items) in earlier (engineering design or component development) tests are favorable, methods addressed in this JOTP are applicable and tailorable to reflect the significance of the change. Evaluate the safe operating temperatures of the explosive materials, as LCEP temperatures may exceed the safe operating conditions of these explosive materials. Figure 6-1 provides an overview of possible data.

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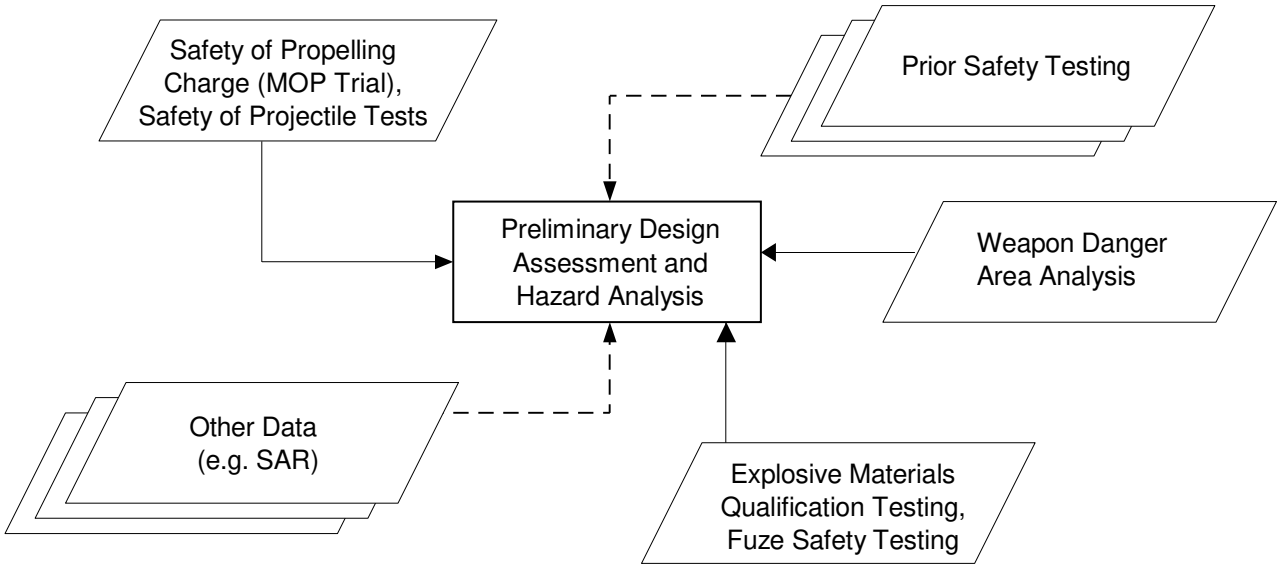


Figure 6-1: Preliminary Design Assessment and Hazard Analysis.

6.3.1. Safety of Projectile.

The pressure generated by the propelling charges during firing shall be in accordance with both the weapon and the projectile safe operating limits to ensure safe firing. Appendix A, Annex 8, Paragraph A.8.3 provides the rationale/background and Appendix D, Annex 4, Paragraph D.4.2 provides the description of the Safety of Propelling Charge (MOP Trial) Test.

6.3.2. Safety of Projectile Tests.

The projectile shall be designed such that it will be able to withstand the forces generated by the pressure of the propelling charge during firing with a suitable margin for safety. The rationale/background for the Safety of Projectile Tests is provided in Appendix A, Annex 8, Paragraphs A.8.4 through A.8.7; with the test descriptions provided in Appendix D, Annex 4, Paragraph D.4.3 through D.4.6. These tests include the Verification of Propelling Charge, Charge Establishment for Projectile Safety Testing, Projectile Strength of Design (SoD) and Projectile Safety in Cannon Test.

6.3.3. Explosive Materials Qualification Testing.

All explosive materials in the munition shall undergo appropriate testing and assessment per NAVSEA Instruction 8020.5C and MIL-STD-1715A to determine whether each possesses properties which make it safe for consideration for use in its intended role.

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6.3.4. Fuze Safety Testing.

This JOTP assumes that the fuze, if applicable, is already qualified and that no further testing is required. The results of fuze qualification are to be provided within the Munition Safety Data Package if the fuze is integral to the projectile. The central objective of S3 of fuzing systems is to confirm and document that the fuzing system is safe and performs as intended in all expected service environments. The design safety requirements standards are MIL-STD-1316 and the test requirements for S3 Assessment are found in MIL-STD-331.

6.3.5. Prior Safety Testing.

Available data from prior safety tests for the munition design or comparable designs can give insight and therefore indicate the need for additional and/or adjusted testing.

6.3.6. Other Data - Safety Assessment Report (SAR).

The SAR is a formal document that summarizes potential hazards to developmental testers that shall be submitted by the material developer to the tester prior to commencement of testing. The SAR shall delineate the safety related characteristics of the munition, identify potential hazards and assess severity and probability of the mishap risk of each identified hazard, and recommend procedures and precautions to mitigate hazards to an acceptable level of risk. Other sources of data are to be considered.

6.3.7. Weapon Danger Area Analysis.

Prior to performing any live firing tests, a weapon danger analysis must be performed to determine the hazardous area that will result from the live firing test. Further guidance may be found in ITOP 05-2-505, AR 385-63, DA PAM 385-63, STANAG 2240 and Allied Range Safety Publication 1 (ARSP-1 VOL II) Weapon Danger Areas / Zones For Unguided Weapons.

6.4. S3 Test Outline.

S3 assessment testing of mortar cartridges requires a series of sequential environmental tests, firing tests, non-sequential climatic tests, system tests, dynamic environmental tests and additional tests. The test flowcharts and test allocation tables are shown in Appendix B. These include sequential and combined environmental tests (i.e., vibration with temperature) to determine if the interaction (synergistic effect) and/or the sequence in which environments are experienced may result in a safety hazard.

6.5. Test Tailoring.

The safety tests recommended in this document are intentionally conservative to account for a wide range of deployment possibilities in NATO service. Test tailoring may be necessary for a variety of reasons including test conduct safety considerations, variation of deployment

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requirements and/or LCEP, the need to address nation specific requirements and/or factors that affect test sample sizes. When nation specific requirements conflict with requirements in this document, the reference tables in Appendix I may be used to assist in the process of cross-referencing the national and international documents. During sequential testing, test levels and/or test configurations may require tailoring such that non-safety hazard related damage in any one (1) particular test does not affect the validity of subsequent test data. If tailoring is determined to be necessary, the tailoring may be carried out in accordance with the following general principles:

a. The tailored environment shall be at least as severe as the expected life cycle environment.

b. Any alternative test standards/methods that are utilized shall be technically equivalent or superior to the referenced standards/methods.

c. The tailored test procedures and severities, along with full justification/rationale shall be documented as part of the S3 assessment report.

d. Tailoring shall be approved by the relevant national authority prior to testing.

Particularly, document the elimination of tests, reduction of sample quantities, or reduction of severities, any of which may result in reduced evidence to fully support the required safety assessment of the mortar cartridges.

6.6. Environmental Test Levels.

The environmental test levels specified in this document are based on the anticipated extreme conditions for storage, transportation, handling, maintenance, and firing of the munition. Natural and induced environmental factors for storage, transportation, and deployment are selected from MIL-STD-810 and MIL-STD-2105. Climatic test levels are based upon climatic categories defined in MIL-HDBK-310 and MIL-STD-810. Dynamic test levels are based on MIL-STD-810; and Electromagnetic Environmental Effects (E3) test levels are based on MIL-STD-464, MIL-STD-461, TOP 01-2-511 JOTP-061, and JOTP-062. Alternative international and national test specifications may be employed to meet the environmental test requirements if it can be demonstrated that the specification is technically equivalent or superior to the MIL-STDs. The deployment (tactical) shock and vibration environments may be tailored based on measured data using tailoring guidance in MIL-STD-810 and MIL-STD-2105. Test time compression in accordance with MIL-STD-810, Method 514 may be acceptable, however the risk of introducing false failure modes should be considered. In addition, the international documents listed in the cross-reference table in Appendix I, Annex 3 may also contain unique test requirements and severities only applicable to the specific nation. Rationale for the specific test levels in this document is provided in Appendix A. Test levels or specification deviations for munitions designated to be deployed to specific areas of the world or on specific transport or tactical vehicles may result in limitations on service use or require use of special procedures.

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6.7. Test Safety Considerations.

Explosive materials can often become less stable with age. This aging is exacerbated by the presence of increased temperature, humidity and vibration/mechanical stressing. It is therefore necessary to review the projected test sequence and determine whether the sequence, including any temperature conditioning and storage, result in an unacceptable hazard. As a minimum this will require an assessment of explosive material stability with respect to extreme temperature exposure durations. It might be necessary to divide the overall test time (shock, vibration, and bounce in particular) into smaller portions to prevent heat build-up within the munition and subsequent unintended energetic reaction potential. It is essential and mandatory to have a log for each round indicating the amount of time that has been spent at extreme temperature for the entire test sequence, including all periods of temperature conditioning.

6.8. Test Sample Quantities.

The test sample quantities are largely dictated by the minimum number of tests (e.g., ballistic firings, Breakdown Test and Critical Analysis (BTCA), hazard classification, and insensitive munitions) to provide sufficient evidence of munition safety. Baseline munition shall be taken from the same pre-production/production lot as qualification munition to allow direct comparison. Specific rationale for the tests in each of the test categories is provided in Appendix A. The following general notes should be considered when assessing the test sample quantities required for an S3 test program:

a. Munitions having more than one (1) functional mode, configuration or operating state and/or operating platform may require increased test sample quantities.

b. Existing safety data may also be reviewed for acceptability with the goal of reducing sample sizes and the number of tests. The degree to which this data can be used depends upon mortar cartridge characteristics, reliability and completeness of the existing safety data, and the adequacy with which it treats hardware configuration, input stress, potential synergistic effects, types and severity of hazards, and the probability of hazard occurrences. However, tests which may interact with each other in a synergistic fashion (e.g., vibration/shock or vibration/climate) must not be removed from the sequence.

c. Additional mortar cartridges beyond those recommended may be needed in the test program for baseline purposes and to replace munitions that become damaged during testing. Also, fully inert munitions may be required for testing to evaluate and certify test procedures. In addition, inert mortar cartridges may be required for rate-of-fire, strength of design, maximum operating temperature, and propelling charge firing tests. Completely functional inert mortar cartridges may also be required to perform powered Hazards of Electromagnetic Radiation to Ordnance (HERO) tests. Additional instrumented inert cartridges and packaging is recommended to permit evaluation of temperature stabilization time.

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d. Completely functional mortar cartridges are only required for test assets designated for the munition level dynamic firing tests. For all other test assets, non-safety critical components may be removed in order to reduce test cost. Any hardware that is removed should be replaced by mass simulants with thermal, structural, and dynamic characteristics similar to the tactical hardware.

e. The test sample quantities or configuration may be modified provided rationale is approved by the appropriate National Authority(ies) or other appropriate authorities. For example, the number of dynamically fired test items may be reduced if:

(1) Previous firing tests of worst case pre-stressed (i.e., previously exposed to the sequence of environmental stresses) and temperature conditioned munitions provide the required fuze arming test data. Data from the previous firing tests are required to be provided with the new S3 assessment file.

(2) The fuze arming tests are not applicable.

6.9. Mortar.

If the mortar cartridge is designed for more than one (1) mortar and/or firing platform combination, distribute the firings across a representative subset of those mortars. The tube is recommended to be in its first quarter of life. This may be necessary where it is not feasible to conduct all munition tests with each intended mortar due to the sample size of the test munition required. Provision must be made to ensure that sufficient cannons with the required wear lives (where applicable) are available for the tests stated in Appendix D, Annex 4.

Inspect the mortar periodically throughout testing in accordance with guidance in ITOP 03-2-802, and whenever there are abnormalities or unexpected occurrences such as projectile breakup, or unexpectedly high chamber pressure.

6.10. Munition Packaging.

The mortar cartridge test configuration should be tailored to the appropriate shipping, handling, storage, and operational deployment (stowage and launch) configuration that the munition will experience during its service life. Test items may be configured as palletized (Figure 6-2), primary packaging (Figures 6-3 and 6-4), secondary packaging (Figure 6-5), or bare munition (Figure 3-3). In the palletized configuration, mortar cartridges are bundled in their primary packaging and tied up with banding to keep them secured during transportation, storage and handling. Mortar cartridges in their primary packaging configuration are packed in their most outer packaging configuration (e.g., wood overpack and/or metal overpack) which keep them protected from different transportation, handling and environmental conditions. Munitions in their secondary packaging are packed in their most basic packaging configuration (e.g., fiber and/or plastic

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container) which is minimally protected from transportation, handling and environmental conditions and readily accessible for firing. Bare munition is a mortar cartridge with no packaging protection for transportation, handling and environmental conditions which are in ready-to-fire mode. Use the appropriate packaging configuration for the transport phase to be tested.

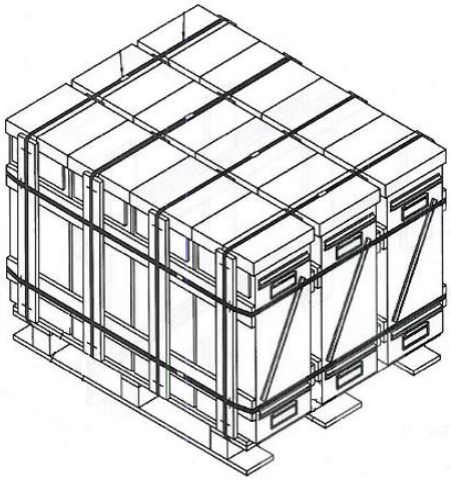


Figure 6-2: Mortar Munition Pallet Configuration.

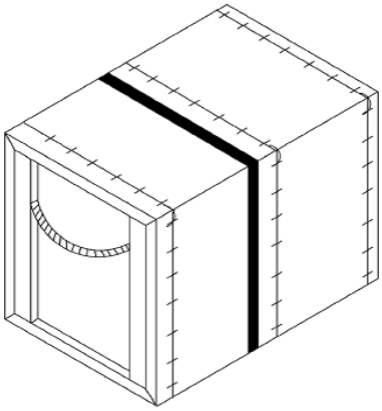


Figure 6-3: Typical Mortar Cartridge Wire-Bound Wooden Box (Primary).

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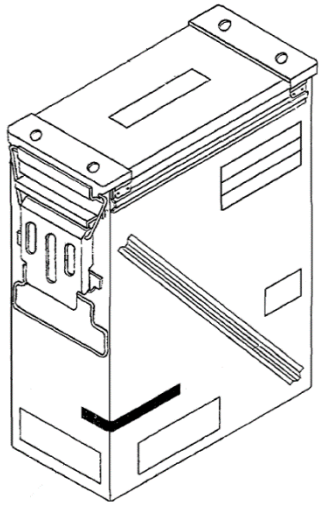


Figure 6-4: Typical Mortar Cartridge Metal Container (Primary Packaging).

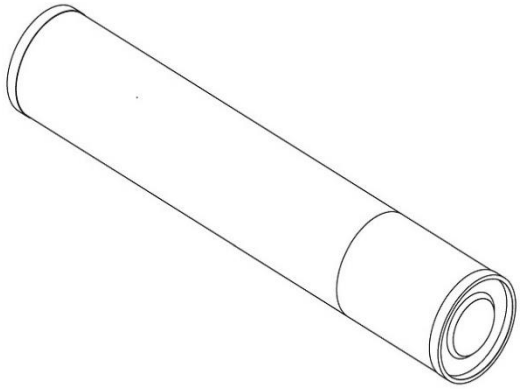


Figure 6-5: Typical Mortar Fiber Tube (Secondary Packaging).

7. PRE AND POST TEST INSPECTIONS.

Munitions and packaging shall be inspected at different moments throughout the sequential environmental trials. Inspections should be conducted in accordance with the following inspection levels:

Initial Inspection	(Baseline)
Level 1 Inspection	(Basic)
Level 2 Inspection	(Intermediate)
Level 3 Inspection	(Full – Breakdown Tests and Critical Analysis (BTCA))

Appropriate inspection methodologies combine visual assessment and detailed chemical and physical tests to fully understand degradation of the munition.

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The flowcharts in Appendix B show an Initial Baseline Inspection and several Level 1 and Level 2 Inspections within the sequential environmental trials. A Level 3 Inspection is optional before (baseline data) and after finishing the sequential environmental trials. The Level 3 Inspection is also known as Breakdown, Test, and Critical Analysis (BTCA).

It is recommended that extra inspections are added when test exposure is considered likely to have adversely affected the munition and packaging. For these additional inspections, the appropriate inspection level (1, 2 or 3) must be chosen. The exact inspection tests need to be determined on a case-by-case basis. Failure modes and life limiting factors identified for comparable munition may assist. Non-destructive techniques utilized shall have the capability to accurately assess condition of the safety critical characteristics.

Note that extra inspections can also be added throughout the complete test scheme as shown in the flowchart in Appendix B including the non-sequential and firing tests. The inspection levels are described in Paragraphs 7.1 to 7.4 below. In Appendix E, additional information can be found concerning tests on Level 3 Inspection.

7.1. Initial (Baseline) Inspection.

An initial inspection, using a sample of the test items, is conducted before the start of the sequential environmental trials. The initial inspections verify conformance of the munition and packaging to build standards (see JOTP-001) against the production drawings and provides an assessment of the baseline condition for subsequent test inspections. Visually inspect munitions and packaging, noting damage and obvious manufacturing defects. In addition to the Level 1 and Level 2 examinations described in paragraphs 7.2 and 7.3, initial inspections should include baseline photographs and the items listed below. Deviations from the build standard should be assessed by the appropriate authorities to determine that the asset(s) is satisfactory for the S3 test program.

The list given below is not intended to be exhaustive, but to indicate the typical aspects to be inspected and reported. However, the specific criteria should be selected on a case-by-case basis depending upon the munition type, construction, its packaging and ancillary items.

a. Type of projectile.

b. Physical characteristics such as weight and all critical dimensions - for both munitions and packaging.

c. Manufacturer, manufacturer's markings and manufacturer's lot/batch number - for both munition and packaging. Note, identify each item under test with a unique number to be used throughout testing.

d. Energetic materials manufacturer, lot/batch number, type, size and charge weight.

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e. Materials of construction.

f. Packaged configuration, type of container, overpack (if any), and number of rounds per container. The individual test item orientation and position within the packaging shall be maintained throughout all testing. Orientation modifications required due to component replacement or testing sequence shall be documented.

g. Software version embedded in munition, cannon, or command launch unit (CLU), where applicable.

7.2. Level 1 (Basic) Inspection.

A Level 1 Inspection is conducted to detect possible degradation due to test exposure. It consists of visual examination of all test munitions and packaging.

The list given below is not intended to be exhaustive, but to indicate the typical aspects to be inspected and reported. However, the specific criteria should be selected on a case-by-case basis depending upon the munition type, construction, its packaging, and ancillary items.

a. Visually inspect munitions and packaging for damage, deterioration, and obvious manufacturing defects. Note the overall condition of munitions and packaging, including damage/degradation (e.g., abrasion, indentations, splits, scratches, corrosion, etc.).

b. Characteristic photograph of bare mortar cartridge.

c. Characteristic photograph of primary and secondary packaging.

d. Physical weight and dimensions of mortar cartridge.

e. Indication of seepage, leaks, or explosive exudation or powdering.

f. State of seals.

g. State of safe and arm (S&A) devices and fuzes.

h. Inspect identification markings on munition and containers (e.g., yellow markings for High Explosive (HE), batch/lot numbers) to ensure they are in accordance with approved drawings and specifications/requirements.

i. Inspect interior of packaging for foreign material, loose propellant, and evidence of moisture. Replace missing or damaged packaging before vibration and rough handling tests on the munitions.

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j. Ignition cartridge condition (e.g., 'cocked', loose, etc.).

k. Looseness of components.

l. Appearance of propelling charge outer surfaces.

m. Proper assembly and condition of obturating band.

n. Check state of internal power sources(s), if applicable.

o. Condition of Built-In-Tests (BIT), if applicable.

7.3. Level 2 (Intermediate) Inspection.

Level 2 (Intermediate) encompasses Level 1, but also consists of radiography and non-destructive examinations (e.g., ultrasonic, tomography, magnaflux, magnetic particle, eddy current) of all munitions and pyrotechnic devices. (Note: Magnetic particle inspection should never be used on live munitions, unless there is proven assurance ahead of time that no hazard exists in doing so to the munition in question. The examination facility should have the capability to conduct radiographic inspection at low temperature extremes within 15 minutes after removal from a cold conditioning chamber (if required). Deviation from this should be recorded and accepted by the appropriate authority. Level 2 inspections should determine the following:

a. Indications of structural damage.

b. Movement of internal components.

c. Radiography of hazardous fillings within the projectile (e.g., High Explosive), if applicable, to determine the presence of voids, cracks and/or any other failure modes identified during the preliminary design assessment. Compare the radiographs to appropriate standards to determine if cracks, voids, and other explosive defects or unusual conditions are present which would contribute to a safety failure. Radiographs are required to examine metal parts for proper positioning, for cracks or damage, for missing parts and to provide evidence that the fuze is in the unarmed (safe) position. For rough handling and transportation tests, radiograph the test munition before and after test as shock, vibration and rough-handling tests may damage munitions. In order to have consistent radiograph inspection, 0º and 90º positions shall be indicated on the cartridge and the x-ray shall be taken in that orientation.

d. Test munitions will be fired if it is judged by the tester that troops in the field would have overlooked or considered the damage negligible by superficial visual inspection and fired the munition. However, the design agency shall be consulted before firing any munition with abnormalities. If such munition is not fired, it shall be considered to have failed that safety test phase.

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e. Inspect munitions for manufacturing defects using the non-destructive techniques (e.g., magnetic particle, ultrasonic or fluorescent particle inspection).

f. State of primers, fuzes (and any other S&A devices if present) to include testing of electrically initiated primers with a certified low current circuit tester or ohm-meter, and electronic fuzes with the appropriate fuze setter to ensure that the munition is safe for handling and continued testing. It is noted that such continuity tests are destructive tests and the primers, fuzes or other S&A devices subject to this test must not be returned to inventory.

g. Physical re-measurements, e.g., diameter, length, fin diameter, fin blade against drawing and specification requirements as necessary.

h. Record physical properties of the munition, to include mass, flight weight, components' weight and, if applicable, center of gravity, and transverse and longitudinal moment of inertia.

7.4. Level 3 (Full - Breakdown Test and Critical Analysis (BTCA)) Inspection.

A Level 3 Inspection using a sample can be conducted before, during and after sequential environmental trials if necessary. This BTCA is conducted to detect possible degradation due to test exposure. It consists of Level 2 Inspection and additional tests.

It is necessary to disassemble the munitions for Level 3 Inspection in order to analyze the munition and its components. Appendix E provides additional information concerning the disassembly and the additional tests.

8. S3 TEST PROGRAM.

8.1. S3 Test Program.

The S3 test program, based on the representative LCEP for mortar cartridges, consists of sequential, and non-sequential tests and assessments. The approach to the S3 test program test sequences is presented in Appendix B.

Figure 8-1 illustrates the total test outline consisting of the Preliminary Design Assessment and Hazard Analysis as described in chapter 6 and the S3 test program itself.

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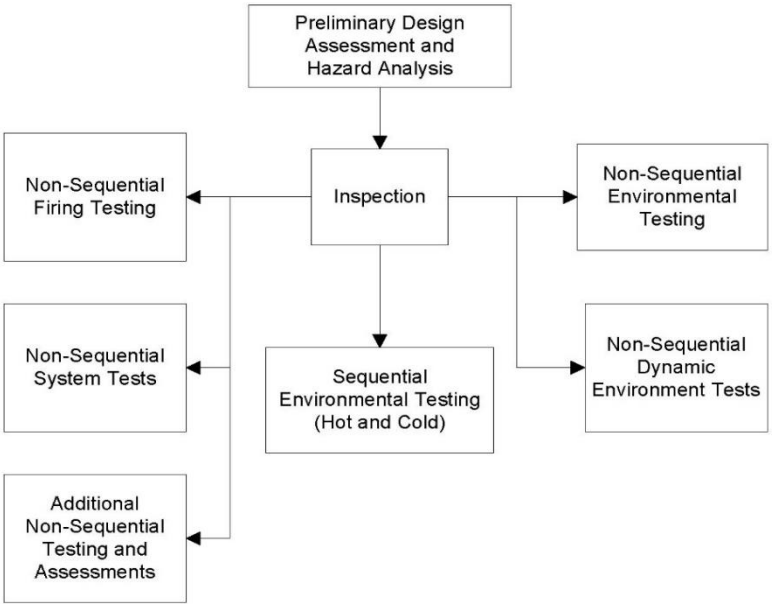


Figure 8-1: S3 Test Program Outline.

In the test descriptions, test details are deferred to MIL-STD-810 documents referenced in the test descriptions whenever possible. Test methods which are not currently covered by MIL-STD are referred to the appropriate national document or ITOP. Conflicts between the referenced test methods and the procedures described in this document should defer to the referenced test method. Background and rationale for these tests are provided in Appendix A. The different parts of the S3 test program are briefly described in the following paragraphs.

The individual test item orientation and position within the packaging shall be maintained throughout all testing. Orientation modifications required due to component replacement or testing sequence shall be documented.

8.2. Sequential Environmental Testing.

Sequential Environmental testing involves the climatic and dynamic tests associated with the life cycle of the munition. Appendix A provides the rationale/background for the particular tests, test sequences are described in Appendix B and Appendix C provides the description of the Sequential Environmental tests. The Sequential Environmental Test has a hot and cold stream. The exposed munitions will be inspected and fired after completion of sequential tests (see Figure B1-1).

8.3. Non-Sequential Climatic Testing.

Appendix C provide descriptions of the environmental tests on munitions separate from the sequential environmental tests, not all of which are mandatory (see Figure B1-1). These are life cycle dependent.

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8.4. Firing Tests.

Appendices C, D, and H also provide descriptions of munition firing tests separate from the sequential environmental tests, not all of which are mandatory (see Figure B1-1). These are life cycle dependent.

8.5. Non-Sequential Systems Tests

These include test and evaluations which are dependent on both the munition and mortar platform for the test planning and execution. It is noted that integration of previously qualified munitions onto a new mortar platform will require reconsideration of these tests and evaluations.

8.6. Non-Sequential Dynamic Environmental Tests.

Non-contact Underwater Explosion (UNDEX), 12 Meter Logistic Safety Drop, and Parachute Resupply tests and evaluations must be considered during development of the S3 package. These tests and evaluations are addressed in Appendix C, Annex 4.

8.7. Electromagnetic Environmental Effects (E3).

E3 assessment testing shall be conducted in accordance with JOTP-061, JOTP 062, MIL-STD-461, MIL-STD-464, MIL-HDBK-240 and TOP 01-2-511. The potential electromagnetic environment, which includes electromagnetic radiation, electrostatic discharge and lightning, could create detrimental effects on mortar electronic fuzes, Electrically Initiated Devices/Electro-Explosive Devices, such as electric primers, and guidance components on guided munitions. This could in turn result in excessive misfires, inadvertent ignition, fuze failure, and failure of the guidance system. The mortar munition should incorporate features to ensure safety and reliability after and during exposure to the E3 environment. This testing must address electrical safety and consists of Hazards of Electromagnetic Radiation to Ordnance (HERO), Electromagnetic Compatibility (EMC), Electrostatic Discharge (ESD), Lightning Tests, and Firing Circuit Analysis. It is noted that performance testing after subjecting the munition to Nuclear Electromagnetic Pulse (NEMP) is not included in this S3 document but may be required by the national S3 authority. For a system expected to have significant changes to its vulnerability with age/use, using environmentally stressed rounds within E3 test and assessment should be considered. Minimum test asset quantities are provided in Appendix B. General guidance is provided in Appendix A, Annex 7 and test details are provided in Appendix H, Annex 2.

8.8. Health Hazards Testing.

Appendix H, Annex 3 describes the testing and analysis to assess potential health hazards posed by the elements or combinations present in munition and by munition use.

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8.9. Additional Tests and Assessments.

Tests and assessments in addition to the sequential and non-sequential testing described above are required as part of the S3 Package. In particular, the Insensitive Munitions Assessment, Munition Software System Safety Assessment, Explosive Materials Qualification Testing, Munition Demilitarization and Disposal Assessment Testing, Render Safe and Disposal Procedure Testing, Range Safety and Sustainability Assessment, Munition Hazard Classification Testing and Fuze Safety Testing are required but the details regarding the series of tests are not provided in this document since they are governed by other standards. Also, a Vertical Replenishment At Sea and a Munition Bare Drop Sensitivity Test should also be considered. General guidance is provided in Appendix A, Annex 7 and test details are provided in Appendix H, Annex 2.

8.10. Mortar Integration/Firing Safety.

Appropriate testing and analysis shall be performed to assess mortar integration for new mortar cartridges entering the inventory. Sufficient evidence should be provided to determine whether the cannon interface and the munition have adequate structural integrity to withstand the anticipated dynamic loading. In addition, live fire testing from applicable cannons will be required to provide sufficient evidence of safe operation and separation, launch/blast effects, and human factors associated with cannon system operation. At a minimum, these tests should encompass the dynamic firing objectives as described in Appendix D, Appendices 2 and 3 and 4, and the operations and maintenance (O&M) objectives as described in Appendix H, Annex 5. General guidance is provided in Appendix A, Annex 7 and test details are provided in Appendix H, Annex 5.

8.11. Operational and Maintenance Assessment.

Appendix H, Annex 5 describes the operational tests required to assess the safety of operational and maintenance procedures and equipment during field handling exercises.

9. MUNITION SAFETY DATA PACKAGE.

As stated in JOTP-001 and AOP-15 Appendix C, the results of the testing and assessments required in this document will be compiled into a Munition Safety Data Package for use by the appropriate S3 approving authority in determining the overall S3 for mortar cartridges.

The safety data package should apply the results of testing (final assessment) to the key areas of launch safety, flight safety, compatibility with cannon design limits, transportation, rough handling, and storage safety, and calculated Life Cycle Standard Deviation (LCSD) of pressure for the design. Data sources will come from component testing, system level testing, and appropriate modeling and analysis.

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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APPENDIX A BACKGROUND/RATIONALE.
ANNEX 1 GENERAL.

A.1.1. GENERAL.

This document was developed within the international community and is written primarily with references to North Atlantic Treaty Organization (NATO) test procedures. Table I3-1 (Appendix I, Annex 3) provides a cross reference of similar national and international test standards.

A.1.2. INTRODUCTION.

This Appendix provides background information and rationale for the sample quantities and test environments recommended by this document. Formal safety testing is required to establish test data, which supports the issuance of the safety certification. The tests may indicate that limitations or restrictions must be imposed when the safety certification is issued. These restrictions may be imposed to limit exposure to certain environments (climatic, dynamic, electromagnetic, etc.), to restrict methods of transportation, or to define special handling and operating procedures. Generally, because of increased severity associated with safety testing, satisfactory performance of the test item is not required. Poor performance after exposure to test environments may indicate a need for further investigation.

A.1.3. LIFE CYCLE ENVIRONMENTAL PROFILE.

During its expected life cycle, munitions will experience transportation from its place of manufacture to a storage facility, transportation to a place of temporary storage in an Operational Theatre, before tactical transportation within that Operational Theatre and finally function or return to storage. At each stage, it will experience exposure to various environments resulting from the local climate, general rough handling and transportation by numerous platforms. It may also experience abnormal environments such as being accidentally dropped.

A.1.3.1. Test Levels.

This appendix gives rationale for the recommended test procedures and test severities. The test levels are credible extreme environments to which the munitions may be exposed as part of the Life Cycle Environmental Profile (LCEP). Conflicts between the recommended test levels and munition specific LCEP environments should be addressed through test tailoring and/or safety release restrictions.

A.1.3.2. Temperatures.

Munitions are required to remain safe and suitable for service at extreme temperatures where personnel are expected to be capable of military operations, namely within NATO climate categories C2 to A1. It would be expected for the munition to remain safe and suitable during and following storage and transportation by various platforms within these climate categories. The extreme temperatures of these climate categories (or the SRE for hot stream munitions) form the basis for the conditioning temperatures for all mechanical environment tests intended to address

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logistic movements. Munitions are also expected to remain safe and suitable following storage at extreme cold conditions of a C3 climate category, but would not necessarily be expected to be moved during the coldest period within this climate zone due to difficulties with vehicles and the temperatures being outside the human comfort zone (i.e., survival as opposed to capable of military operations). For this reason, the cold temperature extreme for mechanical environmental tests has been based on the C2 climate category. Default temperatures to be used for each climate category if no additional data is readily available are shown in Table A-1.

TABLE A-1. CLIMATIC CATEGORY TEMPERATURES [2]

Climatic Category	Storage Temperature Conditioning Limits (ºC)	Firing Temperature (ºC) (see note 1)
A1/B3/M1	71	63
A2/B2/M2	63	56
A3	58	52
C0	-21	-19
C1/M3	-33	-32
C2	-46	-46
C3	-51	-51
M1	69	See note 3
M2	63	See note 3
M3	-32	See note 3

NOTE 1: Firing Temperature relates to firing munitions that have been protected from direct solar exposure. If it has been determined that a potential exists for direct exposure to solar radiation, then it is highly recommended that a solar radiation test be conducted to establish the maximum response temperature. This value should be used as the UFT.

NOTE 2: Prior to using this table, evaluate the safe operating temperatures of the energetic materials, as table temperatures may exceed the safe operating conditions of these energetic materials.

NOTE 3: The moderating effects of shipboard storage vary widely depending on position within the ship, type of ship, and climate control, therefore, the firing temperatures for M1, M2, and M3 are not relevant.

A.1.3.3. Temperature Stabilization.

For environmental tests that require temperature conditioning, temperature stabilization is achieved when the part of the munition considered to have the longest thermal lag is changing no more than 2 °C per hour. Since it may not be practical to monitor the interior parts of live munitions with the longest thermal lag during test without damaging seals, the stabilization time may be determined prior to live munition testing using an instrumented thermally equivalent inert munition. As an alternative, a default duration of 24 hours for unpackaged/bare, 48 hours for primary and secondary packaged, or up to a maximum of 72 hours for palletized may be applied

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after the chamber air around the test article has stabilized to the test temperature. Care should be taken that no munitions exceed the safe life of the energetic material when subjected to multiple exposures to high temperature conditioning.

A.1.3.4. Solar Radiation Equivalent (SRE) Temperature.

As an alternative to installing solar lamps in a test chamber, the SRE temperature is specified in most mechanical environment tests in order to facilitate testing. Determination of this value will require temperature measurement inside the package, and conduct of the full solar test requirement defined in, MIL-STD-810 and MIL-STD-2105. The SRE temperature should be determined for the packaged state and applied for all mechanical environment tests. In the absence of this data, a value of +71 °C should be used in lieu of the SRE temperature since this reflects the maximum value of the A1 Storage and Transit diurnal cycle defined in MIL-HDBK-310 and AR 70-38.

A.1.3.5. Ammunition Movements and Handling.

A list of typical Commercial and Military movements and rough handling modes are shown in Table A-2 below. In particular, Military movements can be subdivided to address Military Logistic movements by land, sea or air, and deployment of munitions on combat platforms (Tactical Operational movements). The former addresses logistic movement into and from a point of entry into Theatre of Operations (such as an airfield or naval vessel) to a bulk storage site. The latter addresses logistic movement beyond a bulk storage site in either a palletized configuration, or as ‘issued equipment’.

TABLE A-2. TYPICAL LCEP MOVEMENTS AND HANDLING

Mode	Paragraph	Restrained/ Unrestrained Cargo	Packaging Configuration
Commercial Logistic Dynamics	A.2.2		
Land Wheeled Vehicle Rail Sea Air	A.2.2.2. A.2.2.2.1. A.2.2.3. A.2.2.4. A.2.2.5.	Restrained	Primary Packaged/ Palletized
Packaged Transit Drop 2.1 m Safety 12 m Drop	A.2.2.1. A.4.3.	Unrestrained	

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ANNEX 1 GENERAL.

TABLE A-2. TYPICAL LCEP MOVEMENTS AND HANDLING (Cont.)

Mode	Paragraph	Restrained/ Unrestrained Cargo	Packaging Configuration
Military Logistic Dynamics	A.2.4.		
Air Fixed Wing Aircraft Jet Aircraft Turboprop Aircraft Helicopter Internal Sea Land Wheeled Vehicle Common Carrier All Terrain Shock Trailer	A.2.4.1. A.2.4.1.1. A.2.4.1.2 A.2.4.1.3. A.2.4.2. A.2.4.3. A.2.4.3.1. A.2.4.3.2. A.2.4.3.3. A.2.4.3.4.	Restrained	Primary Packaged/ Palletized
Air Fixed Wing Aircraft Parachute Delivery Helicopter Underslung and VERTREP	A.2.4.1. A.4.4. A.7.12.2.	Unrestrained	Primary Packaged/ Palletized
Rough Handling Dynamics	A.2.5.		
Loose Cargo	A.2.5.1.	Unrestrained	Primary and Secondary Packaged
Packaged Deployment 2.1 m Drop	A.2.5.2.		Primary Packaged
Tactical Dynamics	A.2.7.		
Land Vehicles Dismounted/Man-Portable Sea Air	A.2.7.1. A.2.9. A.2.7.2. A.2.7.3.	Restrained	Primary and Secondary Packaged/ Unpackaged
Tactical Drop	A.2.8.	Unrestrained	Secondary Packaged/ Unpackaged

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A.1.4. SAMPLE QUANTITIES AND STATISTICAL CONSIDERATIONS.

The sample size recommendations of this document are based on prior tests of similar cannons and munitions, rather than strictly statistical considerations. Serious hazards such as in-bore functioning are observed as binary (pass or fail) events, but the parameters that cause these events are unlikely to be so. For a simple binomial assessment, where the predicted low failure rate couples with a requirement for high statistical confidence, the sample sizes become very large, sometimes in excess of the eventual service population. This is not practical. Therefore, other approaches are required in combination with statistical methods to estimate the residual safety margin based on measured parameters. For sequential environmental testing, confidence is built by ensuring the test environment provides the maximum feasible cumulative stress to the test munition. Statistical methods are used to derive the test severities to ensure as far as practicable they envelope the predicted environment. However, as stated above, the final test quantities presented are a compromise based upon the experience of a large community of subject matter experts.

The objectives of the safety tests are to provide data to demonstrate a high degree of confidence that risks associated with the munition are sufficiently low. Statistical methods and other methodologies such as analogy may be used to derive sample quantities. The sample size must be sufficient to provide reasonable assurance that comparison of test results against safety requirements will be meaningful, and are to be approved by appropriate authorities.

A.1.4.1. Performance Test Data.

Data from additional performance tests (outside the scope of S3 testing) may be used to build the munition safety case. Poor performance may indicate a need for further investigative testing or analysis, as some performance limitations may reveal a hazardous safety condition.

A.1.4.2. Increased-Severity Testing.

In order to yield acceptable confidence in safety test results with a relatively small sample size, increased-severity testing is prescribed in this document. The probability of munition failure resulting in a hazardous condition is increased by testing under conditions, which are representative of credible extremes or slightly above the environments to be encountered in actual munition use. These extreme environments are low-probability environments. Therefore, the test levels recommended in this document are at credible extremes.

A.1.4.3. Sequential and Combined Environments.

Munitions are subjected to environmental testing in a sequential manner, which is representative of the probable LCEP scenario. Testing in accordance with this life cycle sequence and combining environments (e.g., vibration with temperature) is recommended to determine if the interaction

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(synergistic effect) and/or the sequence in which environments are experienced may result in a safety hazard.

A.1.4.4. Inspection for Incipient Failure.

For each test item which fails during test, there may be many items that have not failed, but are approaching the failure threshold. Inspection of the test munition before, during, and after test adds significantly to the confidence of the test data. Radiographic inspections provide particularly useful insight into the condition of the munition including early detection of displaced components as well as cracking or de-bonding of energetic materials. If the inspections indicate that failure is likely to occur, further investigation or testing may be required. If the inspections indicate that a margin of safety exists and that no safety hazard is likely to occur, additional confidence in the data is gained.

A.1.5. VARIABLE TEST DATA.

The use of measured variable data (e.g. pressure, force, etc.) is recommended whenever practical. If margins of safety can be demonstrated between measured test data and measured or analytical failure modes, confidence in the test results are enhanced. If measured variable data indicate only small margins of safety exist, further investigation or testing may be required.

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APPENDIX A. BACKGROUND/RATIONALE.
ANNEX 2. SEQUENTIAL ENVIRONMENTAL TESTS.

A.2.1. GENERAL.

Provided below are the rationale for the Sequential Environmental Tests related to normal usage in typical environmental conditions, or from plausible mishandling during logistic and field operations. The munition should be temperature conditioned at either UCT or the LCT and fired at the peak safety pressure – see Paragraph C.2.8.

The packaging configuration (palletized, primary packaging, secondary packing or bare items) for the specific sequential environmental test should match the most likely packaging configuration that is likely for the mortar cartridge during the LCEP.

Table A-2 in Appendix A describes the typical LCEP movements and handling that forms the basis for the mechanical environmental tests. Life cycle distances for each mode of transport are specified in AECTP 100, MIL-STD-810 and MIL-STD-2105. Each of these environments must be addressed as applicable. However, S3 testing may not require full life cycle distances. The required distance for S3 testing may be based on the number of expected deployments. The test requirements (i.e., distance and time) should be multiplied by the number of expected deployments considered necessary to meet the S3 test program requirements. The National Authority(ies) or other appropriate authorities shall approve the requirements prior to the start of testing.

A.2.2. COMMERCIAL LOGISTIC DYNAMICS.

A.2.2.1. Packaged Transit Drop Dynamics.

The Transit Drop test would normally be conducted in accordance with MIL-STD-810, Method 516, Package Transit Drop, which would simulate an accidental drop encountered in logistical (packaged) handling of the munitions such as a hovering helicopter dropping the munitions from a sling or the unloading of munitions stacked on a truck. The recommended drop height of 2.1 m is based on the scenario of munitions being dropped from the bed of a transport vehicle. Since the 2.1 m packaged drop environment is addressed during the Rough Handing Dynamics portion of the sequential environmental test series, this test is not required to be conducted as part of the Commercial Logistic Dynamics.

A.2.2.2. Land Logistic Dynamics.

A.2.2.2.1. Wheeled Vehicle Dynamics.

The movement of palletized munitions from the point of manufacture to the storage location is usually accomplished by commercial logistic vehicles over improved or paved highways.

Testing of Logistic Dynamics shall be conducted in accordance with MIL-STD-810 and MIL-STD-2105. The AECTP 100 distances are considered the lifetime requirements for the

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APPENDIX A. BACKGROUND/RATIONALE.
ANNEX 2. SEQUENTIAL ENVIRONMENTAL TESTS.

Commercial Logistic Dynamics, Military Logistic Dynamics, Rough Handling and the Tactical Operational Dynamics. Testing using these lifetime distances is not realistic as munitions will not experience all of these LCEP movements and handling scenarios to the maximum extent. Table A2-1 shows default distances for a single deployment on land.

TABLE A2-1. LAND TRANSPORTATION DEFAULT DISTANCES FOR A SINGLE DEPLOYMENT

Lifetime Requirement[1]	Transport Mode[2,3]	Single Deployment Distance[4]		
		Primary Packaging	Secondary Packaging	Total
Commercial Logistic Land Dynamics				
10,000 km Commercial Vehicle	Ground Wheeled Vehicle – Common Carrier	4,800 km (palletized)	n/a	4,800 km minimum
Military Logistic Land Dynamics				
10,000 km Military Vehicle	Ground Wheeled Vehicle – Common Carrier	4,800 km (palletized)	n/a	4,800 km minimum
	Tactical Wheeled Vehicle – All Terrain	800 km (palletized)	n/a	800 km minimum
	Two Wheeled Trailer	50 km (palletized)	n/a	50 km minimum
Tactical Combat Platform Deployment				
5,000 km Military Vehicle	Unrestrained/Loose Cargo	200 km	50 km	250 km minimum
	Dismounted/Man-portable	n/a	250 km	250 km minimum
	Tracked Vehicle[5]	800 km	200 km	1,000 km minimum

NOTE 1: The Lifetime Requirements are taken from AECTP 100 and are provided as an example only. The most current AECTP 100 value should be considered as the default Lifetime Requirement.

NOTE 2: Table A2-1 summarizes the distances assumed for a single deployment of mortar munitions on land combat platforms. This only considers deployment by land platforms and does not address the requirements for deployment on sea and air platforms.

NOTE 3: Table A2-1 provides a set of assumptions regarding the subdivision of distances across multiple land platforms, including deployment on military combat vehicles and the dismounted/man-portable configuration for a single deployment.

NOTE 4: The stated distances are assumed to represent the worst case likely to be experienced during one single deployment cycle. For multiple deployments, additional distance may be added as determined by the munition LCEP

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to give the Lifetime Requirement. In some cases, a follow on surveillance program will address a multiple deployment scenario.

NOTE 5: The combat platform is a complex environment. It is the most likely to require a tailored test approach for the Tactical Operational Dynamics. Where there is no opportunity to tailor then it is considered that the Tracked Vehicle (Materiel Deployed on the Sponson or Installed in Hull of Light Tracked Vehicle) is the most appropriate test levels from the AECTP to use as a default.

A.2.2.2.2 Land Dynamics Testing.

Land dynamics test durations can be addressed by the 'Ground Wheeled Common Carrier' vibration profiles in MIL-STD-810, Methods 514 and 527. No factors of safety need to be applied to the amplitude to AECTP 400 vibration schedules as specified. These vibration schedules have been developed from field data and have conservatism factors built into them. Common Carrier vibration schedule should be applied for a duration equivalent to the distances shown in Table A2-2. This is the first series of tests to be performed in the munition life cycle test sequences described in Appendix B. The intent is to degrade the shipping packaging prior to the climatic environmental tests.

TABLE A2-2. LAND TRANSPORTATION TEST DURATION EXAMPLES BASED ON AECTPS 100-4 AND 400-3

Deployment Phase	AECTP 100 Land Transport Distance (Lifetime)[1]	Transport Mode	Default Distances for Single Deployment	AECTP 400 Test Relation to Field Exposure[3]	JOTP-23 Minimum Test Durations[2]
Logistic (Commercial Vehicle)	10,000 km	Secured Cargo – Common Carrier (packaged or palletized)	4,800 km	Vibration: Vertical Axis 1hr/axis = 4,000 km Longitudinal and Transverse Axes 1hr/axis = 1,609 km	Vertical Axis 24 min/axis Longitudinal/Transverse: 60 min/axis
				Number of Restrained Cargo Shocks to be equivalent to 1,600 km	No requirement if military wheeled vehicle transport shocks are performed.

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TABLE A2-2. LAND TRANSPORTATION TEST DURATION EXAMPLES BASED ON AECTPS 100-4 AND 400-3 (Cont.)

Deployment Phase	AECTP 100 Land Transport Distance (Lifetime)[1]	Transport Mode	Default Distances for Single Deployment	AECTP 400 Test Relation to Field Exposure[3]	JOTP-23 Minimum Test Durations[2]
Logistic (Military Vehicle)	10,000 km	Secured Cargo – Common Carrier (packaged)	4,800 km	Vibration: Vertical Axis 1hr/axis = 4,000 km Longitudinal and Transverse Axes: 1hr/axis = 1,609 km	Vertical Axis 24 min/axis Longitudinal/Transverse 60 min/axis
		Secured Cargo – Tactical Wheeled Vehicle (includes four wheeled trailers) (packaged)	800 km	Vibration: All Axes 40 min/axis = 805 km	40 min/axis
				Number of Restrained Cargo Shocks in Table C2-1 = 800 km	No requirement for packaged if conducted on unpackaged rounds
		Two Wheeled Trailer (packaged)	50 km	Vibration: All Axes: 32 min/axis = 52 km	32 min/axis
Tactical Deployment	5,000 km	Secured Cargo – Tactical Wheeled Vehicle (includes four wheeled trailer) (deployed transport configuration) [6]	800 km	Vibration: All Axes: 40 min/axis = 805 km	40 min/axis
				Number of Restrained Cargo Shocks in Table C2-1 = 800 km	See Table C2-1
		Two Wheeled Trailer (deployed transport configuration)[6]	50 km	Vibration: All Axes: 32 min/axis = 52 km	32 min/axis
		Secured Cargo – Tracked Vehicle[4] (primary packaging)	500 km	Vibration: All Axes: 45 min/axis = 1,600 km	14 min/axis[5]
		Secured Cargo – Tracked Vehicle[4] (secondary packaging)	500 km	Vibration: All Axes: 45 min/axis = 1,600 km	14 min/axis[5]
		Loose Cargo (primary packaging)	125 km	20 min=240 km	10 min[5]
		Loose Cargo (secondary packaging)	125 km	20 min=240 km	10 min[5]

NOTE 1: AECTP 100-4 distances are provided as examples only and reflect potential cumulative life distances. Use distances from the munition LCEP.

NOTE 2: JOTP-23 minimum test requirements reflects the distances likely to be experienced during one deployment cycle. For multiple deployments, additional mileage may be added as determined by the munition LCEP. In some cases, a follow-on surveillance program will address a multiple deployment scenario.

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NOTE 3: AECTP 400-3 time/distance relationships are provided as examples only. Use the most current AECTP 400 values.

NOTE 4: The combat platform is a complex environment. It is the most likely to require a tailored test approach for the Tactical Operational Dynamics. Where there is no opportunity to tailor then it is considered that the Tracked Vehicle (Light Vehicle – Material on Sponson or installed in Hull) is the most appropriate test levels from the AECTP to use as a default.

NOTE 5: Test items may be transported in their primary and/or secondary packaging configurations. If so, divide the test times appropriately among the configurations.

NOTE 6: Tactical Deployment (Tactical Wheeled Vehicle / Two Wheeled Trailer), profile should only be conducted when it is known that munition will be transported outside of its primary packaging.

A.2.2.3. Commercial Logistic Dynamics – Rail.

A.2.2.3.1. Rail Transport Vibration Testing.

Rail transport vibration would normally be conducted in accordance with MIL-STD-810, Methods 514 and 527. Based on an assessment that this environment is relatively benign compared to other S3 test environments, this test was eliminated as a requirement for mortar munitions. Although not required for S3, this test may be required for US military transportation certification.

A.2.2.3.2. Rail Impact Testing.

Rail impact testing would normally be conducted in accordance with MIL-STD-810, Method 526. Based on an assessment that this environment is relatively benign compared to other S3 test environments, this test was eliminated as a requirement for mortar munitions. Although not required for S3, this test may be required for US military transportation certification. Inert assets with similar mass and structural characteristics are typically acceptable for transportation certification.

A.2.2.4. Commercial Logistic Dynamics – Sea.

Based on an assessment that this environment is relatively benign compared to other S3 test environments, testing for this environment is generally excluded. If Sea Transportation Vibration testing is considered necessary, this should be conducted in accordance with MIL-STD-810, Methods 514 and 527, using the distances specified in AECTP 100-4 for Cargo Ships in the Transportation Mode.

A.2.2.5. Commercial Logistic Dynamics – Air.

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Often there is little information on the vibration environment suggested by Commercial Logistic Transportation by air. The nature of Military Logistic Transportation by air suggests it is more severe than the Commercial Logistic alternative. Therefore, it is not normal to test mortars for Commercial Logistic Transportation by air.

A.2.3. LOGISTIC STORAGE.

High and low temperature testing is carried out as part of the sequential environmental trial to simulate the thermal exposure, and aggravate associated degradation mechanisms likely to be encountered during the munition's life cycle.

Throughout their life cycle, munitions will be exposed to hot and/or cold thermal environments during storage, transportation and deployment. Periods of deep storage will most likely assume relatively stable (quasi-static) temperature conditions due to the protection offered by most storage facilities, and may be simulated by constant temperature testing. Periods of transportation, deployment and/or field storage typically follow the daily cyclic temperature variation (diurnal cycles) of the local climate since the level of protection is reduced, and may be simulated by cyclic temperature testing. Note that the temperature cycles likely to occur during transportation, deployment and/or field storage will most probably differ from the prevailing meteorological conditions due to a number of localized factors (e.g., packaging construction, transportation/storage mode, climatic factors such as solar radiation, etc.), and induced temperature cycles are recommended on this basis. Further guidance relating to the climate categories, and their derivation, as used by NATO, can be found in MIL-STD-810 and MIL-STD-2105.

High temperature exposure (cyclic and constant) tends to aggravate chemical degradation (e.g., stabilizer depletion in nitrate-ester propellants) and/or physical processes (e.g., diffusion of chemical substances, such as plasticizers). Cyclic high temperature exposure also aggravates thermo-mechanical degradation (e.g., thermally induced fatigue) due to differential expansion and contraction of materials and interactions between different materials. Little appreciable chemical ageing occurs at cold temperatures, and thermo-mechanical effects predominate – for example, cold strain cracking due to long periods at constant cold temperatures or thermal fatigue for cyclic exposure.

The specific tests that should be applied depend upon the LCEP (e.g., climate category, and protected vs. unprotected storage/transportation), the outcome of the Preliminary Design Safety Assessment and Hazard Analysis. Whatever ageing tests are conducted as part of the sequential trials program, the resulting life predictions must be compared with the results of surveillance to determine how accurate they were and whether any potential failure modes were missed.

It should be noted that laboratory based ageing studies using small samples of material do not take account of the geometry of the component and some potential degradation mechanisms could be missed. Furthermore, it should be noted that conditioning time for mechanical environmental tests

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should not be counted towards life estimates since this is effectively a thermal ramp and it can prove difficult to determine the amount of thermal energy imparted to the munition. Therefore, it is difficult to model the equivalent ageing likely to have occurred within the munition.

A.2.3.1. Cold Constant Temperature Logistic Storage.

Munitions may be exposed to Climate Category C3 conditions during deep storage, although exposure to such extreme conditions is unlikely during transportation and deployment. In Climate Category C3, a constant low temperature of -51 °C is likely to predominate for a significant duration due to a lack of sunlight and solar radiation during the coldest period of the year (MIL-HDBK-310 and AR 70-38), with both meteorological and induced conditions becoming aligned. Category C3 applies to the coldest area of the North American continent and the areas surrounding the coldest parts of Siberia and Greenland. A minimum of 72 hours is recommended since this is considered sufficient duration to thermally stabilize the munition and has proven sufficient to demonstrate short-term safety in cold climates. Note that this duration is unlikely to aggravate degradation mechanisms associated with constant strain during long term storage at cold temperatures, and longer durations may be required (it is recommended that specific guidance should be sought from the munition Design Authority). If it is identified during the Design Safety Assessment that the munition under test could be susceptible to thermo-mechanical stresses due to low temperature fluctuations, the C2 low temperature cycle or that defined in the LCEP is recommended.

A.2.3.2. Cold Cyclic Temperature Logistic Storage.

Munitions may be exposed to Climate Category C2 during storage, transportation and deployment. The induced air temperature diurnal cycle for Category C2 storage and transit conditions given in MIL-STD-810 and MIL-STD-2105 is considered to adequately encompass most conceivable situations that may occur during transportation and ‘field’ storage. Note that at these extreme cold conditions, the meteorological and induced diurnal cycles become aligned. If it is identified during the Design Safety Assessment that the munition under test could be susceptible to cold strain effects due to constant low temperature (such as during storage), C3 constant low temperature exposure or that defined in the LCEP is recommended.

A.2.3.3. Hot Constant Temperature Logistic Storage.

The high temperature cycling test is intended to determine the effects of thermo-mechanical stresses on the munition. The induced air temperature diurnal cycle for Category A1 storage and transit conditions given in MIL-STD-810 and MIL-STD-2105 is considered to adequately encompass most conceivable situations. For other environments, such as naval controlled environments, other storage categories may be considered and are LCEP dependent.

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The high temperature storage test is intended to accelerate chemical and physical based degradation mechanisms via a period of testing using a constant elevated temperature. A constant temperature of +71 °C is the maximum temperature that should be considered since this reflects the peak temperatures likely to be encountered during field storage or deployment in an A1 climate zone. Alternatively, a constant temperature of +58 °C may be more appropriate where the use of +71 °C is thought to generate unrealistic degradation.

It should be noted that laboratory based ageing studies using small samples of material do not take account of the geometry of the component and so some potential degradation mechanisms could be missed. Furthermore, it should be noted that the thermal ramp conditioning time should not be counted towards life estimates since it can prove difficult to determine the amount of thermal energy input to the munition. Therefore, it is difficult to model the equivalent ageing likely to have occurred within the munition. Whatever ageing tests are conducted as part of the sequential trials program, the resulting predictions must be compared with the results of in-service surveillance (ISS) to determine how accurate they were and whether any potential failure modes were missed.

A.2.3.4. Hot Cyclic Temperature Logistic Storage.

For most munitions, 28 hot A1 induced diurnal cycles are considered sufficient to induce thermo-mechanical stressing representative of that which could occur in service. For chemical and/or physical ageing processes (e.g., stabilizer depletion or diffusion of chemical substances) longer durations are necessary to produce sufficient observable change; and 56 hot diurnal cycles have historically provided sufficient confidence to support an initial deployment of up to at least 6 months tactical storage. Chemical and physical processes may be simulated by constant temperature stressing, but care must be exercised since such stressing may induce unrepresentative failure modes or may not adequately exercise potential failure modes. Consideration must be given to the design of the munition and any design limitations. For example, gas cracking, phase changes or changes in the chemical reaction mechanism can occur during constant temperature ageing which may not occur during diurnal cycling or in service. This test should not be conducted instead of high temperature cycling, but may be used to supplement the chemical ageing effects of diurnal cycling tests. If the munition under test could be susceptible to high temperature fluctuations, then the A1 storage and transit (induced) cycle or that defined in the LCEP should be used.

If opting to substitute some of the hot diurnal cycles for fixed temperature stressing, only 28 of the 56 cycles should be substituted (with the remaining 28 cycles being applied along with the constant thermal stressing). Using the Arrhenius kinetic model discussed in MIL-STD-810, MIL-STD-2105 and an activation energy of 70 kJ/mol; constant temperature stressing may be applied for 216 hours (9 days) at +71 °C, or 528 hours (22 days) at +58 °C where unrealistic degradation is anticipated at +71 °C.

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A.2.3.5. Humid Heat Cycling.

The humid heat cycling test is performed to determine the resistance of materiel to the effects of a warm humid atmosphere. Moisture can alter burning characteristics of propellants. High humidity may promote corrosion degradation. Munitions may be exposed to this environment year-round in tropical areas and seasonally in mid-latitude areas. The procedure recommended by this document is an aggravated test. It does not reproduce naturally occurring or service-induced temperature-humidity scenarios. In order to reduce the time and cost of testing, the test munition is exposed to higher temperature and humidity levels than those found in nature; however, the exposure duration is shorter. A minimum of ten test cycles has proven to be effective at inducing degradation/failures that are indicative of long-term effects. Consideration must be taken for the type of seals to ensure that the duration is adequate and therefore longer test durations may be required to obtain a higher degree of confidence that the munition will remain S3 in warm-humid conditions.

A.2.4. MILITARY LOGISTIC DYNAMICS.

A.2.4.1. Air Logistic Dynamics.

Mortar munitions may be subjected to Military Air transportation by either fixed wing transport aircraft (jet and propeller) or helicopters as determined by the LCEP. Distances for each mode of transport are specified in AECTP 100. Each of these environments must be addressed as applicable.

Table A2-3 summarizes the military air transportation dynamics requirements as an example based on the current versions of AECTP 100.

TABLE A2-3. AIRCRAFT CARGO TRANSPORTATION TEST DURATION EXAMPLES BASED ON AECTP 100.

Transport Mode	AECTP 100[1] Flight Durations for Land Vehicle and Naval Ammunition	AECTP 400[2] Test Relation to Field Exposure	JOTP-23 Test Durations[3]
Fixed Wing Cargo Jet	100 hours	1 min/takeoff (10 hr flight/takeoff)	10 min/axis
Fixed Wing Cargo Turboprop	100 hours	1 hr/axis = 20 hrs flight	5 hrs/axis
Helicopter Internal Cargo	20 hours	1 hr/axis = 6 hrs flight	1.5 hrs/axis

NOTE 1: AECTP 100-4 Durations provided as examples only. The most current MIL-STD-810 values should be applied.

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NOTE 2: AECTP 400 Edition D Time/Duration Relations provided as examples only. The most current AECTP 400 values should be applied.

NOTE 3: See text in Paragraphs A.2.4.1.1, A.2.4.1.2, and A.2.4.1.3 for duration rationales.

A.2.4.1.1. Fixed Wing Aircraft – Jet Aircraft Vibration.

The vibration environment associated with cruise is largely addressed by other vibration environments within the LCEP and need not necessarily be tested. The take-off vibration environment is significantly more severe than that for cruise, and can be addressed by the vibration profiles in MIL-STD-810, Methods 514 and 527, for 'Jet Aircraft Cargo – Take-off'. The duration of this test is determined based on the number of take-off events. The number of take-off events in the life of a mortar cartridge may be estimated from the total flight duration defined in AECTP 100 Annex C, for each commodity type transported by 'Jet Aircraft' divided by an assumed average flight duration of 10 hours per flight.

A.2.4.1.2. Fixed Wing Aircraft – Turboprop Aircraft Vibration.

The most common propeller cargo aircraft used throughout NATO is the C130, of which the four (4) and six (6) bladed propeller variants are most typical. The vibration severities for these aircraft are defined in MIL-STD-810, Methods 514 and 527, for 'Propeller Aircraft'. If other cargo aircraft are identified as part of the LCEP, then the blade frequencies for these shall also require consideration. Since it is not always possible to predetermine the specific aircraft types that will be used during transportation, the total test duration based on the total flight duration defined in AECTP 100 for each commodity type transported by 'Propeller Aircraft' should be split between the different blade frequencies identified. For C130, as an example, this will require the test to be divided equally between the two (2) blade frequencies (68 Hz and 102 Hz) as a minimum.

A.2.4.1.3. Helicopter Vibrations.

Mortar munitions may be transported by a variety of helicopters as part of its LCEP. Common cargo helicopter types used throughout NATO can be grouped according to their fundamental blade frequencies as per Table A2-4. The vibration environment for these cargo helicopters can be addressed by the vibration profiles in MIL-STD-810, Methods 514 and 527 for 'Helicopter Cargo'. If other cargo helicopters are identified as part of the LCEP, then the blade frequencies (f_1) for these shall also require consideration but only if they are sufficiently different to the 11 Hz, 17 Hz, and 21 Hz already identified. Since it is not always possible to predetermine the specific aircraft types that will be used during transportation, the total test duration based on the total flight duration defined in AECTP 100 and MIL-STD-810 for each commodity type transported by 'Helicopter' should be split between the different helicopter types identified. In the current release of AECTP 100, mortar munitions are expected to be transported up to 20 hours as internal cargo by helicopter over a lifetime. However, mortar munitions are assumed to be transported no more than 10 hours

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by helicopter in a single deployment. Based on this guidance and the test equivalence of one (1) hour vibration for 6 hours of flight, helicopter cargo transportation would be conducted for 0.5 hour per axis at each of the three (3) blade frequencies for a total of 1.5 hours per axis.

TABLE A2-4: HELICOPTER MAIN ROTOR PARAMETERS

Helicopter[1]	**Main Rotor**			
	Rotation Speed, Hz	Number of Blades	f_1, Hz	S3 Test Frequency, f_1
UH-1 (Huey)	5.40	2	10.80	11 Hz
CH-47D (Chinook)	3.75	3	11.25	
CH-46 (Sea Knight)	4.40	3	13.20	
UH-60 (Black Hawk)	4.30	4	17.20	17 Hz
Sea King/Commando	3.48	5	17.40	
Puma	4.42	4	17.68	
EH101 (Merlin)	3.57	5	17.85	
NH-90	4.26	4	17.04	
CH-53E (Super Stallion)	3.00	7	21.00	22 Hz
Lynx Mk 1, Mk 2, and Mk 3	5.51	4	22.04	

NOTE 1: This list is not exhaustive. Additional helicopter types may be considered based on the LCEP

A.2.4.2. Sea Logistic Dynamics.

Based on an assessment that this environment is relatively benign compared to other S3 test environments, testing for this environment is generally excluded. If Sea Logistics Dynamics testing is considered necessary, this should be conducted in accordance with MIL-STD-810, Methods 514 and 527, using the distances specified in AECTP 100-4 for Cargo Ships in the Transportation Mode.

A.2.4.3. Land Transport Dynamics.

Transportation as secured cargo from a point of arrival into the Theatre of Operations (e.g., friendly port or airfield) up to a bulk storage area may be by Military Transport Vehicles and towed trailers travelling over a variety of terrain such as improved or paved highways, and/or degraded roads.

This mechanical environment is defined by both vibration and shock elements. In particular, a shock element to address minor obstacle negotiation for vehicles travelling in an off-road role is necessary. Wheeled Vehicle testing requires separate vibration and shock elements to fully simulate the environment, and as such neither element can be tailored out.

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A transportation distance of 10,000 km is suggested in AECTP 100-4 for the lifetime requirement. Table A2-1 summarizes the military vehicle transportation distances assumed for only a single deployment of the munition. Tactical (Military) Logistic Transportation by paved roads and improved highways is comparable to Commercial Logistic Transportation by Wheeled Vehicle.

A.2.4.3.1. Wheeled Vehicle Common Carrier Dynamics.

Testing to the 'Ground Wheeled Common Carrier' vibration profiles in MIL-STD-810, Methods 514 and 527 covering the distance given for Tactical Wheeled Vehicle – Common Carrier in Table A2-1 is considered sufficient to cover Commercial and Military wheeled vehicles for a single deployment.

A.2.4.3.2. Wheeled Vehicle – All Terrain Dynamics.

The vibration element for variable terrain such as degraded roads may be addressed by the vibration profiles in MIL-STD-810, Methods 514 and 527 for 'Tactical Wheeled Vehicle'. Testing using the distance given in Table A2-1 is considered sufficient.

A.2.4.3.3. Wheeled Vehicle – Shock.

The Restrained Cargo Transport Shock levels in MIL-STD-810, Methods 516, are not currently considered sufficient to satisfy the intent of this test. The number of shocks specified in this document represents those typical for 800 km of transportation and should be adjusted to match the distance to be covered by the vibration test.

A.2.4.3.4. Two Wheeled Trailer – Vibration.

The vibration element for trailers may be addressed by the vibration profiles in MIL-STD-810, Methods 514 and 527 for 'Two Wheeled Trailer'. Testing using the distance given in Table A2-1 is considered sufficient.

A.2.5. ROUGH HANDLING DYNAMICS.

A.2.5.1. Loose Cargo Bounce.

Munitions may be subjected to unrestrained transportation within its primary packaging and secondary packaging, which may result in sliding or rolling within boundaries of a vehicle. This may be simulated by Loose Cargo testing according to MIL-STD-810, Method 514.

A.2.5.2. Rough Handling Packaged Drop.

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Mortar munitions in their primary packaging may be subjected to accidental drops during handling, such as during resupply. This may be simulated by free-fall impact testing according to MIL-STD-810, Method 514. The drop heights used will be LCEP specific. The minimum recommended drop height is 2.1 m for all munitions (to represent the scenario of munition being dropped from the bed of a transport vehicle), and the munitions should remain safe to fire after being dropped.

A.2.6. Tactical Storage.

Tactical Storage testing in the primary packaging is considered unnecessary because its test procedures would be identical to the Logistic Storage testing, conducted during the logistics phase, and it is considered unnecessary to repeat the identical test. However, it is possible that mortar cartridges may be unpacked from their primary packaging and stored for extended periods within the secondary packaging. It is anticipated that the secondary packaging may be susceptible to the humid heat environment when removed from the primary packaging. This environment is addressed by non-sequential humid heat testing of testing of mortar cartridges packed within the secondary packaging.

A.2.7. TACTICAL DYNAMICS.

Mortar munitions are most commonly deployed on land, either man portable or on a mobile mortar platform. Less commonly, mortar munitions may be deployed on sea platforms. When deployed on platforms, the munitions may be packed within primary or secondary packaging that is stowed on the platform within munition racks or other stowage provision (e.g., straps to restrain the packaged munition). Alternately, the munitions may be stowed on a platform in a ready-for-immediate-use configuration (e.g., stowed within a ready rack or auto loader mechanism). These multiple configurations make it difficult to set pre-determined test schedules for the final elements of the LCEP. Furthermore, the dynamic environment associated with the combat platform is highly specific to that particular platform. Therefore, ‘typical’ test requirements are recommended below to address stowed munitions since this is likely to be the configuration that will be most prevalent before the munitions are consumed.

A.2.7.1. Land – Tracked Vehicles Vibration.

During operational use, mortar munitions may be moved by, and fired from wheeled and/or tracked combat platforms while stationary. When mortar munitions are designated to be fired from a stationary platform, the munitions are typically stowed and secured during movement, such as within platform specific munition stowage racks (e.g., packed within primary or secondary packaging). The munitions may also be secured on the platform, such as within a “ready rack” or within a mortar autoloader magazine, in a ready to fire configuration. Tailoring of the vibration and shock environments, based on measured data, is required accounting for different vehicles, stowage configurations (e.g., in racks, launch tubes, canisters, or rails on a turret). MIL-STD-810 provides guidance regarding deriving tailored vibration test severities, while vibration testing

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should be conducted in accordance with MIL-STD-810, Methods 514 and 527 as appropriate depending upon munition physical size. Shock testing should be conducted in accordance with MIL-STD-810, Method 516. Where platform specific data are not available, then default severities for wheeled and tracked vehicles selected from MIL-STD-810, Methods 514, 516 and 527 may be appropriate for vibration testing, and restrained cargo shocks per MIL-STD-810, Method 516 may be applied using those severities applicable to wheeled vehicles. Both tests should simulate the equivalent operational mode distance specified in Edition 4 of AECTP 100, Annex E, Appendix 1, for a mortar cartridge on a 'Combat Platform' and MIL-STD-810. The ready to fire configuration is assumed to account for at least 20% of the distance travelled on the combat platform. Tailor this distance in accordance with the LCEP.

A.2.7.2. Sea Tactical Dynamics.

Mortar munitions may be deployed on surface ships and small boats either in a stowed and restrained configuration (e.g., packed within primary or secondary packaging), or loaded in mortar magazines (e.g., mortar autoloader mechanism) in a ready-to-fire configuration. Munitions may also be stowed on Naval vessels for subsequent use by an embarked military force (see Paragraph A.2.9 for discussion of Mortar Man Portable Dynamics).

The vibration environment associated with larger surface ships and submarines tends to be relatively benign and is likely to have been addressed by sea logistic dynamics testing and other testing within the LCEP so specific testing is unlikely to be required. Larger surface ships may be subject to relatively severe shock loadings in rough seas, and tailored testing is recommended. Smaller boats tend to demonstrate quite severe vibration and shock loadings due to their size and mass, and tailoring is recommended.

Tailoring of the vibration environment based on measured data should be considered (using guidance provided in MIL-STD-810, Methods 514 and 527. No life equivalence is assigned to such testing but a duration of one (1) hour per axis has been applied historically. The majority of naval shocks will be addressed by other shock tests within the LCEP, although tailored testing of the higher amplitude shocks likely for surface vessels is recommended.

A.2.7.3. Air Tactical Dynamics.

Mortars are not normally operated from aircraft therefore no testing is required.

A.2.8. TACTICAL DROP.

The Tactical Drop test simulates accidental drops encountered during handling of mortar munition during loading and firing operations. The drop heights used are tailored according to the LCEP, but it is recommended that these should be no less than 1.5 m. Since bare munitions and munition packed within secondary packaging may be handled during loading and firing operations, it is

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recommended that drop tests be conducted in both conditions. Munitions dropped in the secondary packaging is expected to remain undamaged after the drop. The munitions dropped in the bare condition should remain safe to dispose after dropping. The munitions may be fired during testing, provided the damage to the test item has not prevented the mortar cartridge from being loaded into the cannon. This would help to assess the possibility that dropped munitions were not disposed and could be inadvertently fired. Drop heights may be tailored taking due account of the maximum muzzle height of the mortar with the tailoring rationale being documented in the S3 Safety Data Package.

A.2.9. MORTAR MAN PORTABLE DYNAMICS.

The LCEP for mortar munitions potentially includes transportation within backpacks that are utilized by soldiers to transport personal items. Soldiers may transport a small quantity of mortar cartridges and mortar components within the backpacks during field operations. Additionally, the loaded backpacks may also be transported as loose cargo within field vehicles. Thus, the loaded backpacks may be exposed to a “Soldier Transport” shock and vibration environment when transported by soldiers and a loose cargo transport environment within vehicles. Additionally, the loaded backpacks may be accidently dropped.

For the purpose of this document, it is assumed that testing to assess the effects of man portable dynamics is covered by loose cargo vibration and 1.5 m drop within secondary packaging.

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ANNEX 2. SEQUENTIAL ENVIRONMENTAL TESTS.

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APPENDIX A. BACKGROUND/RATIONALE.
ANNEX 3. NON-SEQUENTIAL CLIMATIC TESTS.

A.3. NON-SEQUENTIAL CLIMATIC TESTS.

A.3.1. GENERAL.

Provided below are the rationale for the Non-Sequential Climatic Tests related to normal usage in typical environmental conditions or from plausible mishandling during logistic and field operations. If only one (1) test configuration is to be used this must be the most severe configuration, packaged or unpackaged, for the munition.

The following are presented as non-sequential tests. If the design assessment identifies that the munition will be susceptible to specific environments, the relevant non-sequential test(s) should be included in the sequential environmental testing.

A.3.2. SAND AND DUST.

Munitions may be exposed to sand and dust environments on a worldwide basis. The greatest exposure would be expected during operations in desert regions due to vehicle convoys and aircraft/helicopter movements. The movement of military vehicles in hot dry desert regions or in areas where the surface is liable to break up into small particulate is liable to result in dust and sand-laden atmospheres. Material deposited inside the munition may cause short-circuiting, build-up of static electricity, and interference between moving parts. This Joint Ordnance Test Procedure (JOTP) requires the munition to be tested in the most severe configuration; that is outside of its shipping/storage container, using the most severe exposure parameters defined in MIL-STD-810, Method 510. The sand and dust test determines the effects on munitions after exposure to dust and sand atmospheres. Dust consists of particle sizes less than 150 microns. Sand has particle sizes greater than or equal to 150 microns.

A.3.3. MOLD/FUNGUS GROWTH.

Microbial deterioration is a function of temperature and humidity and is an inseparable condition of hot-humid tropics and the mid-latitudes. MIL-STD-810, Method 508 is used to determine if mold growth will occur and, if so, how it may degrade/impact the use of the munition. Twenty-eight (8) days is the minimum test period to allow for mold germination, breakdown of carbon-containing molecules, and degradation of material.

A.3.4. BARE IMMERSION.

The immersion test (MIL-STD-810, Method 512) determines whether water is likely to penetrate the cartridge, affect materials, and/or affect performance of the munition during temporary immersion. This test is also intended to cover the effects of rain. Temperature conditioning the munitions to 27 °C above the water temperature represents exposure to solar heating immediately prior to immersion, and induces a slight negative pressure differential within the munition (on

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cooling) to aggravate potential water ingress. Thirty minutes of immersion at a depth of one (1) meter is required for routine fording operations/maneuvers.

A.3.5. SALT FOG.

The salt fog test (MIL-STD-810, Method 509) provides a set of repeatable conditions to determine the relative resistance of the munition to the effects of an aqueous salt atmosphere. This test helps to identify potential degradation mechanisms within a relatively short period of time, and is required for munition/components that will experience exposure to high levels of salt in the atmosphere. It should be noted that testing at the component level will not address galvanic corrosion.

As a minimum, this JOTP requires alternating wet-dry-wet-dry conditions of 24 hours each to be imposed. Alternating periods of salt fog exposure and drying conditions provides a higher damage potential than does continuous exposure to a salt atmosphere. The munition should be tested in the most severe configuration; that is, outside its shipping/storage container. The number of cycles may be increased if a higher degree of confidence is required to assess the ability of the materials involved to withstand a corrosive environment (e.g., naval-specific environment). Note, there is no relationship between this test and any real world exposure duration but it does provide an indication of potential degradation mechanisms associated with the salt (maritime) environment, nearby water sources, and from salted roads during winter operations.

A.3.6. HUMID HEAT CYCLING – BARE ROUND AND SECONDARY PACKAGED ROUND.

The humid heat test (MIL-STD-810, Method 507) is performed to determine the resistance of materiel to the effects of a warm humid atmosphere. Materiel may be exposed to this environment year-round in tropical areas and seasonally in mid-latitude areas. The procedure recommended by this document is an aggravated test. It does not reproduce naturally occurring or service-induced temperature-humidity scenarios. In order to reduce the time and cost of testing, the test item is exposed to higher temperature and humidity levels than those found in nature; however, the exposure duration is shorter. A minimum of ten test cycles has proven to be effective at inducing degradation/failures that are indicative of long-term effects. For test items incorporating seals which protect moisture sensitive materials, longer test durations may be required to obtain a higher degree of confidence that the munition will remain safe in warm-humid conditions. It is recommended that this test be performed on both bare munitions and munition packaged within secondary packaging.

A.3.7. SOLAR RADIATION.

The solar radiation test (MIL-STD-810, Method 505) is intended to aggravate those thermally induced degradation mechanisms associated with elevated skin temperatures and thermal gradients

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within the munition that are induced due to solar radiation. Since most nations' solar test chambers do not incorporate the ultraviolet element of the spectrum they tend not to aggravate the photo-chemical (actinic) degradation modes associated with solar radiation. If this is of concern (as may be the case for some paints, adhesives and polymers) then a separate ultra-violet exposure test will also be required. A minimum of seven (7) A1 climate category cycles (meteorological temperature and solar radiation) are recommended in order to attain the maximum elevated temperatures throughout the test munition. The solar radiation level of 1,120 W/m² is derived from MIL-STD-810 and MIL-STD-2105. It is also noted that this test is intended to be performed on the bare munition. Solar Radiation testing of sensitive munition packaging materials to ascertain the degradation modes associated with solar radiation should be considered part of the packaging evaluation test and is not addressed herein.

A.3.8. THERMAL SHOCK.

The thermal shock test MIL-STD-810, Method 503 is intended to simulate the rapid temperature transitions that are possible during logistic movements of munition. Two (2) possible approaches are described below. Examine the munition usage scenarios to determine the test item packaging configuration. If feasible, all testing should be carried out on unpackaged items to provide worst case thermal stress conditions. Stabilization at the temperature extremes is required.

A.3.8.1. Phased Thermal Shock.

A.3.8.1.1. Low Temperature Phase.

This test simulates movement of warm munitions from storage or from a transport vehicle in maintenance to an extreme cold environment or vice versa. The low temperature shock test consists of five (5) temperature shock cycles between the temperatures of 21 °C (standard ambient) and -51 °C. In most applications, the munitions will be exposed to the temperature shock environment in its logistic container. However, to address the most severe condition the munition should be tested in its unpackaged configuration.

a. The -51 °C temperature is the low extreme presented in MIL-STD-810 and MIL-STD-2105, for Climate Category C3.

b. Stabilization at the temperature extremes is required. Munitions in storage or in warm buildings associated with vehicle maintenance would likely achieve temperature stabilization. Also, the extremely low temperatures encountered in the natural environment are likely to persist longer than the munition temperature stabilization time.

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A.3.8.1.2. High Temperature Phase.

This test exposes the munition to rapid temperature transition from -5 °C (temperature at an altitude of 8 km, from MIL-HDBK-310 and AR 70-38 to the unpackaged SRE temperature.

a. This test simulates rapid movement of munitions under the following scenarios:

(1) Movement of warm munitions from storage (e.g., magazine or process area) to an extreme cold environment, or vice versa.

(2) Rapid ascent from a desert airfield to high altitude (8 km) in an unheated aircraft compartment or carried externally.

(3) Air delivery or airdrop from high altitude (8 km) to a desert environment.

b. Stabilization at the temperature extremes is required. Munitions in flight prior to air delivery would likely achieve temperature stabilization. Also, the extremely high temperatures encountered in the natural environment are likely to persist longer than the munition temperature stabilization time.

A.3.8.2. Aggravated Thermal Shock.

The handling and transport of munitions between a temperature conditioned storage area and the ambient outdoor environment is the prevailing mechanism for rapid thermal change. This test does not simulate a specific transport scenario but it uses the rapid transition between the extreme temperature values to thermally stress the munition.

In this scenario, a single set of temperature shocks will be required. The temperature shock test consists of ten cycles from a cold extreme temperature of -51 °C to the high extreme temperature of no less than 71 °C. The transfer rate between chambers should be as fast as possible and is dependent upon munition portability requirements (i.e., man vs machine).

A.3.9. CONTAMINATION BY FLUIDS.

Mortar munitions could be exposed to a variety of chemical compounds during storage and use. Some exposures are deliberate (such as decontamination chemicals) and some are inadvertent (such as contact with mortar maintenance fluids and insect repellent used by the User). Select the fluids most commonly encountered throughout the munition's life cycle and apply to the unpackaged munition per MIL-STD-810, Method 504 using the intermittent exposure method. Contamination effects must be analyzed for their immediate or potential (long term) effects on the safety of the munition. A list of typical contaminants can be found in AC/225(LG/3-SG/1) D/14, Table 2.20.

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A.3.10. LOW PRESSURE (ALTITUDE).

Transportation within cargo aircraft may be within pressurized or unpressurised cargo bays depending upon aircraft type. In the case of pressurized cargo bays, most pressurization systems replicate external atmospheric pressure up to a particular altitude, and then maintain a stabilized internal pressure at greater altitudes. This stabilized pressure is known as the 'cabin altitude'. The most common aircraft in NATO service is the C130, for which the cabin altitude may be assumed to be 4,570 m with a corresponding pressure of 56.8 kPa, unless otherwise identified. Other values may be substituted based upon LCEP requirements for transportation by other aircraft. The default test parameters provided in MIL-STD-810, Method 500 are considered appropriate.

A.3.11. CARGO AIRCRAFT DECOMPRESSION.

Rapid decompression can result when cabin pressurization is lost during an accident scenario in a transport aircraft. Rapid decompression may result in bursting of the munition container and this test is required to verify that the packaging does not present a secondary hazard to the munition or aircraft crew. This test should be conducted as part of packaging certification for air transportation and is not necessarily required for munition safety testing. However, some munition components (e.g., seals) may be susceptible to sudden changes in pressure if the container fails to regulate the pressure in a rapid decompression scenario. If the results of the container rapid decompression test or the munition altitude test indicate a potential vulnerability, this test should be conducted on the primary packaged munition. The default test parameters provided in MIL-STD-810, Method 500 are considered appropriate.

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APPENDIX A. BACKGROUND/RATIONALE.
ANNEX 4. NON-SEQUENTIAL DYNAMIC ENVIROMENTAL TESTS.

A.4.1 GENERAL.

Provided below are the rationale for the Non-Sequential Dynamic Environmental Tests related to normal (accidental) handling in typical logistic operations. The packaging configuration (palletized, primary packaging, secondary packing or bare items) for the specific environmental test should match the most likely packaging configuration that is likely for the mortar cartridge during the LCEP.

A.4.2 NON-CONTACT UNDER WATER EXPLOSION (UNDEX).

The shocks likely to occur during non-contact UNDEX cause significant shock amplitudes that exceed those from normal handling. If the munition has HE filling, non-contact underwater explosion must be considered. UNDEX shock testing in accordance with MIL-DTL-901 or appropriate National Standards is a mandatory requirement prior to ship embarkation for some NATO Nations and cannot be tailored out. The overall basis for UNDEX shock is addressed in Allied Navy Engineering Publication (ANEP) 43. Additional guidance may be found in STANAGs 4549 and 4150. The temperature in the ship's storage hold would be expected to be relatively benign, so testing may be performed under standard ambient conditions (+21°C). The typical requirement would be for the munition to remain 'Safe for Disposal' so testing may be conducted non-sequentially. If, however, the requirement is for the munition to remain 'Safe for Use' (as may be necessary for Naval application), UNDEX shock testing must be conducted within the sequence.

A.4.3 12-METER LOGISTIC SAFETY DROP TEST.

This mandatory logistic safety drop test, as described in STANAG 4375, assesses the safety of the munition when exposed to a free-fall drop which may be encountered during ship loading operations. This test is conducted as a non-sequential test since it is representing an accident scenario.

A.4.4 PARACHUTE RESUPPLY.

Mortar munitions are likely to be resupplied by parachutes, whereupon the munition will be expected to remain S3 compliant. Parachutes may also fail, resulting in more severe impact velocities whereupon the munition is only expected to remain safe for disposal. ITOP 4-2-601, the NATO Fuze Qualification manual, gives guidance for each of these scenarios.

A.4.4.1 Low Velocity Parachute Drop.

Low velocity parachute delivery typically results in impact velocities of 8.7 m/s. The 8.7 m/s environment may be replicated by either an aircraft drop per ITOP 07-2-509 or may be simulated

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with a 3.8 m freefall drop unless specific and validated evidence is presented to the contrary. This freefall drop value is consistent with ITOP 4-2-601 and MIL-STD-810.

A.4.4.2 High Velocity Parachute Drop.

Mortar munitions are likely to be re-supplied by parachute delivery and are expected to remain S3 following such an event. High velocity parachute delivery typically results in impact velocities of 27.4 m/s. This environment may be replicated by an aircraft drop per ITOP 07-2-509. This test should be conducted as a non-sequential test on a minimum of three cartridges.

A.4.4.3 Malfunctioning Parachute Drop.

Munitions that may be re-supplied by parachute delivery are at risk of a malfunctioning parachute drop scenario and are expected to remain safe for disposal after such an event. Per ITOP 4-2-601, test E5, malfunctioning parachute systems may result in impact velocities of 45.7 m/s. This environment may be replicated by an aircraft drop per ITOP 07-2-509.

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ANNEX 4. NON-SEQUENTIAL DYNAMIC ENVIROMENTAL TESTS.

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APPENDIX A. BACKGROUND/RATIONALE.
ANNEX 5. NON-SEQUENTIAL FIRING TESTS.

A.5.1. GENERAL.

Provided below are the rationale for the Non-Sequential Firing Tests related to normal firings of the mortar cartridges.

A.5.2. WET EFFICIENCY TRIAL.

Due to the high elevation associated with mortar weapon systems, water may collect in the base of the mortar tube during rain storms if the muzzle cover has been removed for extended periods of time. Additionally, water may collect in the base of the mortar tube due to condensation within the bore.

The purpose of the wet efficiency trial is to check that a mortar cartridge with a propellant charge which has become wet under operational conditions can safely be fired and to assess the effect on muzzle velocity resulting in a range reduction. This test, which includes exposure to bare immersion in fresh water or a blowing rain environment, will determine if additional measures are required (i.e., attempt to disperse water from the cartridge) to safely fire the cartridge after exposure to water.

A.5.3. RESIDUE – DEBRIS TEST.

The muzzle residue – debris test is conducted to determine if the projectile will fire without excessive misfires or hangfires, caused by excess fouling of the cannon. This is to ascertain whether residue, debris and in-cannon burning resulting from firings are likely to cause inadvertent ignition or obstruction of subsequent rounds or any external fire hazard, and to determine the spatial distribution of ammunition debris projected from the cannon. Excessive fouling may also result in high tube stresses while the projectile moves within the tube during firing. The debris distribution has safety implications, such as when firing over the heads of friendly troops, and is also needed for the determination of range safety fans and training scenarios. The test will also be useful to determine or verify the cannon swab frequency required to minimize misfires due to tube fouling.

A.5.4. MUZZLE FLASH TEST.

During firing safety tests, ensure a complete optical radiation hazard evaluation of the ammunition exhaust plume is conducted to protect system operator's and observer's eyes and skin from potential overexposure to high intensity optical radiation to include ultraviolet, visible, and infrared nonionizing radiation.

The muzzle flash test is also conducted to determine the muzzle-flash characteristics (i.e., detectability) of the projectile during firing. Thermal imagery techniques should be used to determine the degree to which the mortar firing position may be detected in both the visible and

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infrared ranges during firing. The results will be compared against a specific set of intensity standards or against another projectile.

A.5.5. FUZE SAFE ARMING TEST.

Perform the fuze safe arming test to verify the correct functioning of the projectile fuzing system. The accidental, premature or incorrect operation of a fuzing system could result in a hazard to personnel either close to the weapon system itself or under the warhead/projectile trajectory, and/or damage to the launch platform and nearby equipment. Therefore, testing of fuzing systems is essential to provide evidence to support the overall safety assessment of the ammunition. Testing shall be conducted in accordance with STANAG 4157 and AOP-20.

A.5.6. WARHEAD ARENA TRIALS.

The safe separation distance for high explosive mortar cartridges is determined by the warhead fragment characteristics (size, mass, velocity, and spatial dispersion). A sample size of at least six is required because only a portion of the total number of fragments produced is collected in the recovery medium. The sample size must be large enough to reliably evaluate fragmentation characteristics in order to determine the average fragmentation spatial dispersion. Note that data from this test is also used to determine range safety parameters (i.e., Weapon Danger Area or "Safety Fan"). This test is conducted on factory fresh assets in order to obtain the maximum fragmentation distance.

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ANNEX 5. NON-SEQUENTIAL FIRING TESTS.

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APPENDIX A. BACKGROUND/RATIONALE.
ANNEX 6. NON-SEQUENTIAL SYSTEM TESTS.

A.6.1. GENERAL.

Provided below are the rationale for the Non-Sequential System Test and Evaluations related to normal firings of the mortar cartridges. These tests categorize the properties of the mortar cartridge and mortar to support the S3 test program. It is noted that these particular tests produce data that is unique to the particular combination of the mortar cartridge and mortar and will likely have to be repeated for each different host mortar for the munition.

A.6.2. FIRING TABLES TEST.

Ballistic performance testing is performed to provide complete range data of munition to develop firing tables and input to mortar ballistic computers for use during employment of mortar systems. Firing tables are normally developed for a specific family of munitions and mortar. The ballistic testing data will also determine the range and deflection precision of the fired munition which will be used to determine the surface danger zone for the munition. As such, the primary purpose of the firing tables test is to categorize performance of the munition but it also produces necessary safety data.

Firing tables provide standard muzzle-to-target firing solutions which consist of the munition's zone or charge, range, mortar elevation, and time of flight. Additionally, the firing tables provide mortar adjustment information to adjust the range, and correction factors to compensate for muzzle velocity differences, wind conditions, air temperature and air density. Also, the firing tables provide supplemental data consisting of range and deflection probable errors, angle of fall, terminal velocity, maximum ordinate, and fuze setting data.

A.6.3. RATE-OF-FIRE TEST.

The objective of the rate-of-fire test is to determine the rapid and sustained firing capabilities of the mortar system. Maximum rate-of-fire is described as the maximum rate at which it is physically possible for a trained gunner to continually fire the mortar; limited by cannon elevation and/or temperature constraints. Sustained rate-of-fire is defined as the rate-of-fire, at the maximum charge for the munition under test, at which the mortar can be continuously fired without exceeding the cannon's design maximum operating temperature.

A.6.4. DOUBLE LOAD TEST.

One (1) of the known hazards with mortars is the effect of firing a mortar cartridge which has been double loaded. In the case of mortars with fixed firing pins, this is only likely to occur when a misfire or hang-fire occurs and a second cartridge is loaded before the first loaded cartridge is fired or removed from the cannon.

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ANNEX 6. NON-SEQUENTIAL SYSTEM TESTS.

Mortar double load tests with high explosive (HE) cartridges are conducted to obtain information on the reaction of the mortar and cartridge in the case that a cartridge is inadvertently loaded into a mortar which already contains another cartridge. This is an extremely dangerous situation which can result in damage or destruction of the mortar, severe injuries and/or fatalities to mortar crew and nearby personnel. The information obtained from this test (mortar and cartridge fragmentation size, velocity and dispersion pattern) will be useful for Subject Matter Experts (SMEs) to aid in evaluating mortar catastrophic events.

It is noted that the results from a double load test are representative only of the double load event for the particular combination of mortar cannon, HE cartridge, and cartridge position on which the test was conducted. If the HE cartridge is used in an alternate mortar with significantly different mortar cannon properties, then an additional test should be considered.

Due to the high cost in conducting a double load test (the probable loss of the mortar and the data acquisition effort), a worst case test scenario should be considered. Such a double load test scenario is described in Appendix D, Paragraph D.3.4. Alternative double load test scenarios are also presented which should be considered but the high expense of conducting the test, resulting from probable destruction of the mortar, must also be considered.

A double load test may also be used to verify the safety of a mortar cartridge which has been designed to withstand certain double load scenarios to improve the safety to a mortar crew. The double load test could verify the design safety of hardened fuzes and other cartridge components.

A.6.5. PROPELLANT, WARHEAD AND FUZE COOK-OFF TESTS.

Propellant, warhead and fuze cook-off testing/assessment is conducted to determine the temperature and time at which the munition is likely to cook-off when loaded into a hot mortar cannon, and the pressures generated in the event of a cook-off. There are three (3) key operational considerations to be evaluated for new munitions: Fuze cook-off, Propelling Charge cook-off after a misfire, and warhead/payload cook-off resulting from a rapid sustained rate-of-fire. Neither the payload fill, fuze energetics nor propelling charge shall function after a misfire occurs when the cannon is near the designated maximum operating temperature. An alternative lower cannon temperature can be used for the assessment only with engineering supportive data and analysis.

Firing munitions for prolonged periods, or a series of engagements at high rates-of-fire may result in sufficient cannon heating in excess of the propellant auto-ignition temperature, potentially causing the propellant to cook-off if left in the mortar cannon for extended periods (such as during a misfire). If the propellant does not cook-off, mortar cartridges filled with HE may cook-off with a low order detonation if left in a hot cannon, destroying the cannon and possibly killing the crew. Additionally, the explosive filling may melt, with the potential for exudation or other effects.

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Hot cannon cook-off is typically tested and assessed based on the results of cannon heating trials, such as rate-of-fire test, and small-scale tests to validate the temperature of ignition of the energetic material(s) used within the munition, which should be known via specification and hazard data sheets prior to the trial.

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APPENDIX A. BACKGROUND/RATIONALE.
ANNEX 7. ADDITIONAL NON-SEQUENTIAL TESTS AND ASSESSMENTS.

A.7.1. GENERAL.

The following additional safety tests and assessments should be performed if data from previous testing and the preliminary assessment indicates that further investigation is required. Selection is based on analysis, previous test results, including evidence of incipient failure modes. Hardware sample sizes depend on the nature of the tests.

A.7.2. ELECTROMAGNETIC ENVIRONMENTAL EFFECTS (E3) ASSESSMENT AND TESTING.

The following electromagnetic environmental effects shall be considered to assess the safety of the munition when exposed to the environment which may be encountered during the LCEP (transportation/storage, staging, handling/loading, firing and immediate post-launch). Levels should encompass sea, land, and aviation storage, usage, and shipment requirements as identified in the LCEP. The potential electromagnetic environment, could create detrimental effects on mortar electronic fuzes, Electrically Initiated Devices (EIDs)/Electro-Explosive Devices (EEDs), such as electric primers, and guidance components on guided munitions. This could in turn result in excessive misfires, inadvertent initiation, fuze failure, and failure of the guidance system. The mortar munition should incorporate features to ensure safety and reliability after and during exposure to the E3 environment. It is noted that performance testing after subjecting the munition to Nuclear Electromagnetic Pulse (NEMP) is not included in this S3 document but may be required by the National Authority. For a system expected to have significant changes to its vulnerability with age/use, using environmentally stressed rounds within E3 test and assessment should be considered. Minimum test asset quantities are provided in Appendix B.

A.7.2.1. Hazards of Electromagnetic Radiation to Ordnance (HERO).

This test provides the data required for HERO classification and assesses the safety of the munition. HERO tests consist of exposing inert instrumented munitions (in accordance with MIL-STD-464, MIL-HDBK-240 and TOP 01-2-511) and its associated weapon platform(s) to its operational electromagnetic environments and monitoring the response of the munition EIDs/EEDs or Electronic Safe and Arming Devices (ESADs) and associated firing circuits when exposed.

A.7.2.2. Electrostatic Discharge (ESD) Tests.

These tests assess the safety of the munition when exposed to ESD phenomenon such as those encountered during helicopter transport, handling and loading. ESD tests are performed in accordance with MIL-STD-464, and include ESD from personnel handling and vertical lift (as applicable).

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A.7.2.3. Lightning Hazard.

These tests assess the safety of the munition when exposed to near and direct strike lightning per MIL-STD-464, which may occur during logistic and field operations.

A.7.2.4. Ordnance Electromagnetic Vulnerability (EMV).

These tests assess the suitability of the munition to operate within the electromagnetic environment (EME) defined in MIL-STD-464 for which they are designed to be used. These tests are performed during simulated normal operation and are designed to assess to what extent the munition not only is affected by the EME in which it is expected to operate but also its electromagnetic effect on other electrical systems it interacts with or is in close proximity to (e.g., on the same platform). Safety and suitability (proper operation and performance) shall be demonstrated by test or analysis. Much of this testing is for reliability purposes however some EMV tests provide safety assurance, for example those designed to monitor for interference carried into the weapon via physical electrical interfaces which may affect the performance of EID and/or ESAD firing circuits.

A.7.3. HEALTH HAZARDS.

Acoustic Energy and Chemical substances are the primary tests aimed at examining blast overpressure, impulse noise, propellant combustion products, and toxic materials produced during operation. The exact conduct of the tests will vary dependent upon the platform and operational requirement (e.g., proximity of crew and other personnel). Health hazard assessment can potentially include musculoskeletal trauma and non-ionizing radiation hazards.

A.7.3.1. Toxic Fumes Test.

Toxicity is also usually measured at standard service pressure, although cold and low pressure charges may also need examination due to the different interior ballistics that may occur at low temperatures and pressures. Propellant gases contain harmful chemical substances such as NH3, CO, CO2, SO2, HCN, NO, and NO2, as well as toxic particulates, for example lead, which may be hazardous to personnel. These toxic substances are usually modeled based upon the charge composition and then, if necessary, the most hazardous substances are measured during firing. These measurements should be taken at standard crew positions; at predetermined distances from the muzzle and breech; and with the platform in all standard firing configurations. If the platform has fume extraction equipment, measurements should be taken with it operating and non-operating. Toxicity may be assessed using a measured rate-of-fire or multiple cartridge rapid fire depending on the system and analysis technique (Note: there may be a build-up of gases following multiple cartridge rapid fire).

A.7.3.2. Impulse Noise and Blast Overpressure Test.

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ANNEX 7. ADDITIONAL NON-SEQUENTIAL TESTS AND ASSESSMENTS.

Firing safety tests should be conducted to measure blast overpressure and impulse noise to determine if the shock wave damages structures and/or injures personnel (especially hearing). Firing will include a range of charges, firing elevations and temperatures. Collection of time-pressure data from munition system testing at each operator or crew positions can be fed into a biomechanical model to compute the probability and severity of lung contusion resulting from single blast exposure within a 24 hour period.

A.7.4. INSENSITIVE MUNITIONS (IM) ASSESSMENT.

IM assessments should be conducted to ensure the munition meets the insensitivity safety requirements which include fire resistant during storage and stowage, bullet impact resistance and sympathetic detonation requirements.

A.7.5. MUNITION SOFTWARE SAFETY ASSESSMENT.

If applicable, munition software shall be designed, assessed and tested to assure its safety and suitability for service. Software itself is not a hazard. However, software can contribute to a system safety risk as a causal or mitigating factor for many types of system hazards. It is key to identify if the software contains functions which could cause a hazard. The presence of these functions indicate the need to conduct software safety testing.

A.7.6. EXPLOSIVE MATERIALS QUALIFICATION TESTING.

All explosive materials in the munition shall undergo appropriate testing and assessment to determine whether each possesses properties which make it safe for consideration for use in its intended role.

A.7.7. MUNITION DEMILITARIZATION AND DISPOSAL ASSESSMENT.

Ensure a verified Demilitarization and Disposal Plan is prepared for the munition in accordance with AOP 4518 or respective national requirements.

A.7.8. RENDER–SAFE AND DISPOSAL.

If required, ensure that verified Explosive Ordnance Disposal (EOD) render-safe and disposal procedures are developed in support of new munitions entering the inventory. This includes those particular courses of action or modes of actions for access to, recovery, rendering safe, and final disposal of explosive ordnance (EO).

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APPENDIX A. BACKGROUND/RATIONALE.
ANNEX 7. ADDITIONAL NON-SEQUENTIAL TESTS AND ASSESSMENTS.

A.7.9. RANGE SAFETY AND SUSTAINABILITY.

While not considered part of the safety test of munitions, an essential safety related analysis which involves the determination of weapon danger areas (also referred to as surface danger zone) that must be provided when the cartridge is employed during training and combat. The basic methods for conducting such a test program involve such things as ricochet area, dispersion patterns, fragmentation area, muzzle debris, noise, maximum ordinates, and ranges for various elevations and propellant weights. Obtain guidance from National Authority(ies).

Additionally, appropriate testing and analysis shall be conducted to assess range safety and sustainability. The potential for individual and cumulative environmental effects of munition use on operational ranges, (e.g., the expected deposition of hazardous substances, pollutants and contaminants, or emerging contaminants) should be assessed. Refer to MIL-STD-882, MIL-HDBK-764 and ITOP 05-1-060 for guidance.

A.7.10. MUNITION HAZARD CLASSIFICATION.

Appropriate munition hazard classification testing shall be conducted in accordance with TB 700-2 and UN ST/SG/AC.10/11.

A.7.11. FUZE SAFETY TESTING AND ASSESSMENT.

Mortar cartridges are provided with the fuze as part of the complete mortar cartridge. This document requires that the fuze be qualified. The results of fuze qualification are to be provided within the Munition Safety Data Package.

A.7.12. OTHER SAFETY TESTS.

The following tests are not required but should be considered if the required tests indicate that additional testing is required to enhance safety information on the mortar cartridge.

A.7.12.1. Munition Bare Drop Sensitivity Test.

The bare drop sensitivity test simulates accidental drops encountered during handling of bare munitions to ascertain the relationship between drop height and the probability of propelling components initiation. This test is intended to complement the bare drop and packaged drop which are performed in a sequential series.

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APPENDIX A. BACKGROUND/RATIONALE.
ANNEX 7. ADDITIONAL NON-SEQUENTIAL TESTS AND ASSESSMENTS.

A.7.12.2. Vertical Replenishment at Sea (VERTREP).

Munitions may be moved as an under-slung load by helicopter over land and/or subjected to VERTREP. The potential impact velocities and effect on munitions must be compared to the 2.1 m packaged drop test levels and determination must be made if additional testing is required.

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APPENDIX A. BACKGROUND/RATIONALE.
ANNEX 8. ADDITIONAL NON-SEQUENTIAL TESTS AND ASSESSMENTS.

A.8.1. GENERAL.

Provided below are the rationale for the preliminary assessments and tests to be conducted to support the preliminary design assessment. These tests which will establish the performance baseline for the mortar cartridge include the Safety of Projectile tests and Safety of Propelling charge tests.

A.8.2. PROJECTILE AND MORTAR PRESSURE COMPATIBILITY.

A mortar is essentially a pressure vessel. As such it is necessary to demonstrate that the cannon can hold and vent the pressurized cartridge gasses, and launch the projectile as intended without presenting a hazard or damage to personnel, the platform, or the projectile itself. An appropriate safety margin must be applied to the design. This is demonstrated through a number of firing tests conducted under various conditions to establish the safe operating pressures for the system. STANAG 4110 defines the pressure terms used for the system, cannon, projectile, and propelling charge, describes their interrelationship and outlines the methods employed in assessing those pressures. It is noted that the System Design Pressure is the lowermost of either the cannon or projectile design pressure. Figure A8-1 gives a graphical interpretation of the pressure terms for cannons and cartridges, with their interrelationships in accordance with STANAG 4110.

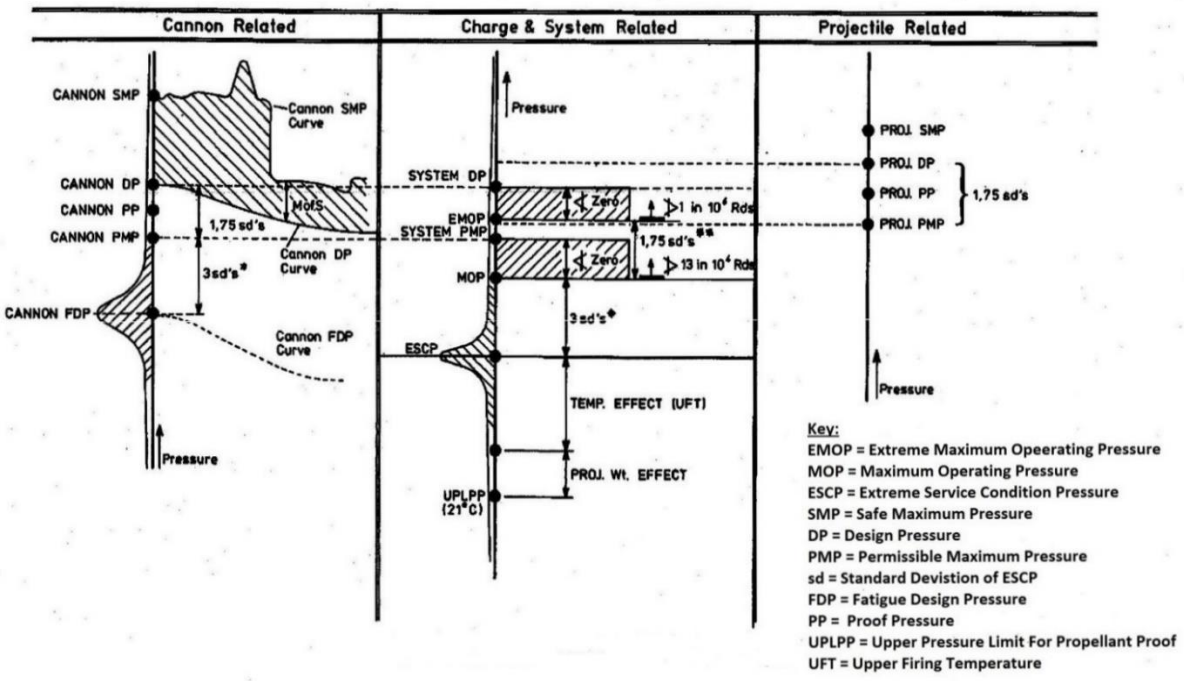


Figure A8-1: Cannon System Pressure Terms and Relationships.

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APPENDIX A. BACKGROUND/RATIONALE.
ANNEX 8. PRELIMINARY DESIGN TESTS AND ASSESSMENTS.

A.8.3. SAFETY OF PROPELLING CHARGE - MAXIMUM OPERATING PRESSURE (MOP) TRIAL.

The purpose of the Safety of Propelling Charge - MOP trial is to validate the internal pressure likely to be achieved in service (for a particular mortar and munition combination). This includes the maximum chamber pressure, pressure/cannon travel characteristics and pressure values at maximum and minimum firing temperatures. This data will be used during the follow-on tests and evaluations to determine if the required in-service combinations of propellant charge and projectiles are safe to fire in the mortar, in respect of the performance of the propellant and the internal pressures generated.

The required trial objectives are as follows:

a. To identify any unsafe or unacceptable feature of the propellant charge pressure-time performance across the range of temperatures required in-service.

b. To verify that the time interval between initiation of the propellant charge and the development of maximum pressure is consistent and within safe limits. Evidence concerning the firing interval is usually obtained during development trials conducted by the Design Authority.

c. No unacceptable differential pressure behind the mortar cartridge. (Note – this is unlikely for mortar systems.)

d. To verify that muzzle velocities are sufficiently consistent.

The data from the trial will be used to develop the Maximum Operating Pressure Curve, which are the values derived from the pressures generated at each point in a specified mortar cannon by the propelling charge.

The above evidence can be collected across all firing trials during development and qualification of the munition. However, the Safety of Propelling Charge – MOP Trial, outlined in Appendix D of this document, has been developed to provide a systematic and unbiased source for the required evidence.

A.8.4. VERIFICATION OF PROPELLING CHARGE.

This test may be required if the MOP trial was not performed on a particular lot of munition propellant that will be included in the S3 assessment. This test is conducted to confirm that the assigned service charge, for a particular lot of propellant, will meet the muzzle velocity and pressure requirements, to verify the temperature coefficient for pressure, and to provide data on round to round velocity and pressure dispersion. For propelling charge safety tests, testing must

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be done with the projectile that produces the highest chamber pressure using the highest charge. For some mortar/munition combinations, an interior ballistics expert should be consulted to determine if the measurement of differential (difference) pressures may be required.

A.8.5. CHARGE ESTABLISHMENT FOR PROJECTILE SAFETY TESTING.

To verify if the munition is safe to fire under all training and combat conditions, the munition must be fired at maximum safety test pressure after exposure to the sequential environmental tests environments. Additionally, the Strength of Design (SoD) test requires the test munition be fired at pressures 5% above the peak safety pressure. This test is required to determine propellant adjustment for Strength of Design and safety pressure tests for a particular lot of propellant. The peak safety pressures must reflect the pressures the munition and mortar are designed to withstand and the extreme spread of pressures likely to be encountered during the munition's life cycle.

A.8.6. PROJECTILE STRENGTH OF DESIGN TEST.

The SoD Test will demonstrate that component casings, the tail unit and structures will not permanently deform or break up under the stresses induced during firing. For the purpose of this test, projectiles developed with energetic or hazardous fill shall be tested with inert fills.

The trial is to determine the strength of the projectile at maximum firing stresses. Therefore, it is important to avoid firing rounds that are inherently stronger than the average population. It is recommended that effort is made to select weak projectiles (e.g., minimum acceptable mechanical properties). However, the critical requirement for the projectiles is that they represent the whole population of manufactured projectiles and that sufficient measurements are taken to ensure they represent that population.

Charge pressure and pressure rise are critical. The maximum stresses must be represented, without exceeding munition DP. If insufficient pressures are achieved, the trial may need to be repeated or a lower munition PMP declared.

The trial is to determine deformation, therefore, as many rounds as possible should be recovered with as little deceleration and impact damage as possible. This may require overwater recovery or other soft recovery methods.

Set-up (expansion of cartridge diameter) should not exceed the difference between the design minimum bore diameter of the cannon and the design maximum cartridge body diameter. Set-down (contraction of the cartridge diameter) should not be more than 0.5% of the caliber. For carrier cartridges, set-down (or set-up of the payload) should not impede payload ejection or damage the payload or ejection system such that it causes a hazard.

Obviously the cartridge should not break up in-bore or in-flight (unless designed to do so, e.g., sabots/fins). However, the post-trial analysis should also examine the cartridges for any signs of

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break up, deformation, cracking, body engraving and dimensional changes. Also, any obturating band shall be examined for correct performance as intended (e.g., retention on projectile, expelled from projectile, or breakup and expelled from projectile). The cannon should also be examined for deformation. Any observations must be recorded and their acceptability assessed.

A.8.7. PROJECTILE SAFETY IN CANNON TEST.

The Safety in Cannon Test is basically a continuation of the Strength of Design Test except that munitions with live energetics are used for the test. The Safety in Cannon Test is conducted to demonstrate that the energetic or hazardous materials from a projectile will not escape or react in-bore or just forward of the muzzle that the munition can safely proceed into the more strenuous sequential environmental tests. This test is conducted on the configuration of the projectile with energetic materials (e.g., explosive fill, booster and burster charges, full propelling charge system, illumination candles) and must be as representative of the overall production population as possible (not specially selected for a particular property). If sufficient data is available from previous tests or developmental test firings, then this specific trial may not be necessary. The safety in cannon test consists of pre-stressing the projectile by exposure to high and low temperature storage, loose cargo shock and vibration, and a packaged drop and firing at the PMP and the high and low extreme firing temperatures. When firing, use modified charges to test the munition against the highest pressures which can occur in service.

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APPENDIX B. S3 TEST PROGRAM FOR MORTAR CARTRIDGES.
ANNEX 1. GENERAL.

B.1 GENERAL.

This appendix provides the overall S3 test programs for mortar cartridges. Each test program is presented in the form of test flowcharts and test asset quantities tables for each of the primary mortar types (60mm, 81mm, and 120mm). It should be noted that several non-sequential test requirements (e.g., hazard classification and IM tests) are considered part of the overall S3 program, but are not governed by this document. For these tests, references are provided for determination of test requirements and quantities. See Chapter 8 of this document for the general description and intended application of the test flow options presented in this appendix.

Figure B1-1 depicts the overall test flowchart of a representative S3 test program.

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APPENDIX B. S3 TEST PROGRAM FOR MORTAR CARTRIDGES.
ANNEX 1. GENERAL.

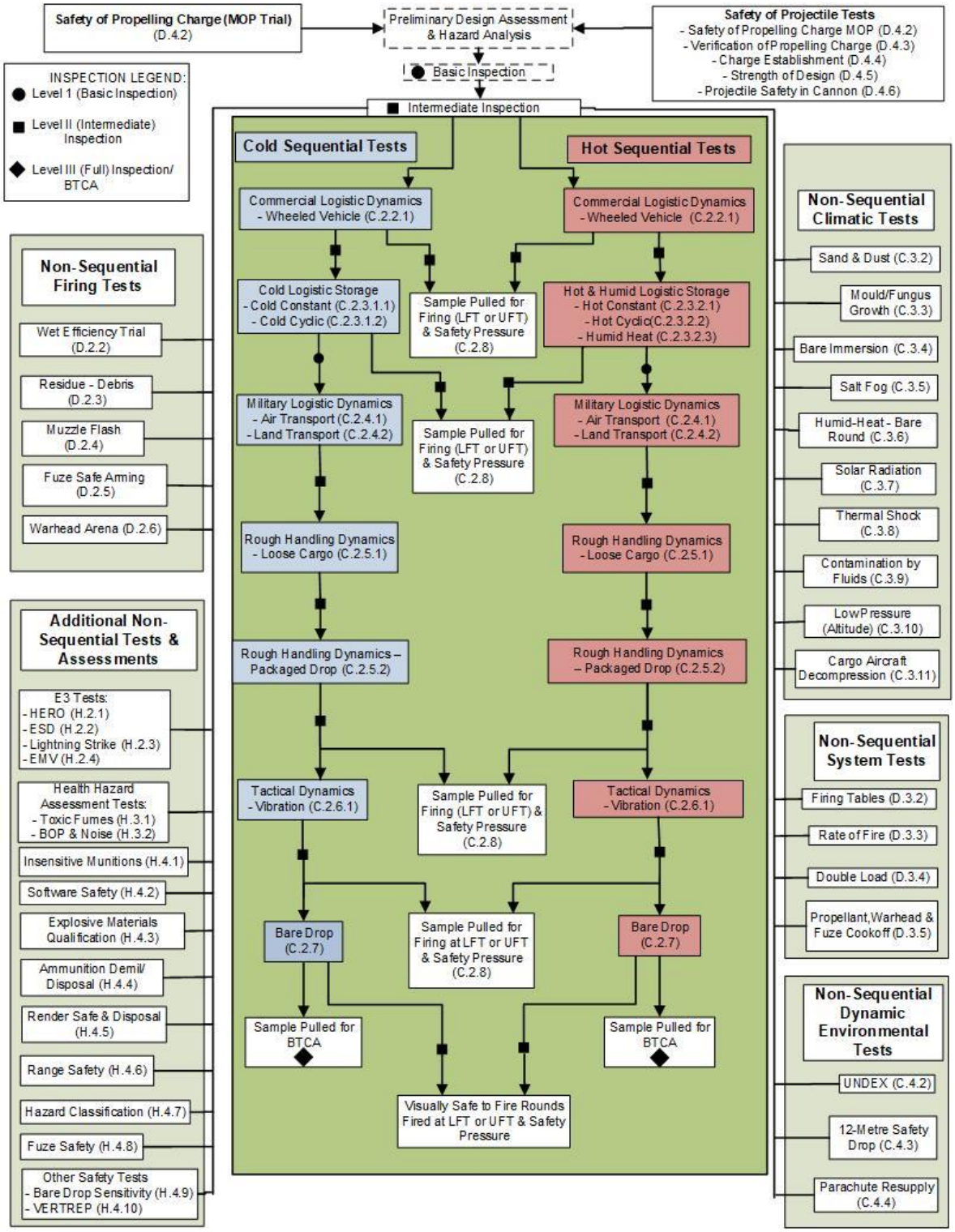


Figure B1-1. Representative S3 Test Program.

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APPENDIX B. S3 TEST PROGRAM FOR MORTAR CARTRIDGES.
ANNEX 1. GENERAL.

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APPENDIX B. S3 TEST PROGRAM FOR MORTAR CARTRIDGES.
ANNEX 2. SEQUENTIAL ENVIRONMENT TESTS.

B.2. SAMPLE QUANTITIES FOR SEQUENTIAL ENVIRONMENTAL TESTS.

The number of assets for Sequential Environmental Testing is based on the packaged unit load, and recommended test quantities and procedures in accordance with the applicable test procedure (e.g., AECTP, ITOP, MIL-STD) Typical packaged unit load quantities are described in Table B2-1. The sample quantities of live munitions recommended for the sequential environmental tests are indicated in Table B2-2. Justification for the test environments and associated test severities can be found in Appendix A. The test methods and severities for the Sequential Environmental Testing can be found in Appendix C. Upon completion of the sequential environmental tests, cartridges are subjected to firing test with the munitions conditioned at LFT and UFT (in accordance with the guidance given in Appendix C) or subject to BTCA. Required sample quantities for Sequential Environmental Testing are given in Table B2-3. During the sequential testing, cartridges may become unsafe. If this occurs, remove them from the sequence, set them aside and document the damage for subsequent assessment. Damaged cartridges may be replaced or repaired via replacement of damaged components (e.g., propelling charges or packaging) but would not count as a full sequential cartridge (i.e., not been subjected to the full sequence). Restrictions on replacement of damaged components must be established prior to commencement of testing. Furthermore, some cartridges might become damaged to such extent that they could not be fired but do not present a direct safety hazard. In such case, these cartridges could complete the environmental sequence and may be used for a Level 3 inspection, which may be warranted by the need to conduct failure investigations.

TABLE B2-1. PACKAGING/UNIT LOADS

MUNITION TYPE	TYPICAL UNIT LOAD		
	ITEMS PER OVERPACK (PRIMARY PACKAGING)	OVERPACKS PER PALLET	TOTAL ITEMS
120mm	2	24	48
81mm	3	21	63
60mm	8	56	448

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APPENDIX B. S3 TEST PROGRAM FOR MORTAR CARTRIDGES.
ANNEX 2. SEQUENTIAL ENVIRONMENT TESTS.

TABLE B2-2. TEST ASSET QUANTITIES FOR SEQUENTIAL ENVIRONMENTAL TESTING

MUNITION TYPE	UCT/UFT			LCT/LFT		
	TOTAL	BTCA	FIRE	TOTAL	BTCA	FIRE
120mm	60	2	58	60	2	58
81mm	63	2	61	63	2	61
60mm	112	2	110	112	2	110

TABLE B2-3. SEQUENTIAL ENVIRONMENTAL TEST PROGRAM

Sub Test	60 mm						81 mm						120 mm					
	High Temp			Low Temp			High Temp			Low Temp			High Temp			Low Temp		
Total Assets For Sequential Tests	112			112			63			63			60			60		
Commercial Logistics Dynamics (Primary Packaging)	112			112			63			63			60			60		
Fire @ Test Temperature	16			16			9			9			8			8		
Cold Logistic Storage (Primary Packaging)				96						54						52		
Hot & Humid Logistic Storage (Primary Packaging)	96						54						52					
Fire @ Test Temperature	16			16			9			9			8			8		
Military Air Transport Dynamics (Primary Packaging)	80			80			45			45			44			44		
Military Land Transport Dynamics (Primary Packaging)	80			80			45			45			44			44		
Loose Cargo Dynamics (Primary Packaging)	48			48			30			30			30			30		
Loose Cargo Dynamics (Secondary Packaging)		32			32			15			15			14			14	
Rough Handling Dynamics (Primary Packaged Drop)	48			48			30			30			30			30		
Rough Handling Dynamics (Secondary Packaged Drop)		32			32			15			15			14			14	
Fire @ Test Temperature	8	16		8	16		9	8		9	8		9	8		9	8	
Tactical Operational Dynamics (Primary Packaging)	16			16			9			9			9			9		
Tactical Operational Dynamics (Secondary Packaging)		24			24			12			12			12			12	
Fire @ Test Temperature	8	8		8	8		4	4		4	4		4	4		4	4	
Bare Drop	8	16	16	8	16	16	5	8	7	5	8	7	5	8	6	5	8	6
Fire @ Test Temperature (If visually safe to fire)	7	16	15	7	16	15	4	8	6	4	8	6	4	8	5	4	8	5
BTCA	1		1	1		1	1		1	1		1	1		1	1		1

SEQUENTIAL ENVIRONMENTAL

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APPENDIX B. S3 TEST PROGRAM FOR MORTAR CARTRIDGES.
ANNEX 2. SEQUENTIAL ENVIRONMENT TESTS.

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APPENDIX B. S3 TEST PROGRAM FOR MORTAR CARTRIDGES.
ANNEX 3. NON-SEQUENTIAL CLIMATIC TESTS.

B.3 SAMPLE QUANTITIES FOR NON-SEQUENTIAL CLIMATIC TESTS.

The sample quantities of mortars recommended for the non-sequential climatic tests depend on different test standards. Justification for the test environments and associated test severities can be found in Appendix A. The test methods and severities to use for the non-sequential environmental tests can be found in Appendices C and D. Upon completion of the non-sequential environmental tests, munitions should be subjected to level 2 inspection. If the design assessment identifies that the munition will be susceptible to specific environments, the relevant non-sequential test(s) should be included in the sequential environmental testing.

TABLE B3-1. TEST ASSET QUANTITIES FOR NON-SEQUENTIAL CLIMATIC TESTS.

Non-Sequential Climatic Tests				
Sub-Test	Assets			Notes
	60 mm	81 mm	120 mm	
Sand and Dust	10	10	10	Conducts on bare rounds. Conduct blowing sand test followed by blowing test.
Mold/Fungus	10	10	10	Conducts on bare rounds. Sub packaging components such as propelling charges may remain in place on unpackaged cartridges for the tests.
Bare Immersion	10	10	10	Immersion test is also required on sealed ammunition packaging as part of packaging qualification.
Salt Fog	10	10	10	Conduct on bare rounds.
10-Day Humid-Heat	10	10	10	Split 10 round test between 5 each on bare rounds and 5 each on rounds in secondary packaging (fiber containers).
Solar Radiation	10	10	10	Conduct on bare rounds.
Thermal Shock	20	24	24	Bare round worst case.
Contamination by Fluids	5	5	5	Conduct on bare rounds.
Low Pressure Altitude	10	10	10	Conduct on rounds within secondary packaging.
Cargo Aircraft Decompression	10	10	10	Conduct on rounds within primary packaging.

Note: All test rounds are live except as noted.

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APPENDIX B. S3 TEST PROGRAM FOR MORTAR CARTRIDGES.
ANNEX 3. NON-SEQUENTIAL CLIMATIC TESTS.

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APPENDIX B. S3 TEST PROGRAM FOR MORTAR CARTRIDGES.
ANNEX 4. NON-SEQUENTIAL DYNAMIC ENVIRONMENTAL TESTS.

B.4. SAMPLE QUANTITIES NON-SEQUENTIAL DYNAMIC ENVIRONMENTAL TESTS.

The sample quantities of munitions recommended for the Non Sequential Dynamic Environmental Tests depend on different test standards.

TABLE B4-1. TEST ASSET QUANTITIES FOR NON-SEQUENTIAL DYNAMIC ENVIRONMENTAL

Non-Sequential Dynamic Environmental Tests				
Sub-Test	Assets			Notes
	60mm	81mm	120mm	
UNDEX	8	3	2	Test one (1) complete box of ammunition.
12 m Drop	96	72	24	Packaging Dependent – three (3) packages dropped at hot and cold – three (3) orientations.
Parachute Resupply - High Velocity	16	16	16	Control rounds are fired with the rounds which were subject to the test environment to determine if there is a performance difference between the two (2) groups.
Parachute Resupply - Low Velocity	16	16	16	
Parachute Resupply - Parachute Malfunction	16	16	16	

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APPENDIX B. S3 TEST PROGRAM FOR MORTAR CARTRIDGES.
ANNEX 4. NON-SEQUENTIAL DYNAMIC ENVIRONMENTAL TESTS.

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APPENDIX B. S3 TEST PROGRAM FOR MORTAR CARTRIDGES.
ANNEX 5. NON-SEQUENTIAL FIRING TESTS.

B.5 SAMPLE QUANTITIES FOR NON-SEQUENTIAL FIRING TESTS.

The sample quantities of munitions for the Non Sequential Firing Tests are provided in Table B5-1. These tests include the Projectile Safety and Safety of Propelling Charge Trial tests which are required to support the Preliminary Design Assessment and Hazard Analysis (see chapter 6).

TABLE B5-1. TEST ASSET QUANTITIES FOR NON-SEQUENTIAL FIRING TESTS.

Non-Sequential Firing Tests & Evaluation				
Sub-Test	Assets			Notes
	60 mm	81 mm	120 mm	
Wet Efficiency Trial	36	36	36	
Muzzle Flash	400	400	400	These rounds are used to collect data for a Residue/Debris test.
Residue/Debris				Recommend to combine with the Muzzle Flash Test. Round count is driven by allowable misfire rate criteria.
Fuze Safe Arming	50	50	50	Quantity required for a cartridge with a single mode (point detonating) fuze.
	100	100	100	Quantity required for a cartridge with a dual/multi-option fuze.
Warhead Arena Test	6	6	6	Also known as Fragmentation Arena Test.

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APPENDIX B. S3 TEST PROGRAM FOR MORTAR CARTRIDGES.
ANNEX 5. NON-SEQUENTIAL FIRING TESTS

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APPENDIX B. S3 TEST PROGRAM FOR MORTAR CARTRIDGES.
ANNEX 6. NON-SEQUENTIAL FIRING TESTS.

B.6 SAMPLE QUANTITIES FOR NON-SEQUENTIAL SYSTEM TESTS.

The sample quantities of munitions for the Non Sequential System Tests are included in Table B6-1.

TABLE B6-1. TEST ASSET QUANTITIES FOR NON-SEQUENTIAL SYSTEM TESTS.

Non-Sequential System Tests				
Sub-Test	Assets			Notes
	60mm	81mm	120mm	
Firing Tables	270	270	270	
Rate-of-Fire	280	220	100	This subset will allow for verification of User requirements for maximum and sustained rate-of-fire.
Double-Loading	4	4	4	One (1) complete double-load, one (1) half-hung. Must include test tube assets or surrogate (destructive).
Propellant Cook-off	7	7	7	
Warhead/Fuze Cook-off	1	1	1	Can be combined with Rate of Fire Test or utilizing artificial heating sources and control rounds to achieve weapon Designated Maximum Operating Temperature (DMOT).

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APPENDIX B. S3 TEST PROGRAM FOR MORTAR CARTRIDGES.
ANNEX 6. NON-SEQUENTIAL FIRING TESTS.

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APPENDIX B. S3 TEST PROGRAM FOR MORTAR CARTRIDGES.
ANNEX 7. ADDITIONAL TESTS & ASSESSMENTS.

B.7 SAMPLE QUANTITIES FOR ADDITIONAL TESTS & ASSESSMENTS.

The sample quantities of munitions for additional tests and assessments are included in Table B7-1.

TABLE B7-1. TEST ASSET QUANTITIES FOR ADDITIONAL TESTS.

Additional Tests and Assessments				
Sub-Test	Assets			Notes
	60mm	81mm	120mm	
HERO	1	1	1	Inert, instrumented EIDs.
ESD	32	32	32	22 rounds for PESD and 10 rounds for HESD.
Lightning Hazard - Indirect Strike Lightning	10	10	10	Conducts on rounds within secondary packaging then conduct on bare rounds. Conduct at same time as Direct Lightning Strike.
Lightning Hazard - Direct Strike Lightning	10	10	10	Conducts on rounds within secondary packaging then conduct on bare rounds. Conduct at same time as Indirect Lightning Strike.
Electromagnetic Vulnerability	5	5	5	Inert, only if munition contains electronics.
Health Hazard Assessment - Toxic Fumes	30	30	30	Round count considerations are finalized by the appropriate health assessment organization.
Health Hazard Assessment - Blast Overpressure and Noise	105	105	475	Ground mounted weapon is assumed for 60mm and 81mm and ground plus vehicle mounted is assumed for 120mm. Round count considerations are finalized by the appropriate health assessment organization.
Insensitive Munitions - Assessment Tests	90	42	44	IM Tests can be harmonized with Hazard Class Tests. Test is packaging dependent. IM Board must concur with quantities.
Munition Software Safety Assessment	0	0	0	Analysis only.
Energetic Materials Qualification	0	0	0	Propellant testing only.
Munition Demil/Disposal	0	0	0	Analysis only.
Render Safe	1	1	1	Requires inert cartridges.
Range Safety/Sustainability	0	0	0	Analysis only.
Hazard Classification Tests	256	61	30	
Fuze Safety	TBD	TBD	TBD	Based on review of previous safety testing.
Bare Drop Sensitivity	50	50	50	May require component testing only.
VERTREP	8	3	2	Test one (1) complete box of munitions.

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APPENDIX B. S3 TEST PROGRAM FOR MORTAR CARTRIDGES.
ANNEX 7. ADDITIONAL TESTS & ASSESSMENTS.

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APPENDIX B. S3 TEST PROGRAM FOR MORTAR CARTRIDGES.
ANNEX 8. PRELIMARY DESIGN TESTS.

B.8 SAMPLE QUANTITIES FOR PRELIMINARY DESIGN TESTS.

The sample quantities of munitions for the Preliminary Design Tests are provided in Table B8-1. These tests include the Safety of Propelling Charge (MOP Trial), Projectile Safety and Safety of Propelling Charge Trial tests which are required to support the Preliminary Design Assessment and Hazard Analysis (see chapter 6).

TABLE B8-1. TEST ASSET QUANTITIES FOR PRELIMINARY DESIGN TESTS.

Preliminary Design Assessment Tests				
Sub-Test	Assets			Notes
	60mm	81mm	120mm	
Safety of Propellant Charge - MOP Trial	108	108	108	
Verification of Propelling Charge	30	30	30	
Charge Establishment	50	50	50	
Strength of Design	30	30	30	
Projectile Safety in Cannon Test	30	30	30	

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APPENDIX B. S3 TEST PROGRAM FOR MORTAR CARTRIDGES.
ANNEX 8. PRELIMARY DESIGN TESTS.

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APPENDIX C. ENVIRONMENTAL TEST DESCRIPTIONS.
ANNEX 1. GENERAL.

C.1 GENERAL.

This appendix provides descriptions of all the environmental (climatic and dynamic) tests required in the Safety Test Programs included in Appendix B. Background and rationale for these tests are provided in Appendix A. Table I3-1 (Appendix I, Annex 3) provides cross reference of similar national and international test standards.

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APPENDIX C. ENVIRONMENTAL TEST DESCRIPTIONS.
ANNEX 1. GENERAL.

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APPENDIX C. ENVIRONMENTAL TEST DESCRIPTIONS.
ANNEX 2. SEQUENTIAL ENVIRONMENTAL TESTS.

C.2.1. GENERAL.

This appendix provides the overall S3 test programs for mortar cartridges. Each test program is presented in the form of test flowcharts and test asset quantities tables for each of the primary mortar types (60mm, 81mm, and 120mm). It should be noted that several non-sequential test requirements (e.g., hazard classification and IM tests) are considered part of the overall S3 program, but are not governed by this document. For these tests, references are provided for determination of test requirements and quantities. See Chapter 8 of this document for the general description and intended application of the test flow options presented in this appendix.

Appendix C, Annex 2, provides the rationale for the Sequential Environmental Tests for which the test descriptions are provided herein. See Figure B1-1, Representative S3 Test Program for the description of the overall sequential test program.

C.2.2. COMMERCIAL LOGISTICS DYNAMICS.

C.2.2.1. Commercial Logistic Transportation – Wheeled Vehicle Dynamics.

Perform vibration testing in accordance with MIL-STD-810, Method 514, using the following test parameters:

a. Munition Configuration: Primary Packaging and/or Palletized (depending upon the LCEP).

b. Test Level: MIL-STD-810, Method 514, Common Carrier, use the appropriate test(s).

c. Test Duration: Equivalent to 4,800 km transportation (Note, this simulates a single deployment).

d. Test Temperature: Temperature condition the test munitions prior to, and during vibration testing. Stabilize all cold munitions to the LCT, and all hot munitions to the UCT. See Table A-1.

Note: Although the Common Carrier vibration environment is relatively benign compared to other wheeled vehicle vibration environments, the test should not be tailored out due to the intent of incurring minor damage to the test munitions and packaging prior to conduct of the remaining sequential environmental tests.

C.2.3. LOGISTIC STORAGE.

C.2.3.1. Cold Logistic Storage.

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APPENDIX C. ENVIRONMENTAL TEST DESCRIPTIONS.
ANNEX 2. SEQUENTIAL ENVIRONMENTAL TESTS.

C.2.3.1.1. Cold Constant Temperature Logistic Storage.

Perform Low Temperature testing in accordance with MIL-STD-810, Method 502, using the following test parameters:

a. Munition Configuration: Primary Packaging.

b. Test Level: Constant temperature of -51 °C.

c. Test Duration: 72 hours (3 days) continuous.

C.2.3.1.2. Cold Cyclic Temperature Logistic Storage.

This test may be performed in addition to or as an alternative to the Cold Constant Temperature Storage Test at the discretion of the National Safety Authority. Perform Cold Temperature testing in accordance with MIL-STD-810, Method 502 using the following test parameters:

a. Munition Configuration: Primary Packaging.

b. Test Level: Temperatures: -37 °C to -46 °C

c. Test Duration: 14 diurnal (24-hour) cycles to be applied.

C.2.3.2. Hot and Humid Logistic Storage.

C.2.3.2.1. Hot Constant Temperature Logistic Storage.

This test may be performed in addition to the Hot Cyclic Temperature Storage Test at the discretion of the National Safety Authority. Perform Hot Temperature testing in accordance with MIL-STD-810, Method 501, using the following test parameters:

a. Munition Configuration: Primary Packaging

b. Test Levels:

(1) Mortar munitions that do not contain temperature sensitive energetic materials: constant temperature +71 °C for 216 hours (9 days).

(2) Munitions that contain energetic material that are temperature sensitive (e.g., explosives based on TNT, or double/triple base propellants): constant temperature of +58 °C for 528 hours (22 days).

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APPENDIX C. ENVIRONMENTAL TEST DESCRIPTIONS.
ANNEX 2. SEQUENTIAL ENVIRONMENTAL TESTS.

C.2.3.2.2. Hot Cyclic Temperature Logistic Storage.

Perform High Temperature testing in accordance with MIL-STD-810, Method 501 using the following test parameters:

a. Munition Configuration: Primary Packaging

b. Test Level: MIL-STD-810, Method 501 'High Temperature Diurnal Cycles' Category A1 Induced Conditions (Temperatures: +33 °C to +71 °C).

c. Test Duration: 28 diurnal (24-hour) cycles to be applied.

If opting to substitute some of the hot diurnal cycles for constant temperature stressing, only 28 of the 56 cycles should be substituted (with the remaining 28 cycles being applied along with the constant thermal stressing). Using the Arrhenius kinetic model discussed in AECTP 300, Method 306 and an activation energy of 70 kJ/mol; constant temperature stressing may be applied for 216 hours (9 days) at +71 °C, or 528 hours (22 days) at +58 °C where unrealistic degradation is anticipated at +71 °C.

C.2.3.2.3. Humid Heat Cycling.

Perform Aggravated Humidity testing in accordance with MIL-STD-810, Method 507 using the following test parameters:

a. Munition Configuration: Primary packaged munition.

b. Test Level: MIL-STD-810, Method 507, 'Aggravated Cycle (cycle 3)'.

c. Test Duration – 10 each 24-hour cycles to be applied.

C.2.4. MILITARY LOGISTIC DYNAMICS.

C.2.4.1. Tactical (Military) Transportation – Air.

These tests address the mechanical environments that may be encountered during military transportation by fixed wing aircraft (propeller and jet) and helicopters. The shocks associated with parachute resupply are tested non-sequentially.

C.2.4.1.1. Fixed Wing Aircraft – Jet Aircraft Vibration.

Perform vibration testing in accordance with MIL-STD-810, Methods 514 and 527 using the following test parameters:

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a. Munition Configuration: Primary Packaging or Pallet Configuration IAW LCEP environment/Test Level: MIL-STD-810, Methods 514 and 527 Materiel Transported in Fixed Wing Jet Aircraft - Takeoff.

b. Test Duration: Equivalent to 10% of the total flight duration specified in AECTP 100-4, Appendix E, Annex 1 for transportation by 'Jet Aircraft'. Assume a minimum test duration of 10 minutes per axis.

c. Test Temperature: Temperature condition the test munitions prior to, and during vibration testing. Stabilize all cold munitions to LCT, and all hot munitions to UCT.

C.2.4.1.2. Fixed Wing Aircraft – Turboprop Aircraft Vibration.

Perform vibration testing in accordance with MIL-STD-810, Methods 514 and 527 using the following test parameters:

a. Munition Configuration: Primary Packaging or Pallet Configuration IAW LCEP environment.

b. Test Level: MIL-STD-810, Methods 514 and 527, Propeller Aircraft for C130K (4-blade, f_0=68 Hz) and C130J (6 blade, f_0=102 Hz), with $L_0 = 1.2g^2/Hz$ for f_0. Other aircraft types may be added if their blade passing frequencies (f_0 component) are known. Use the appropriate test(s).

c. Test Duration: Equivalent to the flight duration specified in AECTP 100-4, Appendix 1, Annex E for transportation by 'Turboprop Aircraft'. The total test duration for a stated axis should be split such that each set of blade passing frequencies are addressed equally. (For C130 only, this would require the total test duration to be divided equally between the two (2) blade passing frequencies of 86Hz and 102Hz).

d. Test Temperature: Temperature condition the test munitions prior to, and during vibration testing. Stabilize all cold munitions to LCT, and all hot munitions to UCT.

C.2.4.1.3. Helicopter Vibrations.

Perform vibration testing in accordance with MIL-STD-810, Methods 514 and 527 using the following test parameters:

a. Munition Configuration: Primary Packaging or Pallet Configuration IAW LCEP environment.

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b. Test Level: MIL-STD-810, Methods 514 and 527, Helicopter Transport. Default blade passing frequencies (f_1 component) of 11 Hz, 17 Hz, and 22 Hz should be used to address most transport helicopter types (see Table A2-4). Other aircraft types may be added if their blade passing frequencies (f_1 component) are known. Use the appropriate test(s).

c. Test Duration: Equivalent to a flight duration of 10 hours. The total test duration for a stated axis should be split such that each set of blade passing frequencies (as stated above) are addressed equally. Assume a minimum test duration of 30 minutes per f_1 component per axis (giving a total test duration of at least 1.5 hours per axis).

d. Test Temperature: Temperature condition the test munitions prior to, and during vibration testing. Stabilize all cold munitions to LCT, and all hot munitions to UCT.

C.2.4.2. Tactical (Military) Transportation – Land.

C.2.4.2.1. Wheeled Vehicle – All Terrain Dynamics.

Perform vibration testing in accordance with MIL-STD-810, Methods 514 and 527 using the following test parameters:

a. Munition Configuration: Primary Packaging or Pallet Configuration IAW LCEP environment.

b. Test Level: MIL-STD-810, Methods 514 and 527, Composite Wheeled Vehicle. Use the appropriate test(s).

c. Test Duration: Equivalent to 800 km transportation (Note, this simulates a single deployment).

d. Test Temperature: Temperature condition the test cartridges prior to, and during vibration testing. Stabilize all cold munitions to LCT, and all hot munitions to UCT.

C.2.4.2.2. Wheeled Vehicle – Shock.

Perform shock testing in accordance with MIL-STD-810, Methods 516, 527 using the following test parameters:

a. Munition Configuration: Primary Packaging or Pallet Configuration IAW LCEP environment.

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b. Test Level: All shocks stated in Table C2-1, applied in each sense of each orthogonal axis. The shocks may be applied as either half-sine pulses or a single decaying sinusoidal pulse encompassing both senses in each axis. Terminal peak saw-tooth pulses or Shock Response Spectrum (SRS) methods may be substituted for the levels specified in Table C2-1 if it can be shown to produce equivalent velocities. MIL-STD-810, Methods 516, 527 provides guidance for SRS methods.

Table C2-1. Restrained Cargo Transport Shock Levels

Half Sine Pulse		**or**	**Decaying Sinusoid**	
Duration 5 ms			**Frequency (F): 100 Hz Duration (T): 0.37 s (No. of Complete Cycles (N): 37) Damping Factor: 3% of critical**	
Amplitude (g pk)	**Number of Shocks**		**Amplitude of First Peak (g pk)**	**Number of Repetitions**
8.0	34		8.0	34
10.0	17		10.0	17
12.0	3		12.0	3

Note: The number of shocks has been tailored from MIL-STD-810, Method 516 Test M3 values to arrive at the equivalent transport distance for each vehicle and munition type. All shocks are to be applied in each sense of each orthogonal axis.

c. Test Duration: Equivalent to 850km transportation.
(Note: Simulates a single deployment.)

d. Test Temperature: Temperature condition the test munitions prior to, and during shock testing. Stabilize all cold munitions to LCT, and all hot munitions to UCT.

C.2.4.2.3. Two Wheeled Trailer – Vibration.

Perform vibration testing in accordance with MIL-STD-810, Methods 514 and 527 using the following test parameters:

a. Munition Configuration: Primary Packaging or Pallet Configuration IAW LCEP environment.

b. Test Level: MIL-STD-810, Methods 514 and 527, Two Wheeled Trailers. Use the appropriate test(s).

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c. Test Duration: Equivalent to 50 km transportation (Note, this simulates a single deployment). Based on the current versions of AECTP 400, the test duration is 32 minutes/axis.

d. Test Temperature: Temperature condition the test munitions prior to, and during vibration testing. Stabilize all cold munitions to LCT, and all hot munitions to UCT.

C.2.5. ROUGH HANDLING DYNAMICS.

C.2.5.1. Loose Cargo Bounce.

Perform loose cargo (bounce) testing in accordance with MIL-STD-810, Method 514, depending upon packaging, using the following test parameters:

a. Munition Configuration: Primary and Secondary Packaging Configuration.

Note: Since it is likely that munitions could be tactically transported either within its primary or its secondary packaging, both configurations should be tested.

b. Test Level: Default values stated in MIL-STD-810, Method 514, Annex A: 300 rpm circular synchronous motion for 20 minutes. Where possible, test in two (2) orientations (horizontal and vertical) dividing total test duration between each. Furthermore, half of the vertical duration should be forward end up and the other half with the forward end down.

c. Test Temperature: Temperature condition the test munitions prior to, and during testing. Stabilize all cold munitions to LCT, and all hot munitions to UCT.

C.2.5.2. Rough Handling Packaged Drop.

Perform drop testing in accordance with MIL-STD-810, Methods 516 using the following test parameters:

a. Munition Configuration: Primary and Secondary Packaging.

Note: Since it is likely that munitions could be tactically transported either within its primary or its secondary packaging, both configurations should be tested.

b. Test Level: 2.1 m onto a concrete supported steel surface.

c. Drop Orientation: Each package is to be dropped twice to impact. Sample sizes should be sufficient to ensure that all orientations are addressed. The following table defines

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the recommended drop order for the various test sample quantities defined in Table B2-3 and the primary and secondary packaged unit load quantities described in Table B2-1. The drop order plan will have to be modified if the test item quantities are different from the Table C2-2.

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APPENDIX C. ENVIRONMENTAL TEST DESCRIPTIONS.
ANNEX 2. SEQUENTIAL ENVIRONMENTAL TESTS.

Table C2-2: Example of Drop Test Orientations vs. Quantities

5 Test Items	Test Item #	1	2	3	4	5											
	1st Drop	A	B	C	D	E											
	2nd Drop	B	C	D	E	A											
8 Test Items	Test Item #	1	2	3	4	5	6	7	8								
	1st Drop	A	A	C	A	B	D	E	D								
	2nd Drop	B	C	E	D	C	B	B	C								
12 Test Items	Test Item #	1	2	3	4	5	6	7	8	9	10	11	12				
	1st Drop	A	A	D	A	A	B	E	B	B	E	C	C				
	2nd Drop	B	C	A	E	C	D	B	D	C	C	A	B				
14 Test Items	Test Item #	1	2	3	4	5	6	7	8	9	10	11	12	13	14		
	1st Drop	A	A	D	A	A	B	E	B	B	E	C	C	D	E		
	2nd Drop	B	C	A	E	C	D	B	D	E	C	A	B	C	D		
15 Test Items	Test Item #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	
	1st Drop	A	A	A	D	E	B	B	B	E	C	C	C	D	D	E	
	2nd Drop	A	B	C	A	A	B	C	D	B	C	D	E	D	E	E	
16 Test Items	Test Item #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
	1st Drop	A	A	A	D	E	B	B	D	B	C	C	C	D	E	E	E
	2nd Drop	A	B	C	A	A	B	C	B	E	C	D	E	D	D	E	B
30 Test Items	Test Item #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	
	1st Drop	A	A	A	A	A	A	B	B	B	B	B	B	C	C	C	
	2nd Drop	A	B	C	D	E	A	B	C	D	E	A	B	C	D	E	
	Test Item #	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	
	1st Drop	C	C	C	D	D	D	D	D	D	E	E	E	E	E	E	
	2nd Drop	A	B	C	D	E	A	B	C	D	E	A	B	C	D	E	
32 Test Items	Test Item #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
	1st Drop	A	A	A	A	A	A	B	B	B	B	B	B	C	C	C	C
	2nd Drop	A	B	C	D	E	A	B	C	D	E	A	B	C	D	E	A
	Test Item #	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
	1st Drop	C	C	D	D	D	D	D	D	E	E	E	E	E	E	A	A
	2nd Drop	B	C	D	E	A	B	C	D	E	A	B	C	D	E	B	C
48 Test Items	Test Item #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
	1st Drop	A	A	A	A	A	B	B	B	B	B	C	C	C	C	C	D
	2nd Drop	A	B	C	D	E	A	B	C	D	E	A	B	C	D	E	A
	Test Item #	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
	1st Drop	D	D	D	D	E	E	E	E	E	A	A	A	A	A	B	B
	2nd Drop	B	C	D	E	A	B	C	D	E	A	B	C	D	E	A	B
	Test Item #	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48
	1st Drop	B	B	B	C	C	C	C	C	D	D	D	D	D	E	E	E
	2nd Drop	C	D	E	A	B	C	D	E	A	B	C	D	E	A	B	C

NOTE: If packaging consists of an alternating mortar orientation, define the package up and down axis and assign orientations based on Table C2-2.

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A - Major axis horizontal

B - Major axis vertical, fuze up

C - Major axis vertical, fuze down

D - Major axis 45 degrees, fuze up

E - Major axis 45 degrees, fuze down

d. Test Temperature: Temperature condition the test munitions prior to testing. Stabilize all cold munitions to LCT, and all hot munitions to UCT. The drop tests should be conducted within the shortest duration possible upon removal from the conditioned environment. The maximum duration should be no longer than 15 minutes. During transport from the conditioned environment to the test site, it is good practice to minimize heat transfer effects through the use of thermal mitigation measures (e.g., insulated transport box or insulating blanket).

C.2.6. TACTICAL DYNAMICS.

In most instances, the mechanical environment is highly specific to the method of deployment and the mortar platform. As described in Paragraph A.2.7, multiple deployment scenarios may have to be considered especially when mortar munitions are deployed on multiple mortar carrier platforms. It is recommended that actual environments be measured and used to develop applicable shock and vibration test profiles criteria whenever possible. The default test severities stated in the following paragraphs are intended to represent minimum test requirements in lieu of actual measured data.

C.2.6.1. Land – Tracked Vehicles Vibration.

Perform vibration testing in accordance with MIL-STD-810, Methods 514 and 527 using the following test parameters:

a. Munition Configuration: Primary and Secondary Packaging.

Note: Since it is likely that munitions could be tactically transported within both its primary and secondary packaging, both configurations should be tested.

b. Test Level: ITOP 01-2-601, Materiel Deployed on the Sponson or Installed in Hull of Light Tracked Vehicles (Low and high speed portions, or as appropriate).

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c. Test Duration: Equivalent to 1,000 km transportation (Note, this simulates a single deployment).

d. Test Temperature: Temperature condition the test munitions prior to, and during vibration testing. Stabilize all cold munitions to LCT, and all hot munitions to UCT.

C.2.7. TACTICAL DROP.

The Tactical Drop test, performed as the last test of the Life Cycle Sequential Test series, simulates accidental drops encountered during handling of bare and secondary packaged munitions. Accidental drops may occur during resupply operations or during loading of munition into the mortars. The drop heights used will be LCEP specific, including consideration of the maximum drop height from the cannon muzzle during loading and maximum drop heights during loading into the combat platform. Minimum drop height is the greater of 1.5 m or the maximum mortar cannon muzzle loading height for all munitions (to represent manual handling); the munition should remain safe to handle and dispose after dropping. Perform drop testing in accordance with MIL-STD-810, Method 516 (Transit Drop) using the following test parameters:

Note: The tactical drop may damage the test munition to such a degree that it is determined that the round should not be loaded into the cannon. If such damage has been observed on two (2) consecutive rounds, at the discretion of the testing authority, a separate subtest using a separate sample should be used to determine a set of reduced-severity conditions of drop orientations and heights which will reduce the damage to the munition to a level where all or nearly all of the test munitions will be capable of being loaded and fired. Usually, the drop height will be reduced in 300 mm increments until a suitable height is established for each orientation. If the limiting factor is chambering the cartridge, an actual cannon should be used, rather than a chamber gauge. A chamber gauge is usually more restrictive. Sometimes munitions which will not fit the chamber gauge can be chambered in the cannon. After the subtest has determined an acceptable height, the remainder of the LCEP rounds will be tested at this height.

a. Munition Configuration: All items shall be dropped in the ready-to-fire configuration.

b. Test Level: Conduct tactical drop in accordance with MIL-STD-810, Method 516 (Transit Drop). Drop each item two (2) times from a minimum height of 1.5 m onto a concrete-supported steel surface.

c. Drop Orientation: Each mortar cartridge is to be dropped twice to impact. Table C2-2 defines the recommended drop order for the various test sample quantities defined in Table B2-3. The drop order plan will have to be modified if the test item quantities are different from Table C2-2.

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d. Test Temperature: Temperature condition the test munitions prior to testing. Stabilize all cold munitions to the LCT and all hot munitions to the unpackaged UCT temperature. Any drop should be conducted within the shortest duration possible upon removal from the conditioned environment. The maximum duration should be no longer than 15 minutes. During transport from the conditioned environment to the test site, it is good practice to minimize heat transfer effects through the use of thermal mitigation measures (e.g., insulated transport box or insulating blanket).

C.2.8. FIRING AT UPPER/LOWER FIRING TEMPERATURE AND SAFETY PRESSURE.

To verify if the munition is safe to fire under all training and combat conditions, the munition must be fired at peak safety test pressure after exposure to the sequential environmental tests environments. The peak safety pressure must reflect the pressures the munition and mortar are designed to withstand, and the extreme spread of pressures likely to be encountered during the munition's life cycle. Normally, the peak testing pressure occurs at UFT; however, a safety-test pressure must also be determined for LFT. The peak testing pressure is preferably determined from the munition PMP, provided by the developer. The PMP of the mortar used for testing should be higher than the munition PMP (refer to STANAG 4110 for pressure terms and how to calculate Maximum Operating Pressure).

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APPENDIX C. ENVIRONMENTAL TEST DESCRIPTIONS.
ANNEX 3. NON-SEQUENTIAL CLIMATIC TEST.

C.3.1. GENERAL.

This annex provides the rationale for the Non-Sequential Climatic Tests for which the test descriptions are provided herein. See Figure B1-1, Representative S3 Test Program for the summary of the overall non-sequential climatic test program. MIL-STD-810, Part One contains general guidance and requirements necessary to conduct the test procedures contained herein and must be used in combination with these test descriptions.

C.3.1.1. Post Test Firing.

The visually satisfactory munitions shall be conditioned at 21 °C for a period of 24 hours and then fired at maximum charge. The following data shall be acquired for the firing test:

(1) Chamber pressure (± 689.5 kPa).

(2) Muzzle velocity (± 0.1 m/s).

(3) Time of flight (± 0.1 seconds).

(4) Range and deflection (± 10 m).

(5) Fuze functioning and/or cartridge functioning observations and height of burst (± 0.5 m), if applicable.

(6) Meteorological conditions aloft.

(7) Illumination candle burn time (± 0.1 seconds).

(8) High speed camera photographs to document projectile integrity.

(9) Radar plots if projectile exhibited parts separation and/or unstable flight.

C.3.1.2. Data Analysis/Procedure.

The results of the adverse environment firing will be compared to those of the control cartridge groups. Calculate and compare the mean velocity, mean chamber pressure, fuze functioning reliability, and the range-and- deflection probable error of the test rounds to the control rounds fired. Make statistical comparisons for equality of means, equality of probable errors, and equality of fuze functioning reliably to the greatest extend possible. Assess the criteria to determine the degree of degradation of the performance parameters due to each adverse environment. Consider a round to have impacted short if its range is less than 80% of the average range of the control group.

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ANNEX 3. NON-SEQUENTIAL CLIMATIC TEST.

Assume velocity, time of flight, pressure, range and deflection to be normally distributed random variables. Assume the number of proper fuze functions per number tested is assumed to be a binomial distributed random variable.

C.3.2. SAND AND DUST.

Perform the Sand and Dust test in accordance with the most severe exposure parameters defined in MIL-STD-810, Method 510, Blowing Dust and Blowing Sand Procedures:

a. Munition Configuration: Unpackaged munition.

b. Blowing Dust Test Level: MIL-STD-810, Method 510, Blowing Dust Procedure, using any of the standard dust small particle compositions, the default particle size distribution and the default average dust concentration in an air velocity of 9 m/s.

c. Blowing Sand Test Level: MIL-STD-810, Method 510, Blowing Sand Procedure, using silica sand with the default structure and hardness. A concentration of 2.2 ± 0.5 g/m3 will be maintained in an air velocity of 18 to 30 m/s.

d. Test Item Orientation: Orient the munition such that all surfaces are equally exposed to the blowing dust or sand.

e. Time of Exposure: Time of exposure for the blowing dust test is 6 hours at standard laboratory ambient temperature and 6 hours at the hot storage temperature for a total of 12 hours. Time of exposure for the blowing sand test is 90 minutes per orientation.

f. Method of Post Test Dust and Sand Removal: Following the test, the munitions will be wiped clean and visually inspected for damage.

g. Post Inspection: After the dust or sand test, the munition shall be inspected for abrasion, loss of protective coatings and blockage of cavities, such as on the fuze or ignition cartridge. Visually satisfactory munitions shall be prepared for firing.

C.3.3. MOLD/FUNGUS GROWTH.

Perform the mold/fungus growth test in accordance with MIL-STD-810, Method 508 to determine if mold growth will occur and, if so, how it may degrade/impact the use of the munition. Twenty-eight (28) days is the minimum test period to allow for mold germination, breakdown of carbon-containing molecules, and degradation of material.

a. Munition Configuration: Unpackaged munition.

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b. Time of Exposure: Twenty-eight (28) days is the minimum test period to allow for mold germination, breakdown of carbon-containing molecules, and degradation of material.

c. Post Inspection: Visually inspect the munition after the test period. Visually satisfactory munitions shall be prepared for firing.

C.3.4. BARE IMMERSION.

Perform the immersion test in accordance with MIL-STD-810, Method 512, using the following test parameters:

a. Munition Configuration: Unpackaged munition.

b. Conditioning Temperature: Munitions are to be preconditioned to a temperature of 27 °C above the water temperature to represent exposure to solar heating.

c. Depth of Immersion: It is unlikely that mortar munitions will be immersed to a depth of 1 m during operation as required by MIL-STD-810, Method 512. Therefore, it is recommended that the munition be immersed in a water bath with a water level sufficient to ensure that the upper surface of the munition is 8 cm below the surface of the water. A depth of one (1) meter, or greater may be required based on the LCEP.

d. Test Duration: It is unlikely that mortar munitions will be immersed for a lengthy time period during operations. Therefore, it is recommended that the immersion time period be 10 seconds. A time period of 30 minutes or greater, as required by the MIL-STD-810, Method 512 may be required based on the LCEP.

e. Post Inspection: After removing the cartridge from the water bath, the cartridges will be fired without shaking. The immersion process will be repeated until all cartridges are fired. If range and velocities are normal, no further bare immersion testing is required. Otherwise, the test will be repeated and cartridges shaken twice prior to firing. Clean/dry the tube between shots.

C.3.5. SALT FOG.

Perform Salt Fog testing in accordance with MIL-STD-810, Method 509 using the following test parameters:

a. Munition Configuration: Unpackaged munition.

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b. Test Levels: Use default parameters as specified in MIL-STD-810, Method 509.

c. Test Duration: Minimum of two (2) wet-dry cycles to maximum of four (4) cycles.

d. Post Inspection: Following exposure, the cartridges will be visually inspected and allowed to dry for a period of forty-eight (48) hours. Cartridges will be re-inspected prior to firing.

C.3.6. Humid Heat Cycling – Bare Round and Secondary Packaged Round.

Perform Aggravated Humidity testing in accordance with MIL-STD-810, Method 506 using the following test parameters:

a. Munition Configuration: Secondary packaging and unpackaged munition.

b. Test Level: MIL-STD-810, Method 506, 'Aggravated Cycle (cycle 3)'.

c. Test Duration – 10 each 24-hour cycles to be applied.

d. Post Inspection: Following exposure, the cartridges will be visually inspected and allowed to dry for a period of forty-eight (48) hours. Cartridges will be re-inspected prior to firing.

C.3.7. SOLAR RADIATION.

Perform Solar Radiation testing in accordance with MIL-STD-810, Method 505 using the following test parameters:

a. Munition Configuration: Unpackaged munition.

b. Test Level: MIL-STD-810, Method 505 (Temperatures: +32 °C to +49 °C. Irradiance: 0 W/m2 to 1120 W/m2.).

c. Test Duration: Minimum of seven (7) 24-hour solar cycles to be applied.

d. Substrate Characteristics: The test items shall be mounted on raised supports during the solar radiation exposure.

e. Post Inspection: Following exposure, the cartridges will be inspected for visual degradation of materials. Cartridges will be re-inspected prior to firing.

C.3.8. THERMAL SHOCK.

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Expose all munitions to the high- and low-temperature phases of the temperature shock tests in accordance with MIL-STD-810, Method 503 and as described below. The aggravated thermal shock cycle, may be substituted for the phased thermal shock approach described below (see Paragraph A.3.8.1).

C.3.8.1. Phased Thermal Shock.

a. Munition Configuration: Unpackaged munition.

b. Low Temperature Phase. Conduct five (5) cycles of the low temperature phase temperature shock test in accordance with MIL-STD-810, Method 503 and the following test parameters:

(1) The high temperature shall be 21 °C and the low temperature chamber shall be -51 °C.

(2) Munitions are to remain in each chamber until temperature stabilization is achieved (24 hours maximum).

c. High Temperature Phase. Conduct five (5) cycles of the high temperature phase temperature shock test in accordance with MIL-STD-810, Method 503 and the following test parameters:

(1) The high-temperature shall be the unpackaged SRE temperature and the low-temperature chamber shall be -5 °C.

d. The munitions are to remain in each test chamber until temperature stabilization is achieved (24 hours maximum for unpackaged munition).

e. Post Inspection: Following exposure, the cartridges will be inspected visually. Cartridges will be re-inspected prior to firing.

C.3.8.2. Aggravated Thermal Shock.

a. Munition Configuration: Unpackaged munition.

b. Test Level: Conduct three (3) cycles of both the low and high temperature shock tests (6 shock in all) in accordance with MIL-STD-810, Method 503 and the following test parameters:

(1) Low temperature chamber: -51 °C.

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(2) High temperature chamber: +71 °C.

c. Test Duration: Three (3) cycles of both the low and high temperature shocks tests (6 shocks in total).

d. The munitions are to remain in each test chamber until temperature stabilization is achieved (24 hours maximum for unpackaged munitions).

e. Post Inspection: Following exposure, the cartridges will be inspected visually. Cartridges will be re-inspected prior to firing.

C.3.9. CONTAMINATION BY FLUIDS.

Perform Fluid Contamination testing in accordance with MIL-STD-810, Method 504, using the following test parameters:

a. Munition Configuration: Unpackaged munition.

b. Test Requirements: To be tailored according to the LCEP.

c. Type of Exposure: Recommend occasional exposure unless the LCEP dictates otherwise.

d. Post Inspection: Following exposure, the cartridges will be inspected visually. Cartridges will be re-inspected prior to firing.

C.3.10. LOW PRESSURE (ALTITUDE).

Perform Low Pressure testing in accordance with MIL-STD-810, Method 500 ‘Storage/Air Transport’ using the following parameters:

a. Munition Configuration: Primary packaged munition.

b. Test Pressure: Maximum defined equivalent altitude of 56.8 kPa (4,570 m equivalent altitude).

c. Rate of Pressure Change: 10.3 kPa/s (equivalent to 10 m/s rate of altitude change).

d. Pressure Stabilization Duration: One (1) hour.

e. Test Temperature: Ambient.

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f. Post Inspection: Following exposure, the cartridges will be inspected visually. Cartridges will be re-inspected prior to firing.

C.3.11. CARGO AIRCRAFT DECOMPRESSION.

Perform Rapid Decompression testing in accordance with MIL-STD-810, Method 500, 'Rapid Decompression' using the following test parameters:

a. Munition Configuration: Primary packaged munition.

b. Test Pressures: Initial pressure = 56.8 kPa. Final (decompression) pressure = 18.8 kPa.

c. Duration of Pressure Change: 10±5 seconds.

d. Decompression Pressure Stabilization Duration: 10 minutes.

e. Rate of Pressure Change to Ambient: 10.3 kPa/s (equivalent to 10 m/s rate of altitude change)

f. Test Temperature: Ambient.

g. Post Inspection: Following exposure, the cartridges will be inspected visually. Cartridges will be re-inspected prior to firing.

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APPENDIX C. ENVIRONMENTAL TEST DESCRIPTIONS.
ANNEX 4. NON-SEQUENTIAL DYNAMIC ENVIRONMENTAL TEST.

C.4.1. GENERAL.

This appendix provides the rationale for the Non-Sequential Dynamic Environmental Tests for which the test descriptions are provided herein. See Figure B1-1, Representative S3 Test Program for the summary of the overall non-sequential dynamic test program.

C.4.2. NON–CONTACT UNDER WATER EXPLOSION (UNDEX).

If national procedures require UNDEX testing, this should be done in accordance with MIL-DTL-901 and ANEP 43using the following test parameters:

a. Munition Configuration: Primary packaged munition.

b. Test Level: Test parameters are to be determined by the National Authority to ensure Safe for Disposal requirements are met. Guidance can be found in North Atlantic Treaty Organization (NATO) publications ANEP-43 and MIL-DTL-901.

c. Test Temperature: Temperature condition the test munitions prior to, and during testing. Stabilize all munitions to +21 °C.

Note: This should be conducted as a non-sequential test on a single tactical transportation package if the criteria is ‘safe for disposal’, or during the LCEP life cycle test sequence on selected munitions if the criteria is ‘safe for use’.

C.4.3. 12-Meter Logistic Safety Drop Test.

This mandatory logistic drop test, as described in MIL-STD-2105 and ITOP 04-2-601 assesses the safety of the munitions when exposed to a free-fall drop which may be encountered during ship loading operations.

a. Munition Configuration: Packaged or palletized (depending upon the LCEP).

b. Test Level: One (1) freefall drop from a height of 12 m onto a concrete supported steel surface.

c. Drop Orientation: The test munition is to be released such that it will approximate an initial impact in one (1) of the following orientations:

(1) Major axis vertical, nose up

(2) Major axis vertical, nose down

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(3) Major axis horizontal

d. Test Temperature: Temperature condition the test munitions prior to, and during testing. Stabilize all cold munitions to LCT, and all hot munitions to UCT.

C.4.4. TACTICAL (MILITARY) LOGISTIC TRANSPORTATION – FIXED WING AIRCRAFT – PARACHUTE RESUPPLY.

C.4.4.1. Low Velocity Parachute Drop.

Munitions may be re-supplied by low-velocity parachute delivery and are expected to remain S3 compliant following such an event. If parachute is an expected mode of deployment, this test should be conducted as a non-sequential test on a minimum of three (3) cartridges. This test is normally required for airdrop certification and specific test requirements are subject to approval by the airdrop certifying agency.

a. Munition Configuration: Conduct this test on individually packaged or palletized munitions with appropriate parachute drop specific padding/crushable material.

b. Test Procedure: This environment may be replicated by either an aircraft drop per ITOP 07-2-509 or may be simulated with a freefall drop. For the simulated drop, conduct one (1) drop in accordance with MIL-STD-810, Method 516 (Transit Drop Procedure), from a height of 3.8 m onto a hard surface, such as steel or concrete, to simulate a Low Velocity Air Drop. The test item is to be released such that it will approximate an initial impact drop orientation of base down. A laboratory shock test may also be applied if it can be demonstrated to produce an equivalent velocity and loading on the munition.

c. Test Temperature: Ambient.

d. Post Inspection: Following exposure, the cartridges will be inspected visually. Cartridges will be re-inspected prior to firing.

C.4.4.2. High Velocity Parachute Drop.

Munitions may be re-supplied by high-velocity parachute delivery and are expected to remain S3 compliant following such an event. High velocity parachute systems may result in impact velocities of 27.3 m/s. If high velocity parachute is an expected mode of deployment, this test should be conducted as a non-sequential test.

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a. Test Configuration: High velocity parachute drops occur in bulk munition (palletized) configuration with appropriate supplemental shock isolation commonly used for parachute drop operations.

b. Drop Height: This environment may be replicated by either an aircraft drop per ITOP 07-2-509 or may be simulated with a freefall drop. For the simulated drop, conduct one (1) drop in accordance with MIL-STD-810, Method 516, from a height of 38 m onto a hard surface, such as steel or concrete, to simulate a High Velocity Air Drop. The test item is to be released such that it will approximate an initial impact drop orientation of base down.

c. Test Temperature: Ambient.

d. Post Inspection: Following exposure, the cartridges will be inspected visually. Cartridges will be re-inspected prior to firing.

C.4.4.3. Malfunctioning Parachute Drop.

Munitions that may be re-supplied by parachute delivery are at risk of a malfunctioning parachute drop scenario and are expected to remain safe for disposal. Per ITOP 4-2-601, Test E5, malfunctioning parachute systems may result in impact velocities of 45.7 m/s. This test should be conducted as a non-sequential test.

a. Test Configuration: Malfunctioning parachute drops occur in bulk munition in the palletized configuration with appropriate supplemental shock isolation commonly used for parachute drop operations.

   Note: Consideration should be given to testing the primary package configuration if the individual primary package containers do not lock together and it is plausible that an individual container will fall instead of the entire pallet.

b. Drop Height: This environment may be replicated by an aircraft drop per ITOP 07-2-509.

c. Number of Drops: It is not expected that munitions would be dropped more than once from this extreme height during its service life; thus, only one (1) drop is required.

d. Test Temperature: Ambient.

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APPENDIX D. ADDITIONAL TESTS AND ASSESSMENTS.
ANNEX 1. NON-SEQUENTIAL FIRING TESTS.

D.1.1 GENERAL.

This appendix provides descriptions of all of the non-sequential firing, system, additional and preliminary design assessment tests in the S3 Test Program. Rationales for these tests are provided in Appendix A. Table I3-1 (Appendix I, Annex 3) provides cross reference of similar national and international test standards.

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ANNEX 2. NON-SEQUENTIAL FIRING TESTS.

D.2.1 GENERAL.

This appendix provides the rationale for the Non-Sequential Firing Tests for which the test descriptions are provided herein. See Figure B1-1, Representative S3 Test Program for the description of the overall test program.

D.2.2 WET EFFICIENCY TRIAL.

A sample of 36 mortar cartridges is fired in order to compare the performance of wet and dry propellant charges. The wet efficiency trial shall be conducted as follows.

D.2.2.1 Test Procedure.

a. Mortar Cartridges: Inert-filled mortar cartridges will normally be used; these will be of a weight representative of service mortar cartridges.

b. Fuzes: These can be live if suitable firing clearance is available. Alternatively they may be inert or plug representing fuzes.

c. Mortar Cannon: Either a service cannon or a pressure cannon which is representative of a service cannon and has a minimum of 75% life remaining may be used.

d. Propellant: In order to afford comparable results, all propellant charges must contain propellant from the same lot which must be representative of production.

e. Propellant Charges: Firings are to be carried out using both the maximum and minimum service charge and also intermediate charges if considered appropriate (see sub-paragraph h below).

f. All firings are to be carried out with the mortar cartridges (including charges) conditioned to 21 °C.

g. Occasions: The complete trial must be fired on one occasion.

h. Firing Matrix. The firing matrix is described in Table D2-1.

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Table D2-1: Test Matrix for Wet Efficiency Trial

Group	Quantity Mortar Cartridges	Charge	Wet/Dry
1a	6	Maximum	Dry
1b	6	Maximum	Wet
2a	6	Minimum	Dry
2b	6	Minimum	Wet
3a	6	Intermediate	Dry
3b	6	Intermediate	Wet

i. Wet Conditioning. Mortar cartridges to be fired wet are to have their tails and propellant charges subjected to either of the following:

(1) Immersion in fresh water at 21 °C for 30 seconds.

(2) Exposure to a meteorological 35 knot wind and simulated 150 mm per hour rain, for 30 seconds.

j. The mortar cartridges are then to be transferred directly to the mortar and fired without any attempt being made to disperse the water from the charge. The cannon will be dry swabbed on completion of each wet series.

D.2.2.2 Records and Observations.

The following data are required:

a. Specifications of munition and weapons fired.

b. Muzzle velocities.

c. Subjective assessment of residue, debris (including external effects) and in-cannon burning.

D.2.2.3 Criteria.

Note: Failure to meet these criteria will require that consideration be given to introducing restrictions in use.

a. The mean muzzle velocity of mortar cartridges with wet propellant charges should not be less than 90%; the muzzle velocity of any individual cartridge with a wet propellant

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charge should be not less than 80% of the mean muzzle velocity of comparable mortar cartridges with dry propellant charges, for each service charge fired.

b. If these criteria are not met, consideration may be given to the adoption of an agreed method of shaking water from the propellant charges after immersion and before loading into the mortar. The firing scheme may then be repeated and the same criteria applied.

c. After firing each wet series there should be no significant quantity of unburnt propellant left in the mortar.

d. In any event, residue and debris should be free from signs of unburnt propellant or significant portions of charge container material, which could cause the mortar cartridge to stick in the cannon, a slow drop, or ignition of subsequent charges.

D.2.3 RESIDUE - DEBRIS TEST.

Conduct the residue-debris test to determine if the mortar cartridges will fire without excessive misfires or whether the propellant leaves residue or debris that could cause a safety hazard. A range with a horizontal surface without any form of debris already present is preferred for the test.

D.2.3.1 Test Procedure.

The following test procedure is provided as guidance.

a. Weapon Setup: Testing will be conducted with the mortar mounted in the host vehicle platform or seated in the ground. The weapon elevation shall be at minimum. To the greatest extent possible the location should be shielded from wind or strong air currents. Collection cloths shall be laid on the ground around the weapon to collect spatial distribution in width and length of debris from the firing event. The number and position of witness cloths depend on the nature of the munition. Ensure collection cloths are free of debris before starting each group.

b. Test Munition: Munitions with inert payloads are recommended for this test.

c. Temperature Conditioning: Temperature condition munitions to the worst case temperature that produces excessive residue for a minimum of 24 hours.

d. Test Firings: Fire munitions in accordance with the test matrix described in Table D2-2.

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ANNEX 2. NON-SEQUENTIAL FIRING TESTS.

e. Conduct bore scope, bore measurements and operator level cleaning of the cannon prior to the start of the test and at the end of each group.

Table D2-2: Residue – Debris Test Firing Matrix

Group	Quantity of Munitions[1]	Charge[1]	Rate-of-Fire[2] (rounds per minute)
1	50	3	2
2	50	2	1
3	50	2	2
4	50	1	1
5	50	0	2
6	50	1	1
7	50	2	2
8	50	1	1

Note 1: The designated charge is based on a 5 zone, charge 0 through 4, propelling charge system. This should be adjusted accordingly for alternate charge systems. Group size shall not be smaller than the intended largest mission profile for the weapon system.

Note 2: The designated rate-of-fire, rounds per minute, is for a slow rate that prevents the barrel from getting extremely hot. Lower charges contribute less heat and likely to experience buildup of residue on bore surface at low charges.

f. Following each group, perform the following tasks:

(1) The weapon should be left to cool to approximately ambient temperature.

(2) Conduct operator level cleaning consistent with procedures to be used during a mission in the field. No use of cleaning aids or solvents should be used that would not be readily available during a mission. Collect all material removed.

(3) Measure bore diameter at multiple locations. Ensure that the minimum diameter of the bore is not smaller than the outside diameter of the projectile.

(4) Perform a borescope inspection. Record the location and amount of coverage of any firing residue on the cannon.

(5) Collect all debris from collection cloths and put aside for analysis.

g. After firing groups 4 and 8, any suspected hang fires, and/or two consecutive misfires, perform the following additional steps:

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ANNEX 2. NON-SEQUENTIAL FIRING TESTS.

(1) Clean the cannon according to post mission or maintenance cleaning instructions. Collect any removed material.

(2) Conduct borescope inspection and compare residue coverage with pre-maintenance cleaning.

(3) Conduct bore diameter measurements in order to assess any gross changes in bore condition.

D.2.3.2 Data Required.

a. Nomenclature and serial number of weapon.

b. Munition, and ignition cartridge and propelling charge nomenclature and lot number.

c. Munition charge or zone number.

d. Cartridge temperature.

e. Ambient conditions (RH and temperature).

f. Wind speed and direction.

g. Video recording of the muzzle exit to record debris.

h. Photos, dimensions and weight of debris collected from collection cloths.

i. Photos, dimensions and weight of debris collected from the cannon.

j. Records of misfires and slow drops.

k. Barrel inspection reports (borescope and bore measurements)

D.2.3.3 Data Analysis.

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a. The collected data shall be analyzed to ascertain the risk for hang fire. Calculate a hang fire risk diameter and based on rate of residue buildup rate (bore measurement inspection reports) develop a cleaning interval.

b. Analyze the effect of firing at various rates of fire and the firing charge on the expelled debris.

c. Measure the distribution of munition debris projected from the cannon. Ensure that larger debris does not enter the crew's operational area. Document the range projected by burning embers. Any ejected debris from the firing event that enters the operator's area or any burning debris that enters the ammo preparation or storage areas should be evaluated for safety purposes.

d. Ascertain the relative difficultly of cleaning the mortar tube after each group.

e. Verify that the cannon maintenance cleaning frequency and procedures are adequate to prevent residue growth in the interior surface of the cannon.

f. Keep track of misfire occurrences exceeding the allowable limits of the firing group after conducting maintenance cleaning. Additionally, any hardened debris collected from the cannon tube that exceeds in any dimension the pin protrusion of the cannon shall be documented.

D.2.4 MUZZLE FLASH TEST.

This test should be conducted concurrently with the residue-debris test.
Install radiometric sensors in the operator's eye positions (including one at the operator's eyepiece and any observer location) and aim them along the flight path of the munition. For observers, install radiometric sensors perpendicular and directed toward the exhaust plume. Deploy calibrated photometric detectors for several firings as above. Radiometric data that contains visible spectrum levels may be reduced to provide photometric data instead. Obtain measurements of radiation capable of causing a thermal injury at the operator's face position.

D.2.4.1 Test Procedure.

The test procedure is identical to the residue-debris test, Paragraph D.2.3.1.

D.2.4.2 Data Required.

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The data required is identical to the residue-debris test, Paragraph D.2.3.2 with the addition of the following:

a. Muzzle flash data for each round, including area, intensity, and measured flash temperature for each round. The data shall be recorded for both the 3 to 5 micrometer region and the 8 to 12 micrometer region.

D.2.4.3 Data Analysis.

The data shall be analyzed to determine if the muzzle flash presents a hazard to the mortar crew and if the emissivity requirements have been met.

D.2.5 FUZE SAFE ARMING TEST.

Perform the fuze safe arming test to verify the correct functioning of the projectile fuzing system.

Determining the minimum fuze arming distance is an essential component to calculate Safe Separation. Mechanical fuze designs commonly require test firing in a horizontal orientation against wooden targets hard enough to make the fuze function on impact mode. The application of the Neyer Strategy or similar strategy is recommended to determine the minimum fuze arming distance. The initial locations for the upper and lower gates (target distance/range from weapon) will be based on design characteristics and should reflect the distances where near zero probability of arming (lower gate) and near 100 percent probability of arming (upper gate) exist. Sufficient number of rounds should be fired at upper and lower operational firing temperature using a high speed camera to record warhead functioning. The target distance at which each function and non-function occurs will be recorded. An assumption will be applied that the probability of arming versus increasing distance is described by a cumulative normal distribution. Maximum likelihood methods will be used to estimate the mean and minimum fuze arming distances. Electro-mechanical fuze design should be assessed if test firing is beneficial to determine the arming distance. All electronic fuze designs that have built-in arming inhibitors may prohibit the value of actually firing rounds to support the analysis of Safe Separation.

D.2.5.1 Test Procedure.

The test shall be conducted in accordance with STANAG 4157 and MIL-STD-331, Test D-2 “Projectile Fuze Arming Distance”. Test D-2 describes four test methods, Probit, Langlie, Weibull One-Shot Transformed Response (OSTR) and Bruceton. The rationale for the selected test method should be clearly stated in the overall detailed test procedure. The chosen test methodology will dictate the number of test units. Typically 50 test units are required. For multi-option fuzes the test quantities may need to be increased based upon the number of options (e.g. a dual action fuze may require the test quantities to be doubled).

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D.2.6 WARHEAD ARENA TRIALS.

Warhead arena trials are performed to provide the spatial distribution of fragments. Modelling using this data is conducted to determine safe separation distances and range safety parameters. These trials should be conducted with non-sequential, factory fresh warheads unless it can be shown that exposure to thermal and dynamic stresses in the Sequential Environmental Test Sequence results in an increase in fragmentation distance. Guidance for this test can be found in ITOP 04-2-813.

a. Perform this test on six individual HE mortar/cartridges at ambient temperature.

b. Warhead arena trials require the use of the warhead only. However, the tester should evaluate whether components directly attached to the warhead or in the immediate area of the warhead, either by design or by inadvertent action, could significantly affect the warhead's fragment dispersion pattern.

c. Place the item in the instrumented arena and detonate the warhead.

d. Determine warhead fragment size, velocity, mass, spatial distribution, and levels of noise and blast pressure.

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ANNEX 3. NON-SEQUENTIAL SYSTEM TESTS.

D.3.1 GENERAL.

This appendix provides the rationale for the Non-Sequential System Tests for which the test descriptions are provided herein. See Figure B1-1, Representative S3 Test Program for the description of the overall test program.

D.3.2 FIRING TABLES TEST.

Perform this test to determine the ballistic performance of munitions when fired from a specific weapon at difference propellant zones and weapon elevations. Data from this test will also be used to determine the range and deflection precision of fired munitions to determine the surface danger zone for the munition.

D.3.2.1 Test Procedure.

Ballistic tests to obtain data for firing tables' preparation shall be performed in accordance with ITOP 03-2-601. Additionally, propellant temperature sensitivity tests shall be conducted in accordance with STANAG 4144 Appendix F, Paragraph 6. The following is provided as general guidance for test plan preparation.

a. Testing will be conducted with the mortar weapon mounted in the host vehicle platform or seated in the ground. The host vehicle location or weapon seating position shall be as close as possible to flat earth.

b. Weigh all test munitions and acquire axial and traverse moments of inertia and center of gravity measurements for a 10 round sample of each munition type.

c. Temperature condition munitions to +21 °C for a minimum of 24 hours except for munitions to be fired for the propellant temperature evaluation.

d. Fire a minimum of 5 round groups of the munitions at each charge or zone and at the weapon minimum elevation, mid-elevation, and maximum elevation. Normal weapon elevations are 800 mils, 1,156 mils, and 1,511 mils although different maximum weapon elevations are not uncommon. Table D3-1 is provided as general guidance for this portion of the test.

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Table D3-1: Firing Table Test Matrix

Charge	Number of Rounds at Quadrant Elevation - mils		
	800	1,156	1,511
0	5	5	5
1	5	5	5
2	5	5	5
3	5	5	5
4	5	5	5

e. Repeat the firing described in Paragraph D.3.2.1.d above on another occasion.

f. For weapons with capability of firing at less than 800 mils, and for munitions intended to be fired at less than 800 mils, fire a minimum of 5 round groups of the munition at each charge or zone and at a minimum of two weapon elevations less than 800 mils (e.g., 356 mils and 533 mils). The low weapon elevations for this portion of the test will be determined by the agency developing the firing tables. Table D3-2 is provided as general guidance for this portion of the test.

Table D3-2: Low Angle Firing Table Test Matrix

Charge	Number of Rounds at Quadrant Elevation - mils	
	356	533
0	5	5
1	5	5
2	5	5
3	5	5
4	5	5

g. Repeat the firing described in Paragraph D.3.2.1.f above on another occasion.

h. For weapons with direct fire capability and for munitions intended to be direct fired, fire a minimum of 5 round groups of the munition to impact target ranges from the minimum to maximum direct fire range. The maximum propelling charge shall be used for this portion of the test. The weapon elevation will be adjusted to the super elevation to impact the targets. The target ranges for this portion of the test will be determined by the agency developing the firing tables. Table D3-3 is provided as general guidance for this portion of the test.

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Table D3-3: Low Angle Firing Table Test Matrix

Charge	Number of Rounds at Range - m						
	160	230	300	475	650	825	1,000
Max	5	5	5	5	5	5	5

i. Condition a sufficient quantity of munitions at the LFT, at a mid-range temperature between the LFT and +21 °C, at another mid-range temperature between +21 °C, and the UFT and conduct the propellant temperature test on two occasions. Rounds will be fired at 800 mils weapon elevation. Tables D3-4 and D3-5 are provided as general guidance for this portion of the test.

Table D3-4: Propellant Temperature Test Matrix Occasion 1

Occasion 1							
Charge	Number of Rounds at Propellant Temperature °C						
	LFT	Mid Cold	+21	UFT	+21	Mid Cold	LFT
0	3	2	3	5	2	3	2
1	3	2	3	5	2	3	2
2	3	2	3	5	2	3	2
3	3	2	3	5	2	3	2
4	3	2	3	5	2	3	2

Table D3-5: Propellant Temperature Test Matrix Occasion 2

Occasion 2							
Charge	Number of Rounds At Propellant Temperature °C						
	UFT	+21	Mid Cold	LFT	Mid Cold	+21	UFT
0	3	2	3	5	2	3	2
1	3	2	3	5	2	3	2
2	3	2	3	5	2	3	2
3	3	2	3	5	2	3	2
4	3	2	3	5	2	3	2

D.3.2.2 Ballistic Match and Similitude Testing.

It may be necessary to determine the degree of ballistic commonality between two or more mortar projectiles with the same external shapes but with different payloads or fuzes. Additionally, it may be necessary to assess the degree of ballistic interchangeability between different weapon and fire control system. Tests to obtain data to assess the degree of ballistic commonality shall be performed in accordance with ITOP 03-2-601 and STANAG 4106.

D.3.2.3 Data Requirements.

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The firing tables test data requirements are defined within ITOP 03-2-601. Ballistic match or similitude test data requirements are defined within ITOP 03-2-601 and STANAG 4106. The following is provided as general guidance for data requirements.

a. Nomenclature and serial number of weapon.

b. Weapon tube length.

c. Munition, fuzes, and propellant nomenclature and lot number.

d. Height of impact area above mean sea level.

e. Munition's axial and traverse moments of inertial.

f. Munition's center of gravity measurements.

g. Metrological Information

(1) Meteorological conditions aloft every hour – bracketing times of firing.

(2) Altitude of meteorological station.

h. Round-By-Round Data

(1) Date and time of firing.

(2) Test round number and tube round number.

(3) Azimuth of line of fire.

(4) Height of weapon above sea level.

(5) Angle of weapon elevation before and after firing.

(6) Charge or zone number.

(7) Propellant, fuze and cartridge temperature.

(8) Cartridge weight.

(9) Time of flight (time to burst for illumination or air burst munitions).

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(10) Range (range to burst for illumination or air burst munitions).

(11) Height of burst for illumination or air burst munitions.

(12) Deflection.

(13) Fuze setting.

(14) Muzzle velocity.

(15) Chamber pressure.

D.3.2.4 Data Reduction.

Data reduction for input to the fire control system shall be in accordance with STANAG 4144. It is emphasized that the data requirements for the fire control system must be discussed and approved by the fire control system developing agency. If the munition is being incorporated for use on an additional mortar, then the data requirements shall be in accordance with STANAG 4106. The data shall include computation of the mean and standard deviation of velocity, range, deflection and probable error in range and deflection.

D.3.3 RATE-OF-FIRE TEST.

The maximum rate-of-fire is described as the maximum rate at which it is physically possible for a trained gunner to continually fire the weapon; limited by tube elevation and/or designated maximum operating temperature (D-MOT) constraints. Sustained rate-of-fire is defined as the rate-of-fire, for a particular charge and munition type, at which the test weapon can be continuously fired without exceeding the tube's D-MOT. This test is normally conducted as part of the weapon safety and qualification testing but it may be necessary to conduct the test if new munition is being qualified for an existing mortar and the new munition has a different weight and/or different propellant system than the munition that was used to qualify the mortar.

D.3.3.1 Test Preparation

a. Affix thermocouples to points along the tube where maximum temperatures may be expected (ignition zone, middle of clamp, near muzzle), or as directed by the weapon designer. Affix thermocouples to the tube so as not to alter the physical properties of the tube.

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b. Optionally, adapt and install an electric timer to measure the time required for a projectile to slide down the tube and strike the bottom for tube elevations of the minimum, mid-range and maximum firing elevation.

c. Place the weapon or test item on a firm soil surface or platform. If placed on a soil surface, fire a minimum of five rounds to seat the weapon.

Note: If a soil surface is selected it should be firm enough to not require re-emplacement of the weapon during the entire rate-of-fire test.

D.3.3.2 Test Procedure.

D.3.3.2.1 Maximum Rate-of-Fire-Test.

a. Record ambient temperature 15 minutes prior to start of test at all locations on tube before firing commences.

b. Fire the weapon at specific elevation(s) and charge(s) specified in the system description as fast as possible for the specified time period or until the tube temperature reaches the D-MOT.

c. Record data as indicated in Paragraph D.3.3.2.1.1 below.

d. Repeat this procedure a minimum of two times at each elevation and/or charge, using different personnel for each repetition.

e. Use appropriate (for metallic or non-metallic tube material or tube condition) non-destructive test (NDT) technique to determine presence of cracks, deformations, etc., on mortar tube, basecap and baseplate.

D.3.3.2.1.1 Data Required.

a. Time of descent of the cartridge at each specified tube elevation (if the weapon was instrumented with electronic timing instruments).

b. For each team member at each specified elevation:

(1) Time between each round fired.

(2) Number of rounds fired to reach D-MOT.

(3) Time to reach D-MOT.

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(4) Maximum rate-of-fire (number of rounds fired/minute, for time to reach D-MOT). Expressed as number of rounds per minute for X minutes.

(5) Video recording of loader movements and effect of blast, smoke, and flash on visibility, operation of fire-control equipment, and on firing team members.

D.3.3.2.2. Sustained Rate-of-Fire-Test

It is recommended that this test be performed immediately after the maximum rate-of-fire test. If not, a sufficient number of rounds must be fired to heat the cannon tube or, alternately, install "heat tape" around the tube in a helical configuration, with a quick-release mechanism near the top to allow rapid removal with minimal manual manipulation, to rapidly heat the cannon but not restrict the tube cooling.

a. Record temperature at all specified locations throughout this test.

b. If necessary, heat the cannon tube with heat tape until a temperature approximately 30 °C below the D-MOT is reached. Immediately following this phase, quickly remove the heat tape and rapidly fire as many maximum service charge rounds as necessary to raise the tube temperature to a pre-determined temperature close to the D-MOT.

c. Alter the rate-of-fire (using one of the charge levels specified in the system description) so that the tube temperature does not exceed D-MOT and is held constant near the D-MOT.

d. Record the rate-of-fire at which the tube temperature is stabilized at a level below D-MOT.

e. Use appropriate (for metallic or non-metallic tube material or tube condition) NDT technique to determine presence of cracks, deformations, etc., on cannon tube, basecap and baseplate.

D.3.3.2.2.1 Data Required.

a. Temperature of mortar tube at all specified, instrumented locations.

b. Sustained rate-of-fire for all applicable charge levels.

c. Report on results of NDT.

D.3.4 DOUBLE LOAD TEST.

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The double load test is conducted to assess the scenario of a second cartridge being loaded into a mortar tube, containing a previously loaded, but unfired cartridge and the previously loaded cartridge fires into the cartridge being loaded. A worst case test procedure is described herein. Mortar double load tests are conducted with complete standard cartridges to assess the reaction of the cartridges and mortar tube. The cartridges' positioning is the same regardless of caliber. Alternative test procedures are also described that should be considered to obtain a more thorough understanding of a double load event.

D.3.4.1 Recommended Double Load Test.

D.3.4.1.1 Test Preparation.

a. The tubes used for this test shall have no material defects, but may be at or near the condemnation criteria for wear. It is noted that the double load test may need to be repeated if the munition is fired from cannons with significantly different build standards (e.g., steel versus Inconel Alloy or composite material). For each test, the tube shall be painted to aid in fragment recovery and identification.

b. Cannon will be modified as necessary to remove the firing pin or rendered safe via the selector switch.

c. Test cartridges must be complete standard issue. The bottom cartridge will be modified with a squib or detonator to initiate its ignition cartridge.

d. Cartridges to be fired should be x-rayed in two planes, 0 and 90 degrees apart, to ensure there are no material or explosive defects, and that the fuze is not armed or defective.

e. Make pre-fire dimensional measurements and inspections (diameters, etc.) on cartridges to be fired. Compare with drawing specification requirements. If satisfactory proceed with the test.

f. Both cartridges will be at the maximum charge. Consideration should be given to testing with reduced charges for certain mortar systems (i.e., 60 mm mortar system in trigger fire/handheld mode).

g. Weapon elevation will be set at 800 mils or other convenient elevation.

h. Use radar to track flight of rounds launched/ejected downrange.

i. If the tube is not destroyed, it should be inspected for defects before re-use (borescope and pullover gage and appropriate material soundness NDT inspection).

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D.3.4.1.2 Test Procedure.

a. The top cartridge will be suspended in such a way so that the suspension apparatus does not prevent the test cartridges from freely exiting the tube. The top cartridge should be located such that only the fuze is outside the top of the tube. The fuze safety/transport pin, if applicable, should be removed. See Figure D3-1.

b. As an alternative to the set-up described in D.3.4.1.2.a and Figure D3.1, locate the mid-point (obturator band) of the second cartridge at the mid-point of the cannon or locate the mid-point of the second cartridge at the tube muzzle. These configurations are not considered the worst case compared with the recommended set-up. They will simulate alternative situations where the first cartridge becomes stuck in the tube before falling and striking the firing pin and subsequently the second loaded cartridge falls and strikes the first cartridge which becomes dislodged, falls and fires and strikes second cartridge.

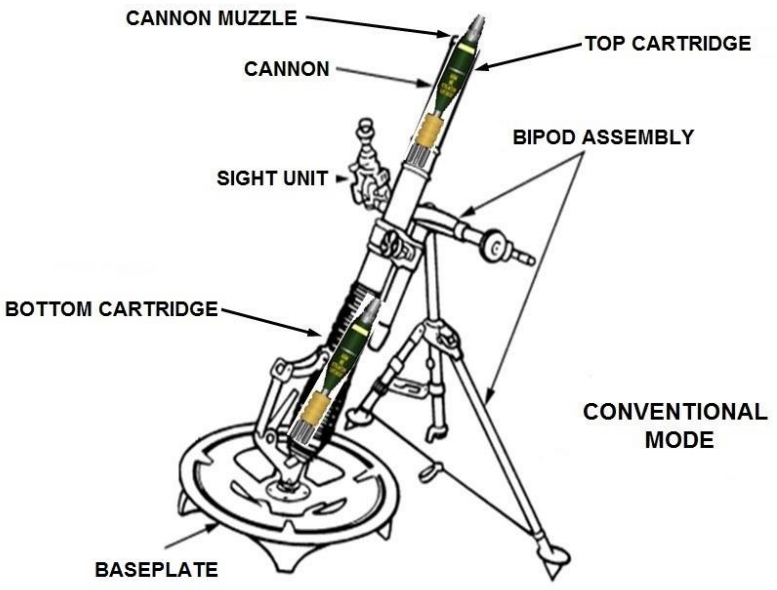


Figure D3-1: Double Load Test Configuration

c. The bottom cartridge is fired into the top cartridge and observations are made.

d. Observe where the cartridges/fragments land and if one or both cartridges function downrange.

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e. Recover weapon fragments, record the distance from weapon emplacement and photograph the location the fragment is found.

D.3.4.1.3. Data Requirements

a. Map, measure, and number all recovered fragments from the gun position.

b. Photograph all fragments found after each shot, including any pieces of explosive found.

c. Weapon configuration and serial number.

d. Munition configuration: Cartridge type for both bottom and top cartridge, cartridge lot number, fuze condition (safe/ partially armed/ armed), propulsion system type (ignition cartridge and propelling charges).

e. Gun tube inspection results (before and after test) and munition history.

f. Munition inspection results.

g. Setup photographs of weapon.

h. Description of method to initiate bottom cartridge.

i. Video of firing event.

j. High speed video focusing on the tube for each shot (with 90 degree mirror to get two views.).

k. Video capturing gun emplacement out to 15 m in front of muzzle (rear view).

l. Cartridge/tube fragments mapped, number of fragments, weight of each fragment, measured, recovered, and photographed, including pieces of explosive.

m. Tube fragments mapped, number of fragments, weight of each fragment recovered, and photographed.

n. Map any cartridges that went downrange, and whether the cartridge functioned at impact.

D.3.4.1.4 Data Reduction and Analysis.

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D.3.4.2 Alternative Test Procedures.

Alternative test procedures should be considered by the national authority and if alternative supplemental procedures are necessary to produce additional data to categorize an accidental double load firing event.

D.3.4.2.1 Alternative Double Load Trial.

The following alternative trial plans are designed to provide a more detailed simulation of the effect of the pressures generated by the propelling charges (without the effect of the high explosive (HE) fill and fuzes), effect of the HE fill (without the effect of the fuzes) , and the total cartridge with live propellant, HE fill and live fuzes. It is noted that the cost for these trials is probably substantial.

D.3.4.2.1.1 Part 1 – Evaluate Pressures Generated by Propelling Charges.

D.3.4.2.1.1.1 Test Preparation.

a. Mortar Cartridges: Inert filled mortar cartridges should be used which are assembled in accordance with the technical specification. Cartridges are to be fitted with inert fuzes. The inert charges inside the projectiles and fuzes must simulate the weight, density and mechanical properties of the explosives they replace. All service charges are to be available.

b. Mortar Tube: A pressure tube capable of withstanding up to 6 times the PMP of a standard service tube should be used. It should be capable of being fired remotely in the horizontal position. It should be fitted with instrumentation to measure pressure at 2 or 4 points:

   (1) At the base of the first round (Mandatory).

   (2) At the base of the second round (Mandatory).

   (3) Just below the obturator ring of each round (Optional).

c. Witness Screens: Witness screens representing the crew at the moment of firing should be positioned at appropriate representative locations.

d. Temperature: Firings are to be carried out with cartridges at ambient temperature and conditioned at the UFT and at the LFT.

D.3.4.2.1.1.2 Test Procedure.

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a. The test is to be repeated 12 times, or until 7 useable results are obtained at each charge and for each temperature. Maximum charges are to be tested first, followed by subsequent lower charges in descending order. If no pressure tube is available to conduct the trial, charges may be fired in ascending order.

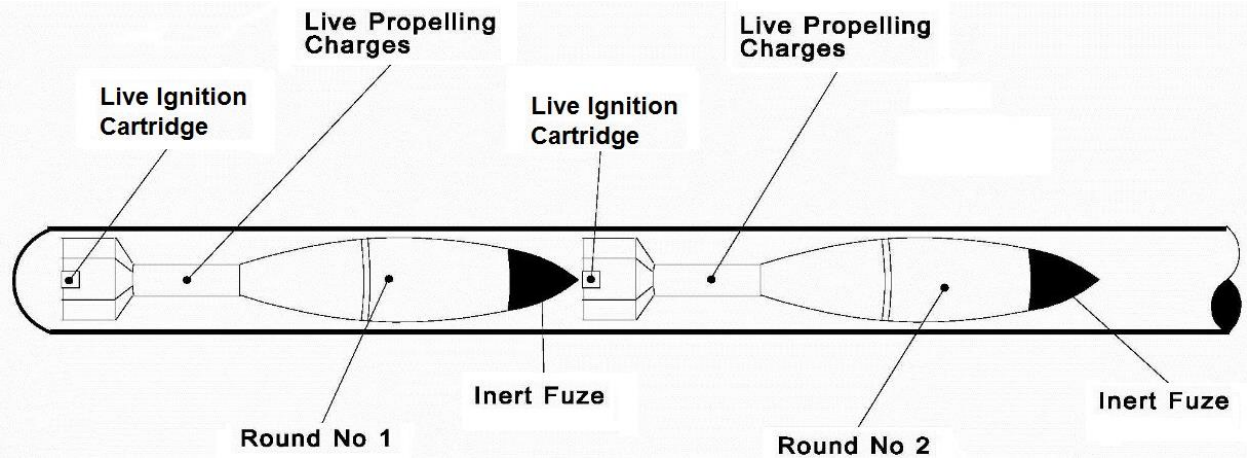


Figure D3-2: Alternative Double Load Test Configuration

Table D3-6: Double Load Test Matrix

	Round Number 1			Round Number 2		
Test	Charge	Filling	Fuze	Charge	Filling	Fuze
Part 1	Live	Inert	Inert	Live	Inert	Inert
Part 2	Live	Live	Inert	Live	Inert	Inert

b. Round Number 1, with the selected charge fitted, is placed at the bottom of the test tube which is then lowered into a horizontal position. Round Number 2 is placed in contact with the first round. See Figure D3-2 and Table D3-6.

c. Round Number 1 is fired remotely.

d. Both rounds are recovered and examined for damage or penetration of gasses at joints in the body.

e. The test is repeated as defined in Paragraph D.3.4.2.1.1.2.a above.

f. The test is to be stopped at the charge for which the pressures recorded are equal to, or just below, the mortar PMP.

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g. Equipment is to be checked between each test series and during tests if any anomalies are found. The tube is to be checked before each firing to ensure it is clear of potential obstructions.

h. The test should be stopped immediately if there is evidence of rupturing of the projectile body, ingress of gas into the body. Impacts on the witness screen likely represent potential for injury to members of the mortar crew.

D.3.4.2.1.1.3 Data Requirements.

a. Specifications of weapons and munition.

b. Pressures/time measurements at each measuring point.

c. Calculation of mean pressures and standard deviations.

d. Details of damage to rounds and evidence of ingress of gas into the body through relevant joints.

e. Evidence of significant impact on the witness screen.

D.3.4.2.1.2 Part 2 – Evaluate Influence of the HE Fill.

This test is only to be undertaken if no defects attributable to the design were identified in Part 1.

D.3.4.2.1.2.1 Test Preparation.

a. Mortar Cartridges: For each firing, one inert filled cartridge, as described in Paragraph D.3.4.2.1.1.1.a is required and one live cartridge. Both cartridges are to be fitted with inert fuzes. The qualifications itemized in Paragraph D.3.4.2.1.1.1.a also apply.

b. Charges: Maximum charge allowed by the Part 1 Test, which is equal to or just below mortar PMP.

c. Mortar Tube: A tube as described in Paragraph D.3.4.2.1.1.1.b should be used.

d. Witness Screens: Witness screens, as specified in Paragraph D.3.4.2.1.1.1.c should be provided.

e. Temperature: Firings are to be carried out at ambient, UFT and LFT.

D.3.4.2.1.2.2 Test Procedure.

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a. The test is to be carried out 3 times at each temperature.

b. Round Number 1, with the maximum charge allowed by the Part 1 Test is placed at the bottom of the test tube which is then lowered into a horizontal position. Round Number 2 is placed in contact with the first round.

c. Round Number 1 is fired remotely.

d. The test is to be stopped immediately if there is evidence of any detonation, initiation or combustion of the live projectile filling, overpressures considered abnormally higher than that recorded during Part 1 or impacts on the witness screen likely to represent potential for injury to members of the mortar crew.

D.3.4.2.1.2.3 Data Requirements.

a. Pressures measured at each measuring point.

b. Details of damage to rounds and evidence of ingress of gas into the body through relevant joints.

c. Evidence of significant impact on the witness screens.

d. Evidence of any detonation, ignition or combustion of the live projectile filling.

D.3.4.2.1.3 Part 3 – Live Full-Up Cartridge Test.

A final test may be undertaken at the discretion of the national authority to assess the potential hazards in the event of worst case double loading event.

D.3.4.2.1.3.1 Test Preparation.

a. Mortar Cartridges: For each firing, 2 live mortar cartridges are required, both fitted with live fuzes and charges.

b. Charges: Firings are to be with maximum charge fitted to both cartridges.

c. Mortar Tube: A standard mortar should be used and set-up with an elevation of greater than 1,067 mils (60 degrees).

d. Witness Screens: Witness screens, as specified in Paragraph D.3.4.2.1.1.1.c should be provided.

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e. Temperature: Firings are to be carried out at the UFT and LFT.

f. Recording Equipment: The following recording equipment is to be used:

(1) High-speed video recorder.

(2) Flash x-ray aligned with the muzzle.

D.3.4.2.1.3.2 Test Procedure

a. Occasions. The test is to be carried out on 2 occasions, once each with mortar cartridges conditioned at each temperature.

b. Round Number 1, with the charge fitted, is placed at the bottom of the test tube. Round Number 2, also with the charge fitted is inserted into the tube so that it is in contact with the first round. The tube is then set at an elevation of greater than 1,067 mils (60 degrees).

c. Round Number 1 is fired remotely.

d. Any cartridge which is expelled from the tube is to be located and safely disposed.

D.3.4.2.1.3.3 Data Requirements.

a. Specifications of weapons and munition.

b. Pressures measured at each measuring point.

c. Functioning of the cartridges, either in the tube, in flight or on impact with the ground.

d. The location, in terms of range and bearing from the firing point, of cartridge impact.

e. Evidence of significant impact on the witness screens.

D.3.5 PROPELLANT, WARHEAD, AND FUZE COOK-OFF TESTS.

Perform this test to determine the tendency for propellant charges, explosive filler, or fuzes with sensitive explosive trains to cook-off in a hot weapon, especially after a period of sustained firing. Cook-off tests may be combined with tests such as sustained rate-of-fire, to take advantage of existing tube temperature in order to eliminate the need for expending additional rounds or using a tube heater to raise the operating temperature of the mortar, so long as individual test objectives

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are not compromised. Guidance for the test can be found in NATO Manual AC/225(LG/3-SG/1) D/14 section 2.14.

A cannon heating trial should have been conducted during its qualification testing. This determines the rate at which the chamber heats up for a stated firing scenario (usually a predetermined number of rounds fired at maximum rate-of-fire) and the rate at which heat is subsequently dissipated. During the cannon heating trial, the temperature of the chamber walls is monitored and recorded (e.g., by using an instrumented munition, thermocouple, or pyrometer).

During qualification of the energetic material(s) used within the munition, the temperature of ignition will have been determined. This temperature should then be compared with that measured for the chamber wall to determine the likelihood that the energetic material(s) will spontaneously ignite. If the temperature of ignition for the energetic material(s) is high compared with the chamber wall temperature, then cook-off in a hot cannon is unlikely. If the temperature of ignition is low, or similar compared with the chamber wall temperature then cook-off in a hot cannon is likely. If it is likely that the energetic material(s) used in the munition could feasibly cook-off, then consideration should be given to the following test methodology:

D.3.5.1 Test Procedure.

A cannon should be instrumented with pressure transducers to provide pressure vs. time data if a cook-off occurs. Also, the cannon should be instrumented with thermocouples at the anticipated high temperature portions of the cannon. Instrumentation should be used to determine cartridge velocity. Position a separate thermocouple to record ambient air temperature.

D.3.5.1.1 Propellant Cook-off Test.

a. Prepare the inert mortar cartridges for the test. The projectiles must be inert fuzed. All projectiles should be assembled with the maximum service charge, and with the ignition cartridge removed, as required.

b. Raise the temperature of the tube to give a chamber wall temperature equal to that determined during the tube heating trial. Heating coils may be used for this. If the cook-off test is not preceded by a rate-of-fire test, it is recommended to heat the mortar tube with a tube heater until a temperature 30°C below the designated maximum operating temperature (D-MOT) is reached. Immediately following this phase, rapidly fire as many maximum service charge rounds as is necessary to bring the tube temperature to the D-MOT.

c. Immediately following the above step, drop one of the prepared cartridges (at maximum service charge) into the tube, and allow time for the round to cook-off. Record the tube temperature at the time the cartridge is loaded.

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d. If the cartridge cooks-off, repeat step c until a temperature is established below which the cartridge will not cook-off. Record this temperature.

CAUTION: Wait at least 30 minutes before attempting to remove an unfired cartridge from the tube.

D.3.5.1.2 Warhead and Fuze Cook-off Test.

a. Instrument an inert loaded projectile (inert payload, inert fuze, no primer, no ignition cartridge, and no propellant) with two thermocouples secured to opposite sides of the inner wall of the projectile next to the bourrelet. Attach two more thermocouples inside the inert fuze; one at the inner wall and one inside the booster cup.

b. Heat the mortar tube with a tube heater until a temperature 30 °C below the D-MOT is reached. Immediately following this phase, rapidly fire as many maximum service charge rounds as necessary to bring the tube temperature to the D-MOT.

c. Insert the test round into the mortar tube and monitor and record the tube and projectile/fuze temperatures for sixty minutes or until the tube cools to 95 °C.

d. Compare the temperatures recorded to the known temperatures at which detonation of the explosive fillers in the fuze and projectile would occur to determine if either the fuze or projectile would have cooked-off.

D.3.5.1.3 Data Requirements.

a. Specifications of weapons and munition.

b. Pressures measured at each measuring point.

c. Time and temperature of cook-off events.

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APPENDIX D. ADDITIONAL TESTS AND ASSESSMENTS.
ANNEX 3. NON-SEQUENTIAL SYSTEM TESTS.

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APPENDIX D. ADDITIONAL TESTS AND ASSESSMENTS.
ANNEX 4. PRELIMINARY DESIGN TESTS AND ASSESSMENTS.

D.4.1 GENERAL.

This provides the rationale for the Preliminary Design Tests for which the test descriptions are provided herein. See Figure B1-1, Representative S3 Test Program for the description of the overall test program.

D.4.2 SAFETY OF PROPELLING CHARGE – MAXIMUM OPERATING PRESSURE (MOP) TRIAL.

The Safety of Propelling Charge - MOP Trial is aimed at providing evidence of regular and acceptable Propelling System performance across the required temperature range and address the objectives identified in Paragraph A.5.3. The trial requires propellant from three manufacturing runs (lots). This addresses the possibility of changes in performance between manufacturing runs. The trial also requires a single tube although firings may be split between two or more tubes to ascertain the effect of firing from multiple weapons. These are usually proofed tubes which have just recently been proofed to give expected maximum pressure. For conventional (non-chromed) tubes, this may be expressed as a tube with a minimum of 80% to 95% wear life remaining.

The tube is drilled at the breech face, and at several locations along its length to accommodate pressure transducers. These pressure transducers provide pressure versus time data for each location, and allow derivation of the pressure versus distance curve for each round fired.

The pressure versus time traces and the data for time to pressure maximum and time to shot exit should be examined for any irregularities. Large time delays which could be classified as a hang-fire (normally in excess of 300 ms for indirect fire and 100 ms for direct fire) or irregular times to pressure maximum or shot exit are considered unacceptable and will require further investigation and possible redesign.

Differential pressures are obtained from simultaneous time-pressure measurements taken at the breech face and just behind the base of the seated projectile. A negative differential pressure occurs whenever the chamber pressure of a cannon at the forward end of the chamber is higher than the pressure at the breech end at some time during the pressure-time history. This is a potentially hazardous condition which is indicative of an oscillating pressure wave in the chamber. In firing a round, the initial occurrence of negative-differential pressure will be observed on a pressure-time trace at a time prior to reaching peak pressure. This may be followed by swings to positive, negative, and then positive differential pressures. The peak of the initial negative differential pressure is usually the most significant because it occurs prior to the movement of the projectile. Projectile movement will permit expansion of the gases and a lessening of any negative differential pressure that may exist.

Rounds are fired conditioned at sufficient temperatures to plot the temperature versus pressure profile. As a minimum this should be UFT, LFT and a nominal ambient (usually 21 °C). This data

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can be used to calculate temperature corrections for the munition. The trial will be fired with rounds conditioned to the temperature which consistently gives the highest pressure. If no temperature consistently gives the highest pressure then the Final Propelling System Safety Test may need to be fired with rounds conditioned at UFT, LFT and nominal ambient.

The MOP trial test is specifically designed to allow Analysis of Variance (ANOVA) techniques to be used to estimate the mean and distribution for the Extreme Service Condition Pressure (ESCP). For a relatively easily calculated unbiased estimate, the data set must consist of rounds split evenly between one or more tube(s), three or more propelling charge lots and four separate occasions. Other conditions such as temperature can be introduced but this will complicate the calculations and require additional cartridges.

Alternatively, a gross mean pressure and variance, assuming a normal distribution, can be estimated. However, statistical confidence cannot be accurately calculated with this alternative method and the effects of the various conditions, particularly propellant lot variation, are harder to detect. Therefore, this alternative calculation requires considerably more rounds to achieve a reliable estimate and could still underestimate variance without well-defined manufacturing controls on lot to lot variability.

The primary function of the estimate of ESCP mean and distribution is to determine if the chance of exceeding PMP is remote and the chance of exceeding System Design Pressure (SDP) is improbable. In terms of risk to the cannon and munition system this equates to; there should be a probability of no more than 13 charge in 10,000 exceeding PMP at extreme service conditions; and there should be a probability of no more than 1 charge in 1,000,000 exceeding System DP at extreme service conditions. The pressures associated with these probabilities are termed the Maximum Operating Pressure (MOP) and Extreme Maximum Operating Pressure (EMOP). Therefore:

$$MOP \leq PMP$$

$$EMOP \leq SDP$$

For the simplistic normal distribution approach, the estimate for MOP is considered to be mean ESCP plus 3 standard deviations (which is approximately 13 in 10,000) and the estimate for EMOP is considered to be mean ESCP plus 4.75 standard deviations (which is approximately 1 in 1,000,000).

The Final propelling Charge Test requires the recorded pressures to represent the Extreme Service Conditions. The charges should be conditioned to the temperature that produces the highest pressure.

D.4.2.1 Pre-Requisite Information and Test Data Requirements.

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a. The Permissible Maximum Pressure Curve for the relevant Mortar (Mortar PMP), established by the developer, is required.

b. The results of development firings should be available, which establish that there is no significant change in cannon pressures or muzzle velocity within:

c. The weight tolerance of the mortar cartridge.

d. The filling weight tolerance of the propulsion system which includes both the ignition cartridge and propelling charges.

D.4.2.2 Mortar Cartridge, Equipment and Test Requirements.

a. Mortar Cartridges: Inert-filled mortar cartridges will normally be used and should be the same weight as the heaviest service projectile, within the weight tolerances, intended to be fired with the propellant charge system under test.

b. Fuzes: These can be live if suitable firing clearance is available. Alternatively they may be inert or plugs representing the fuze (preferred).

c. Mortar: The internal dimensions of the pressure cannon must be as near as possible to those of the service cannon, within manufacturing tolerances. The cannon must be provided with ports along its lengthwise dimension to accept pressure transducers. The number and location of the ports will depend upon the design of the cannon and its critical features. One port must be at the estimated point of maximum pressure, with at least one other port above and one below this position.

d. Propellant: Three (or more) lots of propellant, where possible representative of production, are required for testing in the form of propellant charges. These charges must be within the specified filling weight tolerance. Ideally, at least one propellant lot should be at the top of the pressure specification.

e. Charges: The maximum propelling charge shall be used for all firings.

f. Temperature: Firings are to be carried out at three temperatures having been conditioned for the durations specified within Paragraph A.1.3.3 of this document:

(1) The UFT.

(2) +21 °C.

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(3) The LFT.

g. Occasions: Firings are to be carried out on not less than four separate occasions. An occasion is defined as one continuous period of firing during which the mortar is not taken out of action, nor is there any significant change in ambient conditions. A new occasion is defined as when the mortar has been taken out of action for a period of at least two hours combined with at least one of the following circumstances:

(1) Another day.

(2) A new location.

(3) A significant change in ambient conditions.

h. Cannons: If sufficient development trials evidence has been gathered to establish that cannon-to-cannon variation can be regarded as negligible for the system, then a single cannon can be used for the firings.

i. Range and Consistency: The fall-of-shot will be recorded to verify the range and consistency for the Quadrant Elevations being used, taking account of the effect on range of a fixed pressure cannon.

D.4.2.3 Firing Plan.

a. Mortar cartridges are to be fired in accordance with the scheme set out in Table D4-1. It should be noted that:

b. Two warmer rounds of the same build standard as the cartridges under test should be fired prior to each occasion.

c. If warmers of the same build standard be unavailable, one anti-interference cartridge of the same build standard as those under test should be fired prior to each occasion, in addition to 2 warmers.

d. In certain circumstances, less than three (3) propellant lots might be fired. In this case, the mortar cartridges allocated to lots B and/or C will be allocated depending on three (3), two (2), or one (1) propellant lot is available. If more than 3 lots are used, extra mortar cartridges will be required to ensure that each firing has a minimum of three (3) mortar cartridges.

Table D4-1: MOP Trial Firing Plan

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Temperature		LFT				21 °C				UFT			
Occasions		1	2	3	4	1	2	3	4	1	2	3	4
Propellant Lot	A	3	3	3	3	3	3	3	3	3	3	3	3
	B	3	3	3	3	3	3	3	3	3	3	3	3
	C	3	3	3	3	3	3	3	3	3	3	3	3
Totals		36				36				36			
		108											

D.4.2.4. Data Requirements.

The following are required:

a. Specifications of munition and weapons fired.

b. Pressure-time recordings for each transducer.

c. Muzzle velocities.

d. Ranges and deflections of fall-of-shot.

e. Meteorological data.

f. Subjective assessment of residue, debris (including external effects) and any in-cannon burning resulting from firings.

g. Weights of projectiles as fired.

D.4.2.5 Data Reduction.

From the pressure time recordings obtained for each transducer location in the cannon and for each round fired, the maximum pressure is recorded. Statistical procedures are then applied to these values to determine, for each transducer location, that value of pressure which, it may be asserted with 95% confidence, will not be exceeded by more than 13 in 10,000 mortar cartridges under extreme service conditions, (i.e., the MOP). The statistical procedures to be applied are described in Paragraph D.4.2.7. A curve connecting the single values of MOP for each transducer location can be drawn for the whole cannon. This curve is described as the MOP curve.

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D.4.2.6 Criteria.

a. The relationship between pressures observed at each transducer location at the LFT and corresponding pressures observed at 21 °C and at the UFT must be acceptable and free from anomaly. The highest pressure observed in the cannon usually occurs at the UFT for conventional propelling charges, but this may not be the case for propelling charges consisting of moderated propellant.

b. The intervals between initiation of the propellant charges and the development of maximum pressure must be consistent and within safe limits.

c. The MOP Curve must not exceed the PMP Curve in value for any point in the mortar cannon.

d. Muzzle velocities and fall-of-shot must show good consistency and meet requirements. A statistical inspection of all relevant development rounds must show that the incidence of short ranging directly attributable to low muzzle velocities is acceptable.

e. Residue, debris and in-cannon burning resulting from firings should be at a level not likely to cause inadvertent ignition or obstruction of subsequent rounds, or any external fire hazard.

D.4.2.7 Statistical Derivation of MOP.

D.4.2.7.1 Introduction.

This section describes:

a. The initial analysis to determine homogeneity of variances among temperatures. This is necessary in order to guard against erroneous tolerance limits on MOP.

b. The general statistical model for mortar pressure that underlies the subsequent analysis.

c. The method of determining a value m such that the estimator of the variance of the pressure at a given temperature is approximately distributed as a scalar multiple of a chi-squared variate with m degrees of freedom.

D.4.2.7.2 Initial Analysis.

For each temperature level (UFT, 21 °C, LFT) a total of 6 mortar cartridges will be fired for each combination of propellant lot (3 levels) and occasions (2 levels). By temperature level, calculate

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the variance of each 6 round group and then pool the variances, thereby obtaining a common variance for each temperature level. Use the Bartlett-Neyman-Pearson test for homogeneity of 3 population variances, with 30 degrees of freedom in each sample (Reference 1 in D.4.2.7.10.). The test is to be performed at the 0.05 level of significance. If a significant result is not obtained, the analysis may continue as outlined in the statistical model below. If a significant result is obtained, then loge transform may be tried to stabilize the variance. If the loge transform still gives a significant result, then each temperature will have to be analyzed separately. These analyses could be accomplished using the procedure outlined below, after omitting the temperature factor. This would result in computing an upper tolerance limit as the critical limit. A flow chart showing in more detail the specific steps in this initial analysis is given in Figure D4-1.

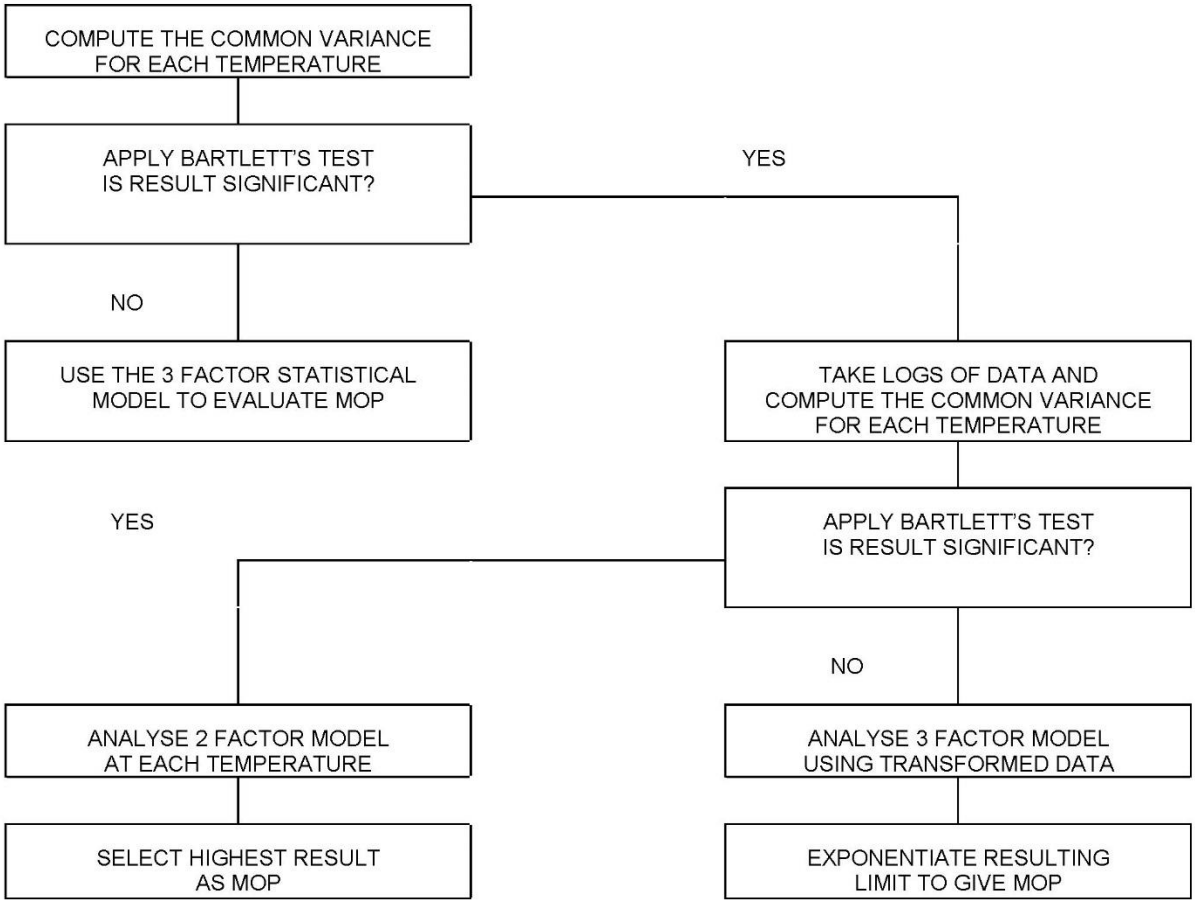


Figure D4-1: Flow Chart for Initial Analysis

D.4.2.7.3 Statistical Model.

The general statistical model for the mortar pressure test of the safety of propellant is of the form:

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$$x_{ijkl} = A + d_i + b_j + (db)_{ij} + T_k + (dT)_{ik} + (bT)_{jk} + (dbT)_{ijk} + z_{l(ijk)}$$

with $I = 1, 2, \ldots\ldots h$
$j = 1, 2, \ldots\ldots p$
$k = 1, 2, \ldots\ldots q$
$l = 1, 2, \ldots\ldots n$

where: A is a constant
d_i represents the (random) effect occasion i
b_j represents the (random) effect of 411 j
T_k represents the (fixed) effect of temperature k

and where x_{ijkl}, $zl_{(ijk)}$ represent respectively the pressure of the mortar, and the random error for trial l on occasion i with batch j at temperature k. All interactions are taken to be random effects; parentheses are used to denote interactions in the model, e.g. the interaction between batch j and temperature k is denoted by $(bT)_{jk}$.

D.4.2.7.4. Expected Mean Squares

The expected mean squares can be shown to be as follows:

Factor	Expectation of Mean Square	Degrees of Freedom
1. Occasions	$\sigma^2 + n\sigma^2_{ijk} + pn\sigma^2_{ik} + qn\sigma^2_{ij} + pqn\sigma^2_{i}$	(h-1)
2. Batches	$\sigma^2 + n\sigma^2_{ijk} + hn\sigma^2_{jk} + qn\sigma^2_{ij} + hqn\sigma^2_{j}$	(p-1)
3. Occasions.Batches	$\sigma^2 + n\sigma^2_{ijk} + qn\sigma^2_{ij}$	(h-1)(p-1)
4. Temperature	$\sigma^2 + n\sigma^2_{ijk} + hn\sigma^2_{jk} + pn\sigma^2_{ik} + hpn\sigma^2_{k}$	(q-1)
5. Occasions.Temps	$\sigma^2 + n\sigma^2_{ijk} + pn\sigma^2_{ik}$	(h-1)(q-1)
6. Batches.Temps	$\sigma^2 + n\sigma^2_{ijk} + hn\sigma^2_{jk}$	(p-1)(q-1)
7. Occasions.Batches.Temps	$\sigma^2 + n\sigma^2_{ijk}$	(h-1)(p-1)(q-1)
8. Residual Error	σ^2	hpq(n-1)

D.4.2.7.5 Estimators of Variance Components.

It follows that estimators of the variance components are:

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$$\hat{\sigma}_i^2 = \frac{MS_1 - MS_3 - MS_5 + MS_7}{pqn}$$

$$\hat{\sigma}_j^2 = \frac{MS_2 - MS_3 - MS_6 + MS_7}{hqn}$$

$$\hat{\sigma}_{ij}^2 = \frac{MS_3 - MS_7}{qn}$$

$$\hat{\sigma}_k^2 = \frac{MS_4 - MS_5 - MS_6 + MS_7}{hpn}$$

$$\hat{\sigma}_{ik}^2 = \frac{MS_5 - MS_7}{pn}$$

$$\hat{\sigma}_{jk}^2 = \frac{MS_6 - MS_7}{hn}$$

$$\hat{\sigma}_{ijk}^2 = \frac{MS_7 - MS_8}{n}$$

$$\hat{\sigma}^2 = MS_8$$

where MS_r is the mean square for the rth factor in the previous table.

D.4.2.7.6 Negative Valued Individual Estimated Variance Components.

When stating individual estimated variance components, those that turn out to be negative shall be replaced by zero.

D.4.2.7.7 Estimator of a Sum of Variance Components.

The variance of x_{ijkl} for the kth level of temperature is estimated as:

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$$V(x_{ijkl}|k) = \hat{\sigma}_i^2 + \hat{\sigma}_j^2 + \hat{\sigma}_{ij}^2 + \hat{\sigma}_{ik}^2 + \hat{\sigma}_{jk}^2 + \hat{\sigma}_{ijk}^2 + \hat{\sigma}^2$$

$$= \underset{(h-1)d.f.}{\frac{MS_1}{pqn}} + \underset{(p-1)d.f.}{\frac{MS_2}{hqn}} + \underset{(h-1)(p-1)d.f.}{\frac{MS_3}{hpqn}}(hp-h-p) + \underset{(h-1)(q-1)d.f.}{\frac{MS_5}{pqn}}(q-1)$$

$$+ \underset{(p-1)(q-1)d.f.}{\frac{MS_6}{hqn}}(q-1) + \underset{(h-1)(p-1)(q-1)d.f.}{\frac{MS_7}{hpqn}}(q-1)(hp-h-p) + \underset{hpq(n-1)d.f.}{\frac{MS_8}{n}}(n-1)$$

D.4.2.7.8 Effective Degrees of Freedom of the Sum of Variance Components.

$$\hat{m} = \frac{\hat{V}^2(x_{ijkl}|k)}{\sum_{t=1}^{8}\frac{(a_t MS_t)^2}{r_t}}$$

D.4.2.7.9 Upper Tolerance Level on Mortar Pressure.

Suppose that we require a confidence 1-α, that not more than a proportion β of rounds exceeds the upper pressure tolerance limit. Then statistical theory (References 4 and 5 in D.4.2.7.10.) shows that the upper pressure tolerance limit is:

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$$\bar{x}_{..K.} + (1.05)\lambda.s$$

where $\bar{x}_{..K.} = \max_k \bar{x}_{.k.}$ the highest of the sample mean pressures among the different temperatures;

$$\lambda = \sqrt{\frac{m}{\chi^2_{m,\alpha}}}.r$$

where $\chi^2_{m\alpha}$ is the lower 100.α % point of χ^2_m and *r* is the solution of

$$\int_{\frac{1}{\sqrt{N}}-r}^{\frac{1}{\sqrt{N}}+r} \phi(t)dt = 1-2\beta$$

where $\phi(t)$ is the standard normal density; $N=hpn$ the number of pressure readings at temperature level *K*

and $s = \sqrt{\hat{V}(x_{ijkl}/k)}$.

As *m* is unknown it is replaced by $\hat{m}$ in the formula for λ.

D.4.2.7.10 References.

1. Johnson, N L and Leone, F C (1968) Statistics and experimental design in engineering and physical sciences, Volume 1 pp 221-223. Wiley. (1968 2nd corrected printing.)

2. Idem Vol II pp 143-147 (1968 2nd corrected printing).

3. Satterthwaite, F E (1946). An approximate distribution of estimates of variance components. Biometrics, Vol 2, No 6 pp 110-114.

4. Kendall, M G and Stuart, A. The advanced theory of statistics, Vol 2 pp 128-130. (1967 2nd edition.)

5. Owen, D B (1958) Tables and factors for one-sided tolerance limits for a normal distribution. Sandia Corporation Monograph under AEC Contract AT - (29-1) - 789 (Appendix A).

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D.4.3. Verification of Propelling Charge

D.4.3.1 Preparation

Prepare weapon for measuring chamber pressure as a function of time using pressure gauges.

D.4.3.2 Test Procedure.

D.4.3.2.1 Firing at Ambient Temperature.

Using the maximum service charge, fire 10 mortar cartridges (with previously-verified charge weight) conditioned at 21°C. Following the 21°C firing, compute the mean and standard deviation of the projectile muzzle velocity, and compare it with the required muzzle velocity. Examine the pressure-time records in conjunction with the developer. If the muzzle velocity and pressure-time records meet requirements, proceed with testing in Paragraph D.4.3.2.2 below. If the pressure-time does not conform to expectations, proceed with testing only after consultation with the developer.

D.4.3.2.2 Firing at UFT and LFT.

Using the verified charge, fire ten rounds each at the UFT and LFT.

D.4.3.2.3. Data Required

a. Date and time of firing.

b. Component identification

c. Projectile weight

d. Propelling charge weight

e. Munition temperature.

f. Weapon identification and elevation.

g. Muzzle velocity.

h. Weapon chamber pressure using crusher gauges and/or electrical transducers.

i. Pressure-time traces, as applicable.

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j. Smear camera recording near muzzle (12 m).

k. Fuze functioning (and burst height where appropriate).

l. Range and deflection of impact.

D.4.3.2.4 Data Examination.

a. For each test temperature, calculate the mean and standard deviation of the peak chamber pressure and muzzle velocity.

b. Plot chamber pressure versus temperature. Compare the result with data provided by the manufacturer and previous developmental tests.

D.4.4 Charge Establishment for Projectile Safety Testing.

D.4.4.1 Preparation.

Prepare weapon for measuring chamber pressure as a function of time using pressure gauges.

D.4.4.2 Test Procedure.

a. With the munitions uniformly conditioned at 21°C, fire ten rounds using 110% of the verified service charge weight, and ten rounds at 120% of the verified service charge weight. The percentages of the verified service charge weight may be adjusted from 110% and 120% if prior test experience indicates more appropriate percentages to use. Use the average peak chamber pressures obtained during the 110% and 120% charge weight firing tests, and the average peak chamber pressures measured during the ten round, 21°C, verification of propelling charge test (Paragraph D.4.3 above) to establish a second order least squares prediction curve for chamber pressure as a function of propellant weight at 21°C.

b. The chamber-pressure/charge-weight relationship at LFT and UFT is established by constructing parallel curves using the pressure data obtained during the LFT and UFT verification of propelling charge test as reference points as see in Figure D4-2. Basic data for the lower and upper extreme temperature firings at 100% of the verified service charge are obtained from the verification of propelling charge firings. The lower and upper extreme temperature chamber pressure/propellant weight relationships are prerequisite to estimating the adjusted propelling charge weight necessary to achieve the peak safety test pressure for each temperature extreme (PTEST(UFT) and PTEST(LFT)).

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c. $P_{TEST(UFT)}$ and $P_{TEST(LFT)}$ are determined in accordance with Paragraph C.2.8. The LFT and UFT safety test charge weights are determined using $P_{TEST(UFT)}$ and $P_{TEST(LFT)}$ and the LFT and UFT curves of Figure D4-2.

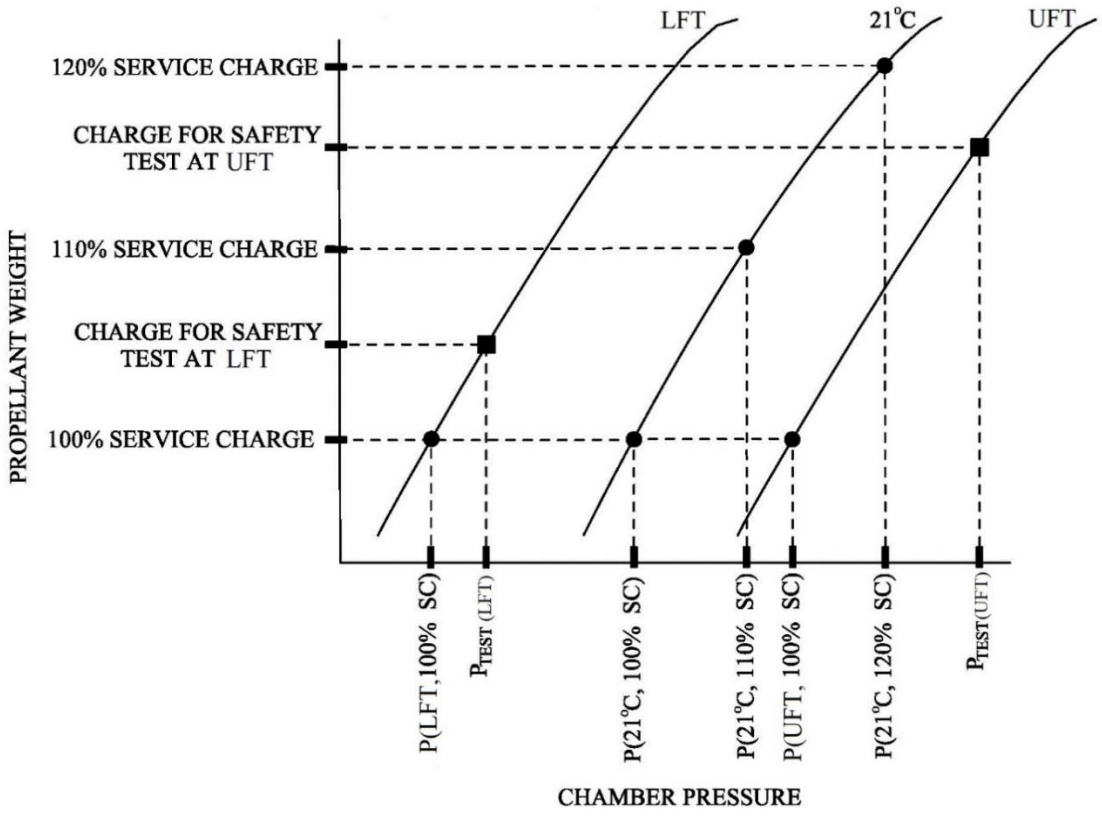


Figure D4-2: Estimation of Adjusted Propelling Charge Weight for Extreme Temperature, Weapon Firing Safety Tests

d. Once the charge weights for high and low temperature safety test pressures are calculated, conduct a limited firing test to confirm ballistic performance. Fire ten rounds conditioned at UFT, at the propellant weight determined above. The average peak chamber pressure should be within ± 2% of the desired peak chamber pressure. If the pressure is outside this tolerance, adjust the propellant weight and repeat the process. Repeat this process at the lower firing temperature to confirm the charge weight for achieving the low temperature safety pressure.

e. Use these adjusted propelling charges for all subsequent high and low temperature weapon firing phases of the safety tests.

D.4.4.3 Data Required.

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Collect data in accordance with Paragraph D.4.3.2.3.

D.4.4.4 Data Reduction.

a. Determine the mean and standard deviation of the chamber pressure of all munition groups.

b. Provide a set of three curves based on Figure D4-2, but provided with quantitative scales of pressure and propellant weights.

c. Identify the propellant weights to be used in the UFT and LFT safety tests.

D.4.5 Projectile Strength of Design Test.

D.4.5.1 Pre-Requisite Testing.

It is necessary that a projectile PMP shall have been established by the developer or a MOP curve calculated as described in Paragraph D.4.2 (Safety of Propellant - MOP Trial), when the former is not available. It is desirable that this trial should be conducted at pressures based upon PMP, but the use of pressures based on the calculated MOP value is the minimum requirement and is the level quoted in this text. The following information must therefore be available before this trial can be carried out:

a. The maximum value of PMP or MOP.

b. The calculated value of MOP at the LFT (i.e., the MOP derived from firing at the LFT only).

D.4.5.2 Trial Outline.

A sample of 30 cartridges, representative of manufacturing production standards (less explosives), is split into 2 equal sub-samples. One sub-sample is preconditioned to the UFT. It is then fired with a modified propelling charge which will produce a pressure equivalent to the maximum value of MOP plus 5% but not exceeding the munition DP. The second sub-sample is preconditioned to the LFT. It is then fired with a modified propelling charge which will produce a pressure equivalent to the LFT MOP plus 5%. Weights of mortar cartridges fired, chamber pressures, and muzzle velocities shall be recorded. Assessments are based on visual inspection and measurements of cartridges before and after firing, on high-speed photography of mortar cartridge leaving the muzzle of the mortar, and on radiographic or other techniques of crack detection applied to mortar cartridges which have been fired and recovered.

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D.4.5.3 Test Procedure.

a. Thirty (30) mortar cartridges are selected which are representative of the production standard.

b. Inert fillings (including inert exploders) and inert-filled fuzes (or suitable plugs) are normally used.

c. Fuzes are not being tested in this Strength of Design trial. However, inert fuzes (or plugs) must represent the weight of the heaviest fuze to be used in service with the mortar cartridge under test.

d. External measurements are made of the mortar cartridge body diameter at three (3) separate locations on the body: at each location the diameter is measured in two (2) directions at right angles. External measurements are also made of the length of the mortar body, and of the diameter of the tail boom at a central point. Measurement locations are to be marked so that corresponding measurements can be made after firing. Measurement locations for the mortar cartridge body diameter may be made at the bourrelet, obturator location non-inclusive of obturator, and rear end of the body, or alternatively at other locations considered critical.

e. Visual examination of each mortar cartridge is to be made for damage and for misalignment of fins. Mortar cartridges so damaged as to affect assessment of deformation after firing are to be rejected for test purposes.

f. Minimum conditioning times to be used in relation to the mortar cartridge caliber are specified within Paragraph A.1.3.3 of this document.

g. The tube used for this test shall have a minimum of 75% wear life remaining.

D.4.5.4 Sub-Sample Firings.

D.4.5.4.1 First Sub-Sample Firings.

a. The first sub-sample is to be conditioned to the UFT.

b. A modified propellant charge is fitted to the sub-sample mortar cartridges which is designed to give the mortar cartridge under test a level of pressure equivalent to the value of MOP (Paragraph D.4.2. above) plus 5% and shall not to exceed munition DP.

c. Mortar cartridges (additional to the sub-sample) should be fired in order to warm the cannon of the test mortar.

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d. The sub-sample is to be fired at the UFT for recovery.

D.4.5.4.2 Second Sub-Sample.

a. The second sub-sample is to be conditioned to the LFT.

b. A modified propelling charge is fitted to the sub-sample mortar cartridges which is designed to give the mortar under test a level of pressure equivalent to a theoretical LFT MOP plus 5%. This MOP is derived by adding three (3) standard deviations to the LFT mean value of maximum pressure (Paragraph D.4.2 above).

c. The sub-sample is to be fired at the LFT for recovery.

D.4.5.5 Records and Observations.

a. The following are required:

b. Maximum pressure of each mortar cartridge fired.

c. Muzzle velocity of each mortar cartridge fired.

d. A high-speed photographic record of each mortar cartridge as it leaves the muzzle.

e. Measurement and inspection of each mortar cartridge recovered after firing, as detailed in Paragraphs D.4.5.6 and D.4.5.7 below.

D.4.5.6 Measurements after Firing.

Measurements made before firing (Paragraph D.4.5.3.d. above) are to be repeated after firing and recovery, at the same measurement locations on each mortar cartridge as were used for measurements before firing.

D.4.5.7 Examination after Firing.

Mortar cartridge bodies and tail assemblies will be examined visually and radiographically if possible, for cracks or other evidence of failure. Photographic records of damage will be made. Damage which is clearly attributable to impact should be recorded separately.

D.4.5.8 Criteria.

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APPENDIX D. ADDITIONAL TESTS AND ASSESSMENTS.
ANNEX 4. PRELIMINARY DESIGN TESTS AND ASSESSMENTS.

Maximum pressures and muzzle velocities recorded on firings must show, that for the mortar cartridge weight fired, the required test conditions were met. Mortar cartridges will be considered to have met the requirements for strength of design between the stated UFT and LFT limits if the following criteria are met:

a. There must be no evidence of mortar cartridge break-up, cracks in metal structures, or any other failure which is attributable to launch.

b. Corresponding measurements before and after firing must not differ by more than can be attributed to instrumental error, or must be within the limits prescribed in the Technical Specification when applicable (Paragraph A.8.6 provides additional guidance).

D.4.6 PROJECTILE SAFETY IN CANNON TEST.

D.4.6.1 Test Preparation.

Make pre-fire dimensional measurements (diameters, hardness, weight) on projectiles to be fired. Compare with drawing requirements. If satisfactory, proceed with the test. The cannon used for this test shall have a minimum of 75% wear life remaining. Examine the cannon bore with a borescope and record its condition.

The number of test samples needed depends heavily on the type of test and what the test results are expected to accomplish or support. The number of rounds to be fired depends on the required reliability and confidence level. If no guidance is given, then a sample of 120 rounds shall be used.

D.4.6.2 Test Procedure.

Subject all test samples in sequence to the following environments:

a. High temperature storage test.

(1) Munition Configuration: Primary Packaging

(2) Test Level: 7-day temperature cycle using the Climatic Category A1 MIL-STD-810.

b. Low temperature storage test.

(1) Munition Configuration: Primary Packaging

(2) Test Level: 7-day temperature cycle using the Climatic Category C3 of MIL-STD-810.

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c. Level 1 Inspection.

d. Loose Cargo Test.

(1) Munition Configuration: Primary Packaging

(2) Test Level: Default values stated in MIL-STD-810, Method 514: 300 rpm circular synchronous motion for 20 minutes.

e. Level 2 inspection.

f. Packaged Drop test.

(1) Munition Configuration: Primary packaging

(2) Test Level: One drop in accordance with MIL-STD-810, Method 516 for each round at a height of 1.5 m into a concrete supported steel surface.

(3) Drop orientation: 50% of the test quantity shall be dropped in the major axis horizontal orientation and 50% of the test quantity shall be dropped in the nose down/bottom up orientation.

(4) Test Temperature: Ambient temperature.

g. Level 2 inspection.

h. Bare drop.

If using mature and qualified fuze, this test may be omitted.

i. Test Firing

50% of the test quantity is conditioned at the UFT and fired at the hot temperature MOP derived from the MOP trial.

50% of the test quantity is conditioned at the LFT and fired at the cold temperature MOP derived from the MOP trial.

D.4.6.3. Data Required

a. Date and time of firing.

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b. Component identification

c. Projectile weight

d. Propelling charge weight

e. Munition temperature.

f. Weapon identification and elevation.

g. Muzzle velocity.

h. Weapon chamber pressure using crusher gauges and/or electrical transducers.

i. Pressure-time traces, as applicable.

j. Smear camera recording near muzzle (12 m).

k. Fuze functioning (and burst height where appropriate).

l. Range and deflection or impact.

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APPENDIX D. ADDITIONAL TESTS AND ASSESSMENTS.
ANNEX 4. PRELIMINARY DESIGN TESTS AND ASSESSMENTS.

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APPENDIX E. BREAKDOWN TEST AND CRITICAL ANALYSIS (BCTA).
ANNEX 1. GENERAL INSPECTION.

This document is primarily written with references to test procedures. Table I3-1 (Appendix I, Annex 3) provides a cross reference table of similar national and international test standards. The following BTCA procedures must only be conducted by suitably qualified and experienced personnel.

E.1 GENERAL INSPECTION.

Prior to disassembly for BTCA, conduct a thorough review of Level 1 (basic visual) and Level 2 (radiography) inspection results and non-functioning test results obtained throughout the sequential environmental trial. Any anomaly should be carefully considered with regard to the safety of the munition disassembly and BTCA processes.

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APPENDIX E. BREAKDOWN TEST AND CRITICAL ANALYSIS (BCTA).
ANNEX 1. GENERAL INSPECTION.

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APPENDIX E. BREAKDOWN TEST AND CRITICAL ANALYSIS (BCTA).
ANNEX 2. BREAKDOWN AND ANALYSIS REQUIREMENT.

E.2. BREAKDOWN AND ANALYSIS REQUIREMENT.

E.2.1. APPLICABILITY.

The following tests are broadly applicable to the mortar cartridge primary components including the cartridge body and propulsion components (ignition cartridge and propelling charges).

E.2.2. REQUIREMENT CONSIDERATIONS.

The exact requirements for BTCA need to be determined on a case by-case basis taking into consideration the degree of novelty and/or complexity of the munition. They will be determined by known failure modes and life limiting factors for comparable munitions.

E.2.3. BASELINE TEST CONSIDERATIONS.

Prior to commencement of all trials, at least one (1) munition from the same batch/lot as those undergoing the sequential environmental trial should be disassembled and analyzed to identify potential failure modes that may occur. This sets the baseline for comparison against the environmentally stressed munition. There should also be baseline munitions for the functioning (dynamic and static firing) tests. It may also be possible to use the results from material Qualification tests (to AOP-7 and NAVSEAINST 8020.5C) for baseline purposes, or data from material manufacturers batch/lot acceptance tests provided these give data equivalent to that from the Qualification tests. Furthermore, firing data from development trials may be used for baseline purposes provided the munition is of the same build standard as the test munition and provides the required data. However, it should be noted that none of these latter options will permit comparison against the physical condition of the munitions following the sequential environmental trial.

E.2.4. TEST CONSISTENCY CONSIDERATIONS.

It is essential to ensure that the same test procedures used to determine the baseline properties of materials are used during BTCA.

E.2.5. CONTAMINATION CONSIDERATIONS.

During disassembly and material extraction, care must be taken to ensure that the extracted samples do not become contaminated (by structural materials or other matter) and/or physically damaged/changed (e.g., compressed, cracked, abraded).

E.2.6. FUNCTIONAL TEST CONSIDERATIONS.

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APPENDIX E. BREAKDOWN TEST AND CRITICAL ANALYSIS (BCTA).
ANNEX 2. BREAKDOWN AND ANALYSIS REQUIREMENT

Small items such as primers, booster charges, etc. pose particular difficulties during disassembly, and it may not be possible to extract sufficient material without damaging the material contained within. In such cases, it is acceptable to perform just visual and radiographic inspection followed by functioning tests (at extremes of service temperature). This should include electrical resistance checks and tests performed during lot acceptance such as performance tests during functioning. In some cases, it may be possible to extract sufficient material to perform small scale tests such as volatile content determination or differential scanning calorimetry (DSC).

E.2.7. BTCA TEST REQUIREMENTS.

The aspects below are provided as an indication of the types of testing required.

E.2.7.1. Inspection and Disassembly.

a. Prior to the disassembly, perform physical integrity checks on the munition, sub-systems, energetic materials and structural materials. This can be achieved through visual examination (including photography as required), radiography, Computerized Tomography Scan, Dye Penetrant, Bore-scope (for base bleed conduits and other cavities), Ultrasonic inspection and/or Fluoroscopy both prior to, and following disassembly. Some techniques may be more applicable to structural materials which must also be assessed. Dimensional checks should assess physical dimensions and mass of the complete munition, sub-systems and energetic materials to demonstrate compliance with specifications/drawings.

b. During disassembly and material extraction, care must be taken to ensure that the extracted samples do not become contaminated (by structural materials or other matter) and/or physically damaged/changed (e.g., compressed, cracked, abraded, etc.). Furthermore, caution must be exercised when unscrewing threaded joints where explosive crystals may be present as a result of environmental exposure, leading to the potential for their initiation from pinching and/or friction. Processing (unpacking and disassembly) should only be undertaken in well ventilated areas using appropriate personal protective equipment (respirators, etc.). Phosphine is likely to be present in the packaging of smoke munitions containing Red Phosphorous, which in quantity exceeding human exposure limits, can be lethal. Pyrotechnic compositions containing metal fuels (e.g., Magnesium) are prone to oxidation resulting in the production of Hydrogen gas, which if present in sufficient quantity, is explosive. Therefore, relevant safety procedures must be implemented when processing munitions containing these compositions.

c. During disassembly, pay particular attention to signs of cracking, surface crystallization/dusting, exudation, corrosion, discoloration, wear, missing components, dislodged components, and other damage (e.g., splits in combustible cases).

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d. During disassembly, examine specifically: plastics, rubbers, foams, seals, etc., for signs of degradation or uptake of plasticizer. 'O' rings should be examined for compression set and that they still meet their specification requirements.

E.2.7.2. Chemical Tests.

a. Chemical composition, including total volatile matter and moisture content, must be assessed to demonstrate compliance with specifications/drawings.

b. Chemical stability must be assessed for all energetic materials, although the tests used will be material dependent. The vacuum stability test is particularly applicable for main charge explosives. Chemical stabilizer depletion testing (refer to AOP-48) is applicable for nitrate-ester propellants, with a preference for multi-temperature aging since this gives both stabilizer content and chemical kinetics.

c. Pyrotechnic compositions having a metal fuel are prone to oxidation (e.g., illuminating candle compositions). This is brought about by the reaction of the metal fuel with moisture which results in the production of hydrogen. To assess levels of the fuel degradation it is recommended that the free (after oxidation) and total (oxidized and un-oxidized) metal contents of compositions should be determined using suitable analytical techniques.

E.2.7.3. COMPATIBILITY TESTS.

a. Chemical/explosive compatibility between all components of construction with the explosives they will be in communication with (both in physical contact and by gas/vapor path) should have been assessed during material qualification and/or design of the munition. This compatibility data shall be presented as a matrix that lists the materials, and for each explosive declares whether there is communication or not with evidence to support the claim of compatibility where communication is expected.

b. During BTCA, any material incompatibilities and/or migration of explosive species are likely to become evident during inspection. Any such anomalies observed shall be noted and assessed further to address whether the munition remains safe as defined in JOTP-001. An example is the migration of species from one (1) energetic material to another (e.g., burn-rate modifiers from one (1) propellant to another).

E.2.7.4. Physical Properties – Explosives.

a. Assess flow properties and particle size distribution for granular materials (such as granular propellants and some pyrotechnic compositions), checking for coagulation of

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granular materials, 'slump' (particularly in propellants), bulk cracking, and surface cracking/crazing.

b. Thermal analysis methods, especially Differential Scanning Calorimetry, are useful tools that may indicate changes in the material over time and are particularly suited to subsequent comparison during In-Service Surveillance. They are applicable to most energetic materials, especially pyrotechnics, since they can be performed on small samples of material.

E.2.7.5. Mechanical Properties.

Mechanical properties (such as tensile/compressive/ shear strength and hardness) of explosive materials must be assessed at the full range of working temperatures for the munition. It will also be necessary to test structural materials at temperature extremes for safety critical items, such as projectile bodies and tail boom, in order to verify design safety margins. Typical methods will include uniaxial tensile test to STANAG 4506 and Dynamic Mechanical Thermal Analysis (DMTA) to STANAG 4540. It may also be necessary to assess fatigue crack growth for some structural materials. The types of testing will ultimately be determined by the type of material being tested.

E.2.7.6. Hazard Properties.

a. Repetition of the small scale tests to assess hazard properties must be undertaken. These may include, but are not limited to, methods to determine ease of initiation by impact, friction and electrical spark, along with temperature of ignition. Explosive material testing and assessment should be conducted in accordance with AOP-7 and NAVSEAINST 8020.5C.

b. Normally the small scale tests will be sufficient but larger scale tests may also be required if an issue is identified. The exact methods used would depend upon the type and quantity of material available for the tests but may include 'gap tests' and tests to assess Velocity of Detonation. However, they may ultimately require full scale (i.e., complete round) tests to assess the IM properties of the munition following environmental exposure.

E.2.7.7. Electrical Components.

a. When the munition contains electrical sub-assemblies (e.g., electronic safe/arm device, weapon controller, seeker) these should be removed during BTCA for inspection and functional checks. Functional checks should be performed initially on the initial sub-assembly, using the factory test specification. Where this is not possible or does not

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allow full testing, then the sub-assembly may require further disassembly to permit such testing.

b. Following this, full disassembly should be conducted for detailed component level inspection. Specific points to observe are broken/loose joints (connectors and solder), damaged/broken components, damaged/broken circuit board tracks, abraded/broken cables/wiring, corrosion, dendritic growth (e.g., 'tin whiskers'), condition of 'potting' compound (if present), and burst batteries.

c. Electrical resistance of igniters/ Electrically Initiated Devices (EIDs)/ Electro-Explosive Devices (EEDs) should be checked along with EIDs/EEDs functioning, using a normal firing pulse.

E.2.7.8. Fuze (Mechanical) Components.

a. When the munition contains a mechanical fuze this should be removed during BTCA for inspection where possible.

b. If there is any doubt regarding the safe and reliable function of the fuze, or it cannot be demonstrated by alternative means, it may be necessary to carry out tests that simulate the various external stimuli required to arm the fuze (e.g., acceleration, spin).

c. The fuze (either armed or safe) should be disassembled to determine its internal physical condition and verify its safe condition.

E.2.7.9. Tests – Other.

E.2.7.9.1. Primer Sensitivity.

Mortar primers must be sensitive enough to function reliably in the cannons in which they are used; yet they must also be sufficiently insensitive to allow safe handling and loading. The majority of mortar munitions use a percussion primers that function upon physical impact, although some may use electrical primers that function when the correct voltage and current are applied. This test determines, and/or confirms, the sensitivity of the primer to the correct stimulus.

E.2.7.9.2. Percussion Primers.

There is no single specific test procedure that covers all percussion primers, although the test equipment and methods required, along with the "go/no go" threshold are usually stated in the munition and/or primer specifications. The procedure and equipment used should be reported along with the results and observations.

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APPENDIX E. BREAKDOWN TEST AND CRITICAL ANALYSIS (BCTA).
ANNEX 2. BREAKDOWN AND ANALYSIS REQUIREMENT

E.2.7.9.3. Electrical Primers.

Each type of electrical primer is designed to function by specific conditions of voltage and current. There is no single specific test procedure that covers all electrical primers, although the test equipment and methods required, along with the "go/no go" threshold are usually stated in the munition and/or primer specifications. The procedure and equipment used should be reported along with the results and observations. Electrical resistance should also be measured and recorded using a low current circuit tester/squib tester.

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APPENDIX F. FACILITIES AND INSTRUMENTATION REQUIREMENTS.
ANNEX 1. GENERAL

This document was written with primarily references to NATO test procedures. Table I3-1 (Appendix I, Annex 3) provides cross reference of similar national and international test standards.

Table F-1: Facility Requirements

ITEM	REQUIREMENT
Inspection and Non-Destructive Test Facility	Material inspection equipment such as ultrasonic and radiographic must be available to determine the condition of the munition and its components before and after exposure to environmental tests. Facility should have the capability to conduct radiographic inspection of munitions at low temperature extremes or within 15 minutes of removal from a conditioning chamber.
Climatic Test Facility	• Climatic chamber equipment capable of temperature conditioning live munitions to the extremes of -55 °C to 75 °C and relative humidity from 5% to 95%. • High temperature chamber equipped with solar lamps capable of at least 1,120 W/m^2 output. • Altitude chamber capable of testing live munitions to at least 56.8 kPa (4,570 m equivalent altitude). • Equipment capable of conducting Sand and Dust, Salt Fog, Rain, and Immersion tests on live munitions.
Rapid Decompression Test Facility	Chamber capable of pressure change from 60 kPa to 18.8 kPa within 15 seconds. Must be suitable for packaged, live munitions.
Dynamic Test Facility	Equipment suitable for simulating the full range of dynamic environments (e.g., transportation shock and vibration, loose cargo, tactical shock and vibration, drop test) expected during the munition's lifetime. Facility should have the capability to conduct shock and vibration tests at temperature extremes and drop tests within 15 minutes of removal from a conditioning chamber.
Firing Range	Selected to suit munition test requirements and to provide adequate protection for personnel and equipment. Facility should have the capability to conduct firing tests at temperature extremes or within 30 minutes of removal from a conditioning chamber.
Warhead Test Area	Test area must have an adequate surface safety danger zone, including overhead air space for open field testing.
Munition Disassembly	Facility suitable for disassembly of live munitions for detailed inspection and component level testing.

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APPENDIX F. FACILITIES AND INSTRUMENTATION REQUIREMENTS.
ANNEX 1. GENERAL

Table F-1: Facility Requirements (Cont.)

ITEM	REQUIREMENT
Energetic Material Extraction	Equipment suitable for the extraction of energetic material samples for chemical analysis.
Chemistry Laboratory	Equipment suitable for the conduct of the chemical analysis tests set out in AOP-7, NAVSEAINST 8020.5C and Paragraph E.1.4.1 of Appendix E (Level 3 Inspection).
Electromagnetic Radiation Test Facility	Facility suitable for the generation of the specified field intensities with an adequate test volume for the test of the munition and launcher as required by the stockpile to launch configuration.
Electrostatic Discharge Test Facility	Facility suitable for the generation of the required ESD environments and large enough for the munition and launcher as required by the stockpile to launch configuration.
Lightning Test Facility	Facility capable of conducting the required lightning strike test on live munitions.
Data Collection/Processing Facility	Test data shall be recorded on Digital Recorders for post-test processing. The data processing system shall edit, display, and print out the desired data plot for analysis and reporting purposes.
Video/Photographic	Closed circuit video is required for personnel safety to permit observation of munition tests. Video Camera/Recording Systems having a sufficient frame rate to record and playback desired events. High speed digital cameras and/or UV/IR cameras may also be required.

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APPENDIX F. FACILITIES AND INSTRUMENTATION REQUIREMENTS.
ANNEX 1. GENERAL

Table F-2: Measurement Tolerances

DEVICES FOR MEASURING	MEASUREMENT TOLERANCE
Pressure	± 5 percent of the value or ± 200 Pa, whichever is greater.
Strain	± 1 percent of highest expected value
Heat Flux	± 1 percent of highest expected value
Resistance (Low Current Circuit Tester/ Squib Tester)	± 0.05 ohms
Firing Pulse (Automatic Fire Control System)	As required for the firing and automatic sequencing of the data collection systems.
Motor Ignition Events (Video)	Frame rate sufficient to record desired event.
Time	± 1 percent
Temperature Climatic Temperature Measurements Firing Temperature Measurements	 ± 2 °C ± 5 °C
Relative Humidity	± 5 percent
Solar Radiation	MIL-STD-810, Method 505
Vibration and (Acceleration)	MIL-STD-810, Method 514 and (513)
Acoustic Sound Pressure Level	MIL-STD-1474
Mechanical Shock	MIL-STD-810, Method 516
Toxic Gas (NO, NO_2, NO_x, CO, CO_2, SO_2)	2 percent of full scale
Particulates (0.5-15 microns)	2 percent of full scale
Pyrolysis products (fluoride, chloride, bromide, cyanide, aldehydes)	2 percent of full scale
Length	± 1 percent
Weight	± 1 percent
Meteorological Conditions Temperature Relative Humidity Barometric Pressure UV Radiation Potential Lightning/Severe Weather Wind	 ± 2 °C ± 3 percent ± 0.25 mm of Hg ± 20 W/m^2 > 2 km ± 3 km/hr

Note: Copper Crusher gauges, in accordance with Allied Engineering Publication (AEP)-23, are sometimes used to measure firing chamber pressure or verify performance of piezo electric pressure gauges. These gauges should not be attached to the test munitions until ready to be fired.

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APPENDIX F. FACILITIES AND INSTRUMENTATION REQUIREMENTS.
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APPENDIX G. RESERVED FOR FUTURE USE
ANNEX 1. GENERAL

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APPENDIX G. RESERVED FOR FUTURE USE
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APPENDIX H. HEALTH HAZARDS.
ANNEX 1. ADDITIONAL NON-SEQUENTIAL TESTS/ASSESSMENTS.

This appendix provides descriptions of all of the non-sequential required and optional tests in Appendix B. Rationales for these tests are provided in Appendix A-7.

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APPENDIX H. HEALTH HAZARDS.
ANNEX 1. ADDITIONAL NON-SEQUENTIAL TESTS/ASSESSMENTS.

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APPENDIX H. HEALTH HAZARDS.
ANNEX 2. ELECTROMAGNETIC ENVIRONMENTAL EFFECTS (E3)

H.2 GENERAL.

Conduct an evaluation of the E3 conditions affecting the munition in accordance with MIL-STD-464 and MIL-HDBK-240 and perform E3 testing in accordance with TOP 01-2-511. Munitions shall be packaged and unpackaged as required.

H.2.1 HAZARDS OF ELECTROMAGNETIC RADIATION TO ORDNANCE (HERO).

Conduct the HERO test using guidance in MIL-STD-464, TOP 01-2-511, MIL-HDBK-240, JOTP-061, OP3565 Vol. 2, and the parameters found in MIL-STD-464 MIL-for all LCEP configurations. HERO tests are performed using one complete inert cartridge with instrumented inert or live Electrically Initiated Devices (EIDs) and/or ESADs. The HERO tests generally use an electric measuring chain (instrumented EIDs) that will collect measured induced current data. The explosively loaded EIDs are replaced with fiber optic instrumented versions of the inert EID. In cases where instrumentation of the device is not feasible, reasonable results can be obtained with a go/no-go technique but a considerably higher number of units and a theoretical analysis will be required.

H.2.2 ELECTROSTATIC DISCHARGE (ESD) TESTS.

H.2.2.1 Personnel Handling.

a. Personnel handling ESD tests are performed using inert munitions which contains inert or live EIDs/ESAD's. A minimum of 22 complete sets of EIDs/ESADs are required (see Appendix B).

b. Conduct personnel handling ESD tests using guidance in JOTP-062, MIL-HDBK-240 and MIL-STD-464. The discharge is applied to all connectors (protective covers removed) and electronics accessible during system checks and/or field assembly. ESAD's shall be tested while in the functional mode.

c. Inspect and test all EIDs/ESADs for activation or duding.

H.2.2.2 Helicopter-Borne Transportation.

a. Helicopter-borne transportation ESD tests are performed using inert munitions, which contain inert or live EIDs/ESADs. A minimum of 10 complete sets of EIDs/ESADs are required (see Appendix B).

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b. Conduct helicopter-borne transportation ESD tests using JOTP-062, MIL-HDBK-240 and MIL-STD-464.

c. Inspect and test all EIDs/ESADs for activation or duding.

H.2.3 LIGHTNING HAZARD.

a. The tests are performed with the weapon in the worst case configuration based on analysis of the LCEP scenario.

b. Direct or indirect (or both where appropriate) lightning tests shall be performed using inert weapons with instrumented inert or live EIDs/ESADs. A minimum of 20 complete sets of EIDs/ESADs (10 for indirect lightning strike and 10 for direct lightning strike) are required to provide adequate data when instrumented components are not available (see Appendix B Annex 7). In addition, Nation specific requirements may necessitate direct and/or indirect lightning tests on one complete live cartridge.

c. Perform the lightning strike tests using the parameters found in MIL-STD-464.

H.2.4 ORDNANCE ELECTROMAGNETIC VULNERABILITY (EMV).

a. Evaluate and test in accordance with MIL-HDBK-240 and AECTP 508 Leaflet 1.

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APPENDIX H. HEALTH HAZARDS.
ANNEX 2. ELECTROMAGNETIC ENVIRONMENTAL EFFECTS (E3)

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H.3 GENERAL.

Health hazard data is to be collected during all required safety tests. The tests to assess toxic fumes, and blast overpressure (BOP) and noise hazards are described below.

H.3.1 TOXIC FUMES TEST.

Perform the test to measure the concentrations of toxic gases produced during firing to determine if hazardous substances will potentially build up at the crew and, if applicable, at the vehicle occupant positions during firing. Also, perform the test to determine if the mortar crew can fire mortar rounds at the required maximum and sustained firing rate of the mortar weapon without exposing the crew to an unacceptable level of hazardous substances. Refer to MIL-STD-1472, TOP 01-1-015, TOP 01-2-610 and MIL-HDBK-46855A for test guidelines. Follow national guidelines to determine the maximum allowable toxic gas concentrations. (e.g., US standards are delineated by MIL-STD-1472, Paragraph 5.13.7.4).

Note: The weapon's physical rate-of-fire capability will be ascertained in a separate rate-of-fire test to be conducted prior to this test.

H.3.1.1 Test Procedure.

The following test procedure is provided as guidance.

a. Munition Firing: The test will be conducted by requiring the gun crew to fire the main mortar at the required rate-of-fire to evaluate the toxic gas concentrations during mortar firings and variation of platform operating conditions if applicable (e.g., hatches open/closed, ventilators and heaters on/off). Data collection will be conducted in accordance with ITOP 05-2-502 and TOP 02-2-614.

b. Instrumentation: The test requires instrumentation to measure concentrations of basically six gaseous compounds. These include: ammonia (NH3), carbon monoxide (CO), carbon dioxide (CO2), sulfur dioxide (SO2), oxides of nitrogen (NO and NO2) and hydrogen cyanide (HCN). Infrared (IR), ultraviolet (UV), chemiluminescent, and pulsed fluorescence analyzers will be used to continuously monitor the concentrations of toxic gases produced at the crew/ occupant positions in the test vehicles. Air sampling tubes should be located in the breathing zones of the crew and vehicle passenger members, if applicable. All positions will be continuously analyzed for toxic gas concentrations. The CO and NH3 samples should be analyzed on nondispersive infrared analyzers. The NO samples should be analyzed on chemiluminescent

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analyzers. The ambient Relative Humidity (RH) and temperature data as well as current wind conditions during the time of test shall be measured.

Table H3-1 contains the recommended test instrumentation specifications for the specific gases measured.

Table H3-1: Test Instrumentation Specifications

Gas	Range	Accuracy
NH_3	0 to 300 ppm	6 ppm
CO	0 to 1,000 ppm	20 ppm
CO	0 to 2,500 ppm	50 ppm
CO	0 to 10,000 ppm	200 ppm
CO_2	0 to 5%	0.1%
SO_2	0 to 25 ppm	0.25 ppm
NO_2	0 to 200 ppm	4 ppm
NO	0 to 100 ppm	1 ppm
HCN	TBD	TBD

c. Crew Positions: Crew positions should be downwind of the muzzle, to the maximum extent feasible.

d. Munition Conditioning: Inert mortar cartridges, conditioned to 21 °C ± 3 °C for 24 hours minimum, will be fired.

e. Weapon Elevation: The weapon elevation shall be at minimum standard firing elevation.

f. Phase 1 Firings: Phase 1 firings should be conducted as indicated in Table H3-2. Also, the platform configuration, if applicable, should be as indicated in Table H3-2.

g. Toxic Gas Measurement: Data acquisition will begin 5 seconds (minimum) prior to the beginning of the firing scenario. During the firings and immediately following the 16th round, the toxic fume levels should be measured. The crew should evacuate the weapon platform during this measurement.

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Table H3-2: Phase 1 Firing Matrix

Scenario	Charge	Weapon Azimuth (mils)	Weapon Elevation (mils)	Number of Rounds	Rate -of- fire	Platform Configuration (if applicable)			
						A/C	Heaters	Hatches	Ventilators
1	0	0	800	16	Max	off	off	closed	off
2	1	0	800	16	Max	off	off	closed	off
3	2	0	800	16	Max	off	off	closed	off
4	4	0	800	16	Max	off	off	closed	off

h. Respirators: The operating crew should be equipped with respirators supplied with bottled air or other type of appropriate respirator for protection from overexposure in the event the concentrations of toxic gases exceed the allowable limits during testing.

i. Testing Restriction: Testing should be avoided when the ambient RH exceeds 85 percent or the average wind speed exceeds 16 km/hr or wind gusts exceed 32 km/hr.

j. Phase 2 Firings: If applicable for mortar platform configuration, phase 2 of the test will determine the effect of the heaters, air conditioner, hatches, and ventilation on the concentration of toxic fumes. Firings will be conducted and platform configuration will be as indicated in Table H3-3. This phase will most likely require tailoring of the matrix provided. Follow Paragraph H.3.1.1, steps c through i above for test procedures, data acquisition and restrictions.

Table H3-3: Phase 2 Firing Matrix

Scenario	Charge	Weapon Azimuth (mils)	Weapon Elevation (mils)	Number of Rounds	Rate -of- fire	Platform Configuration (if applicable)			
						A/C	Heaters	Hatches	Ventilators
5	WC	0	800	16	Max	off	off	closed	on
6	WC	0	800	16	Max	off	off	open	on
7	WC	0	800	16	Max	off	off	open	off
8	WC	0	800	16	Max	off	off	closed	off
9	WC	0	800	16	Max	off	on	open	off
10	WC	0	800	16	Max	off	on	closed	off

NOTE: WC - Worst Case charge that produces the highest concentration of toxic fumes from previous phase.

k. Phase 3 Firings: If deemed necessary by the material developer and evaluator, phase 3 of the test will determine the effects of weapon orientation on the concentration of toxic fumes. Firings will be conducted and platform configuration will be as indicated in

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Table H3-4. This phase, if conducted, will most likely require tailoring of the matrix provided. Follow Paragraph H.3.1.1, steps c through i above for test procedures, data acquisition and restrictions.

Table H3-4: Phase 3 Firing Matrix

Scenario	Charge	Weapon Azimuth (mils)	Weapon Elevation (mils)	Number of Rounds	Rate -of- fire	Platform Configuration (if applicable)			
						A/C	Heaters	Hatches	Ventilators
11	WC	1,600	800	16	Max	WC	WC	WC	WC
12	WC	3,200	800	16	Max	WC	WC	WC	WC
13	WC	4,800	800	16	Max	WC	WC	WC	WC
14	WC	0	1,167	16	Max	WC	WC	WC	WC
15	WC	0	Max	16	Max	WC	WC	WC	WC

NOTE: WC - Worst Case charge that produces the highest concentration of toxic fumes from previous phase.

H.3.1.2 Data Required.

a. Weapon type and serial number and platform identification (if applicable).

b. Platform orientation.

c. Ambient conditions (RH and temperature).

d. Wind speed and direction, mph.

e. Platform configurations (see Platform Configuration in the tables above).

f. Munition type(s) and lot number(s).

g. Number of rounds fired, firing rate, and interval.

h. Charge fired.

i. Effects of blast, smoke, and visibility on members of firing crew.

j. Concentration (ppm) versus time data for each gas for all crew member positions.

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k. The predicted percent carboxyhemoglobin (COHb) level for each crew member/occupant position sampled.

H.3.1.3 Data Analysis.

a. The concentration versus time data for each gas will be analyzed. The peak and time-weighted average (TWA) concentrations and doses (product of concentration and time) will be determined for each gas at each crew position. The data will be summarized and presented in tabular form by trial number and crew position. Tables H.3-2, H.3-3 and H.3-4 include the vehicle operation conditions, number of rounds of specific munition fired, and charge. Concentration versus time plots will be prepared for all gases present in significant concentrations.

b. For each trial, the blood carboxyhemoglobin (COHb) levels will be estimated for each crew member in accordance with MIL-STD-1472. An estimate of the maximum number of allowable consecutive exposures (MACE) will be determined by iteration: the COHb concentration at the completion of the first exposure will be the initial COHb concentration for the second exposure, etc., until the final COHb value exceeds 10 percent. MACE is one less than the number of exposures required to reach 10 percent. A work stress level of 4 will be used.

H.3.2 IMPULSE NOISE AND BLAST OVERPRESSURE TEST.

Perform the test to determine the impulse noise and muzzle blast overpressure (BOP) levels of the mortar munition, determine impulse noise and BOP levels at the mortar crew locations within the host vehicle platform for mounted mortar configurations, and at mortar crew locations for dismounted/ground mortar configurations. Refer to MIL-STD-1474 and ISO 10843 for test guidelines. In addition, TR-HFM-090-ANN-H contains a compilation of different blast overpressure methodologies and analysis. In particular, the Stuhmiller and the Blast Overpressure - Health Hazards Assessment (BOP-HHA) V2 models are currently used by many organizations for occupational exposure limits. The test data shall be provided to the appropriate Health Assessment Office or the appropriate health assessment organization for determination of impulse noise and BOP-related firing restrictions. Follow national guidelines for limits of impulse noise and BOP with no hearing protection, single or double hearing protection (e.g., US standards are delineated by MIL-STD-1474).

H.3.2.1 Test Procedure.

The following test procedure is provided as guidance.

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a. Weapon Mounting: Testing should be conducted with the mortar weapon mounted in the host vehicle platform or seated in the ground. If the mortar has both a vehicle mount and ground seated configuration, testing in both configurations may be necessary to determine the worst case condition for impulse noise and BOP.

b. Weapon position: The vehicle or ground firing position should be in a flat area with no extraneous reflecting surfaces (except those required to conduct testing), within 10 m of both the test item and the transducers, sound level meters, and blast test devices (BTDs).

c. Transducer Placement – 140 dB Pressure Contour: An array of transducers or sound level meters will be placed around the vehicle or seated mortar, including behind the vehicle and to the sides, to determine the 140 dB pressure contour. The transducers or sound level meters shall be oriented to receive grazing incidence sound in accordance with MIL-STD-1474.

d. Munition Firing Pressure. During firing, peak chamber pressures will be measured using pressure transducers.

e. Metrological Conditions. Ambient temperature, barometric pressure, and surface-wind velocity and direction will be measured. This test will not be conducted when wind speeds exceed 20 km/hr or during precipitation.

f. Platform Configuration. For platform mounted testing, hatches and ramps shall be open or closed in accordance with the test plan and operational guidance for the platform.

g. Blast Test Device (BTD) Placement. Appropriate BTDs shall be positioned at the crew positions and appropriate transducers or sound level meters shall be positioned at the crew head positions. The BTDs should be aligned to face the muzzle of the weapon. The BTDs and head position transducers or sound level meters should rotate along with the weapon as crew positions change with firing azimuth. The transducers or sound level meters shall be oriented to receive grazing incidence sound in accordance with MIL-STD-1474.

h. Transducer Placement – Crew Positions. For ground seated weapon testing, an array of transducers or sound level meters shall be placed rear of the gun position. For 120 mm mortars, recommended placement is along 180 degrees, 150 degrees, and 120 degrees radials rear of the gun position, at 1.0 m and 2.0 m distance from the gun muzzle and

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0.8 m and 1.2 m height above ground. For 81 mm and 60 mm mortars, recommended placement is along 180 degrees, 150 degrees, and 120 degrees radials rear of the gun position, at 0.5 m and 1.0 m distance from the gun muzzle and 0.7 m height above ground. The transducers or sound level meters shall be oriented to receive grazing incidence sound in accordance with MIL-STD-1474. Transducer placement should be discussed with the Health Assessment Office or the appropriate health assessment organization.

i. Munition. Normally, only inert cartridges will be fired to assess the BOP and Noise characteristics of a family of cartridges. If there are weight, propulsion or body profile differences between the High Explosive, Smoke, Illumination or inert (training) cartridges, consideration must be given to including the tactical cartridges in the test plan.

j. Test Firings. A three or four step approach is recommended to determine the worst case munition charge and temperature and weapon quadrant elevation (QE), azimuth (for vehicle-mounted weapon where azimuth adjustment is allowed), and vehicle configuration (hatches and ramps opened and closed).

k. Firing Groups. Normally for each specific munition, weapon, and vehicle condition, a five round group of mortar firing is conducted. Additional rounds may be required if the number of rounds in a group must exceed the maximum dB spread per MIL-STD-1474.

l. Phase 1 Firings. Determine worst case munition temperature and charge for impulse noise and BOP.

Recommended test will consist of firing 5 round groups at maximum, minimum and ambient (+21 °C) temperatures and top charge, top charge less one and possibly top charge less two. The weapon will be set to a mid QE position, normally 1,156 mils. The vehicle configuration for ramp and hatches opened or closed will be the predicted worst case for impulse noise or BOP. This test will establish the worst case munition temperature and charge to be used for the remainder of the test. The estimated number of rounds is 5 (round groups) x 3 (temperature conditions) x 3 (charges) = 45 rounds. This test may be repeated to determine to effect of the addition or deletion of a weapon component, such as a Blast Attenuation Device (BAD) which may have significant effect on selection of munition worst case temperature and charge.

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m. Phase 2 Firings. Determine effect of the addition or deletion of a weapon component, such as a Blast Attenuation Device (BAD).

The phase 1 test may be repeated to determine the effect of the addition or deletion of a weapon component, such as a BAD which may have significant effect on selection of munition worst case temperature and charge. The recommended test will be identical to the Phase 1 test requiring 45 rounds. Dependent on test results, a decision will be made to conduct the remaining test steps with or without the weapon component.

n. Phase 3 Firings. Determine worst case weapon QE and azimuth (in the case of vehicle-mounted mortars), for impulse noise and BOP at the munition worst case temperature and charge.

The recommended test will consist of firing 5 round groups at maximum, minimum and midpoint weapon QEs, and various azimuths (for vehicle-mounted mortars) at the worst case previously determined worst case munition temperature and charge. This test will establish the worst case weapon QE and azimuth to be used for the remainder of the test. The number of different azimuths to be evaluated is design dependent. The vehicle configuration for ramp and hatches opened or closed will be the predicted worst case for impulse noise or BOP. Typically a minimum of three and a maximum of seven azimuths will be evaluated. The estimated number of rounds for a ground seated weapon is 5 (round groups) x 3 (QEs) = 15. The estimated maximum number of rounds for a vehicle-mounted is 5 (round groups) x 3 (temperature conditions) x 7 (azimuth settings) = 105.

o. Phase 4 Firings (for vehicle-mounted configurations only). Determine effect of various vehicle configurations for impulse noise or BOP at previously determined worst case munition and weapon QE conditions.

The recommended test will consist of firing 5 round groups at the previously determined worst case weapon QE and munition conditions with vehicle ramps and hatches opened and closed. Since the resultant effect of weapon azimuth and vehicle configuration on impulse noise or BOP is interdependent, adjustment of weapon azimuth should be done for this step. The number of different vehicle configurations is design and operationally dependent. Typically no more than five vehicle configurations will be evaluated. The estimated maximum number of rounds for a vehicle configuration is 5 (round groups) x 7 (azimuth settings) x (5 vehicle configurations) = 175.

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p. Test Summary. In summary, a maximum of approximately 45 + 45 + 15 = 105 rounds may be required for ground seated mortar testing. A maximum of approximately 45 + 45 + 105 + 175 = 370 rounds may be required for vehicle-mounted mortar testing.

H.3.2.2 Data Required.

a. Cartridge Temperature (± 1 °C).

b. Chamber pressure (± 700 kPa) and pressure gauge number using pressure transducers.

c. Cartridge charge.

d. As fired, in-flight projectile weight (± 0.005 kg).

e. Meteorological data.

f. Weapon elevation.

g. Weapon azimuth.

h. Vehicle configuration (hatches, ramps etc. configuration) (if applicable).

i. Data collected from transducers, sound level meters and BTDs.

H.3.2.3 Data Analysis.

a. Data reduction and presentation will be in accordance with ITOP 04-2-822. Worst case pressure data will be included in the report. Impulse noise data will be analyzed in accordance with MIL-STD-1474. BOP data will be analyzed with appropriate software developed by the US Army Medical Research and Material Command (MRMC) or the appropriate NATO country organization.

b. The appropriate national authority (e.g., The U.S. Army Health Hazard Assessment Program for US programs) will develop firing restriction criteria for each zone charge. The restrictions are based on worst-case crew position exposure for each zone. These restrictions are based on factors such as worst-case munition temperature, mortar configuration, elevations, azimuth, and other factors. In the event that an appropriate amount of test rounds cannot be acquired, the system developer and combat developer have the option of adhering to absolute worst-case firing restrictions for the highest zones for all

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use scenarios based on analysis completed by the appropriate national authority (e.g., U.S. Army Public Health Command, Health Hazard Assessment Program for U.S. Programs). The most stringent (i.e., lowest) allowable number of rounds would be used for all subsequent zones and all firing scenarios for the duration of the system life. Possibly, certain vehicle configurations (e.g., hatches or ramps positions and/or weapon azimuths and elevations) may be disallowed during operational firing based on excessive impulse noise or BOP data for those configurations.

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APPENDIX H. HEALTH HAZARDS.
ANNEX 4. ADDITIONAL TESTS AND ASSESSMENTS.

H.4 GENERAL.

Appendix A, Annex 7 provides the rationale for the Additional Tests and Assessments for which the test descriptions are provided herein. See Figure B1-1, Representative S3 Test Program for the description of the overall test program. It is noted that the details of most additional test and assessments are not provided in this document since they are governed by other standards.

Insensitive Munitions (IM) Assessment, Munition Software Safety Assessment, Explosive Materials Qualification, Munition Demilitarization, Render Safe and Disposal, Range Safety/Sustainability, Hazard Classification and Fuze Safety Assessment are required as part of the S3 package but the details regarding the series of tests required are not provided in this document since they are governed by other standards. Additionally, munition bare drop sensitivity and Vertical Replenishment at Sea (VERTREP) testing are not required but recommended to enhance safety information on the munition.

H.4.1 INSENSITIVE MUNITIONS (IM) ASSESSMENT.

The IM assessment testing shall be conducted in accordance with MIL-STD-2105, STANAG 4439 and AOP-39. For a system expected to have significant changes to its vulnerability with age/use, using environmentally stressed munition within IM vulnerability test and assessment should be considered.

H.4.2 MUNITION SOFTWARE SAFETY ASSESSMENT.

If applicable, munition software shall be designed, assessed and tested to assure its safety and suitability for service in accordance with ITOP 01-1-057, QAP-268 and Joint Software Systems Safety Engineering Handbook.

H.4.3 EXPLOSIVE MATERIALS QUALIFICATION TESTING.

All explosive materials in the munition shall undergo appropriate testing and assessment per STANAG 4170 and AOP-7 to determine whether each possesses properties which make it safe for consideration for use in its intended role.

H.4.4 MUNITION DEMILITARIZATION AND DISPOSAL ASSESSMENT.

Prepare a verified Demilitarization and Disposal Plan for the mortar cartridges in accordance with STANAG 4518 or respective national requirements.

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H.4.5 RENDER-SAFE AND DISPOSAL.

If required, ensure that verified Explosive Ordnance Disposal (EOD) render safe and disposal procedures are developed in support of new munition entering the inventory. This includes those particular courses of action or modes of actions for access to, recovery, rendering safe, and final disposal of explosive ordnance (EO). Render safe and disposal test plans are normally develop by the national EOD organization.

H.4.6 RANGE SAFETY AND SUSTAINABILITY.

Prior to performing any live firing tests, a weapon danger analysis must be performed to determine the hazardous area that will result from the live firing test. Further guidance may be found in STANAG 2240, STANAG 2401 and Allied Range Safety Publications (ARSP-1 VOL I and ARSP-1 VOL II).

Additionally, appropriate testing and analysis shall be conducted in accordance with AOP-15 to assess range safety and sustainability. The potential for individual and cumulative environmental effects of munition use on operational ranges, (e.g., the expected deposition of hazardous substances, pollutants and contaminants, or emerging contaminants) should be assessed.

H.4.7 MUNITION HAZARD CLASSIFICATION.

Appropriate munition hazard classification testing shall be conducted in accordance with STANAG 4123 and AASTP-3.

H.4.8 FUZE SAFETY TESTING AND ASSESSMENT.

The central objective of safety testing of Fuzing Systems is to confirm and document that the fuzing system is safe and performs as intended in all expected service environments. The design safety requirements standard is MIL-STD-331 and the fuze qualification test procedures are within ITOP 4-2-601. Overall test requirements for S3 Assessment are established in ITOP 04-2-806, which is based on the principles of AOP-15. Review of existing fuze safety, design, and test data is mandatory, and should be conducted prior to development of the mortar cartridges safety test plan. This will allow personnel to determine what additional testing is required to qualify the fuze on the specific configuration.

H.4.9 MUNITION BARE DROP SENSITIVITY TEST.

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This test should be performed on bare munitions in the orientation considered most likely to result in critical damage, degradation or initiation and should be conducted as a non-sequential test. The purpose of the test is to determine the 50% initiation drop height and the associated initiation drop height standard deviation. This data can be used to determine the initiation and safety probabilities at various drop heights in an analytical manner. This test is intended to complement the bare drop and packaged drop which are performed in a sequential series.

a. Munition Configuration: Unpackaged.

b. Test Level: The drop test will be structured and conducted using sensitivity test and analysis methods such as a Bruceton analysis and/or Neyer D-Optimal test methods. The test will be structured to determine the 50% initiation level (the drop height at which 50% of the sample will initiate) and the standard deviation. The impact surface shall be a concrete supported steel surface.

c. Drop Orientation: Each test cartridge is to be dropped once to impact in the orientation determined to represent the most likely to result in munition initiation as determined by the National Authority. This is probably in one of the following two orientations:

   A - Major axis vertical, nose up/base down
   B - Major axis vertical, nose down/base up

   Consideration should be given to using guided drop equipment to decrease the variability of the impact orientation.

d. Test Temperature: The test should initially be conducted at +21 °C to establish a common ambient temperature baseline for drop sensitivity characteristics. The test can be repeated at the maximum hot storage temperature and minimum cold storage temperature to determine the drop sensitivity at those temperature extremes. The drop tests should be conducted within the shortest duration possible upon removal from the conditioned environment. The maximum duration should be no longer than 15 minutes. During transport from the conditioned environment to the test site, it is good practice to minimize heat transfer effects through the use of thermal mitigation measures (e.g., insulated transport box or insulating blanket)

e. Post Firing: Since this is a test to determine bare drop height initiation sensitivity, there shall be no post firing of munitions except to dispose of test munitions that are visibly undamaged.

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H.4.10 VERTICAL REPLENISHMENT AT SEA (VERTREP).

Munitions may be moved as an under-slung load by helicopter over land and/or subjected to Vertical Replenishment at Sea (or VERTREP). The potential impact velocities and effect on munition must be compared to the 2.1 m packaged drop test levels and determination must be made if additional testing is required. The vibration environment associated with under-slung movement is less severe than internal carriage and can be addressed by other tests in the environmental sequence. In the case of over land movement, the shock associated with set-down will typically be addressed by other tests in the environmental sequence.

For VERTREP, the ship's motion affects the impact velocity and therefore impact is related to sea-state. The values provided in Table H4-1 are based on those from Def-Stan 00-035, Part 3, Issue 5. The impacts at lower sea-states (up to sea-state 5) are addressed by the 2.1 m packaged drop test in the environmental sequence so there will be no requirement to specifically test for these. VERTREP at sea-state 6 is considered unlikely for mortar munitions. If required, VERTREP is commonly replicated by a freefall impact in accordance with MIL-STD-810, Methods 516, 527 and should be conducted as a sequential test if required.

Table H4-1: Impact Test Severities for VERTREP

Sea State	Total Impact Velocity (m/s)	Equivalent Drop Height (m)
3	3.3	0.6
4	4.0	0.8
5	5.6	1.6
6	6.9	2.4

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APPENDIX H. HEALTH HAZARDS.
ANNEX 5. OPERATIONAL AND MAINTENANCE (O&M) ASSESSMENTS.

Operational tests assess the safety of operational and maintenance procedures and equipment during field handling exercises. Human factors engineers (HFE) shall be involved in the planning, conduct, and evaluation of the following tests.

H.5.1. Operational and Maintenance Simulation

Soldiers using inert munitions and non-maintenance support items will perform tactical transportation, system handling, and firing operations tests under simulated battlefield conditions. Human factors engineering tests during simulated firing missions include setup, munition loading, simulated firings, and disassembly. Training exercises are performed with the complete training package. The operator manuals are reviewed and followed during all of the above. Operators will wear temperate weather and arctic clothing and nuclear, biological, chemical (NBC) masks and clothing. The tester will consider performing a low-temperature (cold room) operational test to assess the soldier's ability to operate the cannon with protective gear. Live munitions may be used once enough testing has been completed to satisfy the safety authorities that the system is safe for use. Human factors engineers will review and exercise the system support package and assess the safety of preventive and corrective maintenance operations up to depot level. Simulated system faults may be used to exercise test sets and other test, measurement, and diagnostic equipment. Maintenance manuals will be used for these exercises and evaluated in terms of safety.

H.5.2. Human Error Checklist

Develop a checklist of "Common Sources of Human Error" to categorize human errors that occur during operational tests and to suggest potentially hazardous human errors that apply to the system. Develop additional safety checklists to address electrical, mechanical, and miscellaneous safety items. Information for developing this checklist is specified in MIL-STD-1472, TOP 01-1-015, TOP 01-2-610 and MIL-HDBK-46855A.

H.5.3. Operational and Maintenance Report

Record, describe, and score actual and potential unsafe operations and maintenance practices by using observations, video records, checklists, measurements, and operator and maintainer debriefings. Note, the experience and impressions gained by the test persons during handling of the equipment should be recorded during and/or immediately after the tests. This could be done best in the form of standardized interviews made by persons who are experienced in social sciences (e.g., Human Factors Engineers) using a catalog of previously determined questions. The interview results shall be evaluated based on social science criteria (statistical evaluation, etc.).

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APPENDIX H. HEALTH HAZARDS.
ANNEX 5. OPERATIONAL AND MAINTENANCE (O&M) ASSESSMENTS.

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APPENDIX I. ABBREVIATIONS/REFERENCES AND RELATED DOCUMENTS.
ANNEX 1. GENERAL.

This document was written with references to test procedures to provide a framework for procurement and test programs. Table I3-1 (Appendix I, Annex 3) provides detailed comparison of similar national and international test standards. While each test standard often has unique requirements, the table does not imply the standards are the same or interchangeable. If tailoring is determined to be necessary, the tailoring may be carried out in accordance with the following general principles:

1. The tailored environment shall be at least as severe as the expected life cycle environment:

2. Any alternative test standards/methods that are utilized shall be technically equivalent or superior to the referenced standards/methods.

3. The tailored test procedures and severities, along with full justification/rationale shall be documented as part of the assessment report.

4. Tailoring shall be approved by the relevant national authority prior to testing.

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APPENDIX I. ABBREVIATIONS/REFERENCES AND RELATED DOCUMENTS.
ANNEX 1. GENERAL.

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APPENDIX I. ABBREVIATIONS/REFERENCES AND RELATED DOCUMENTS.
ANNEX 2. ABBREVIATIONS.

AAS3P	Allied Ammunition Safety and Suitability for Service Publication
AASTP	Allied Ammunition Storage and Transport Publication
AECTP	Allied Environmental Conditions Test Publication
AEP	Allied Engineering Publication
ANEP	Allied Navy Engineering Publication
ANOVA	Analysis of Variance
AOP	Allied Ordnance Publication
AP	Allied Publication
ARSP	Allied Range Safety Publication
BAD	Blast Attenuation Device
BIT	Built In Tests
BTD	Blast Test Device
BOP	Blast Over Pressure
BTCA	Breakdown Test and Critical Analysis
°C	Degrees Centigrade
CLU	Command Launch Unit
CO	Carbon Monoxide
CO2	Carbon Dioxide
COHb	Carboxyhemoglobin
dB	Decibels
Def Stan	Defense Standard
D-MOT	Designated Maximum Operating Temperature
DMTA	Dynamic Mechanical Thermal Analysis
DP	Design Pressure
DSC	Differential Scanning Calorimetry
E3	Electromagnetic Environmental Effects
EED	Electro-Explosive Devices
EID	Electrically Initiated Devices
EMC	Electromagnetic Compatibility
EME	Electromagnetic Environment
EMOP	Extreme Maximum Operating Pressure
EMV	Electromagnetic Vulnerability
EO	Explosive Ordnance
EOD	Explosive Ordnance Disposal
ESAD	Electronic Safe and Arming Devices

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APPENDIX I. ABBREVIATIONS/REFERENCES AND RELATED DOCUMENTS.
ANNEX 2. ABBREVIATIONS.

ESC	Extreme Service Conditions
ESCP	Extreme Service Condition Pressure
ESD	Electrostatic Discharge
°F	Degrees Fahrenheit
FDP	Fatigue Design Pressure
G2/Hz	Power Spectrum Density
HCN	Hydrogen Cyanide
HE	High Explosive
HERO	Hazards of Electromagnetic Radiation to Ordnance
HFE	Human Factors Engineers
hr(s)	Hour(s)
Hz	Hertz
IM	Insensitive Munitions
ISO	International Organization for Standardization
ISS	In-Service Surveillance
ITOP	International Test Operations Procedure
kg	Kilograms
kJ/mol	Kilojoule per Mole
km	Kilometer
kPa	Kilopascals
LCEP	Life Cycle Environmental Profile
LCSD	Life Cycle Standard Deviation
LCT	Lower Conditioning Temperature
LFT	Lower Firing Temperature
m	Meter
MACE	Maximum Number of Allowable Consecutive Exposures
m/s	Meters per Second
mil(s)	Angular Measurement Equal to 1/6400 of a Circumference
MIL-HDBK	Military Handbook
MIL-STD	Military Standard
Min	Minute
mm	Millimeter

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APPENDIX I. ABBREVIATIONS/REFERENCES AND RELATED DOCUMENTS.
ANNEX 2. ABBREVIATIONS.

MOP	Maximum Operating Pressure
MRMC	Medical Research and Material Command
NASA	National Aeronautics and Space Administration
NATO	North Atlantic Treaty Organization
NBC	Nuclear, Biological, and Chemical
NDT	Non-Destructive Test
NEMP	Nuclear Electromagnetic Pulse
NH3	Ammonia
NO	Nitric Oxide
NO2	Nitrogen Dioxide
NSO	NATO Standardization Office
NTMS	NATO Terminology Management System
O&M	Operations and Maintenance
OSTR	One-Shot Transformed Response
Pa	Pascals
PMP	Permissible Maximum Pressure
PP	Proof Pressure
ppm	Parts per Million
QAP	Quality Assurance Provision
QE	Quadrant Elevation
RH	Relative Humidity
s	Second(s)
S3	Safety and Suitability for Service
S&A	Safe and Arm
SAR	Safety Assessment Report
SD	Standard Deviation
SDP	System Design Pressure
SME	Subject Matter Expert
SMP	Safe Maximum Pressure
SO2	Sulfur Dioxide
SOD	Strength of Design
SRE	Solar Radiation Equivalent
SRS	Shock Response Spectrum

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APPENDIX I. ABBREVIATIONS/REFERENCES AND RELATED DOCUMENTS.
ANNEX 2. ABBREVIATIONS.

STANAG	Standardization Agreement
TOP	Test Operating Procedure
TWA	Time Weighted Average
UCT	Upper Conditioning Temperature
UFT	Upper Firing Temperature
UNDEX	Underwater Explosion
UPLPP	Upper Pressure Limit for Propellant Proof
US	United States
W/m2	Watts per Meter Squared
VERTREP	Vertical Replenishment

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APPENDIX I. ABBREVIATIONS/REFERENCES AND RELATED DOCUMENTS.
ANNEX 3. REFERENCES AND RELATED DOCUMENTS.

Note: Table I3-1 is a cross-reference of similar international and national test standards. It should not be assumed that the various methods listed in Table I3-1 are technically equivalent, or that methods other than those specified in this document will be deemed acceptable by the relevant National Authority. In addition, the various documents listed may also contain unique test severities that are only applicable to the specific nation. Alternative national or international test methods may only be used if it can be demonstrated that they are technically equivalent or superior to the referenced methods; and following the guidelines in Paragraph 1 of this document. Further advice should be sought from the relevant National Authority before alternative test methods to those specified in this document are used. Revision identifiers have been intentionally removed since the latest version of the referenced documents should be referred to.

Table I3-1: Cross-Reference Table

	SHORT TITLE	NATO	US	UK	FR	GE
1	Ammunition Safety Testing	STANAG 4629	ITOP 04-2-504(3); MIL-STD-2105; MIL-STD-882		ITOP 04-2-504(3)	ITOP 04-2-504(3)
2	System Safety	AOP-15	MIL-STD-882; MIL-HDBK-764; ITOP 05-1-060	Def Stan 00-56	AOP-15	VG 95373, DIN EN 61508; ITOP 05-1-060
3	Safety Assessment	AOP-15	MIL-STD-882	AOP-15; Joint Services Publication-520		
4	Hazardous Material Classification	STANAG 4123; AASTP-3	TB 700-2; UN ST/SG/AC.10/11	Joint Services Publication 482 Chapter 4; UN ST/SG/AC.10/11	UN ST/SG/AC.10/11	STANAG 4123; AASTP-3
5	Hazardous Material Classification (Thermal Stability)	UN ST/SC/AC.10/11	TB 700-2; UN ST/SG/AC.10/11	Joint Services Publication 482; UN ST/SG/AC.10/11	UN ST/SG/AC.10/11	
6	Insensitive Munitions Tests	STANAG's 4240, 4241, 4382, 4396, 4496	MIL-STD-2105	STANAG's 4240, 4241, 4382, 4396, 4496; UN ST/SG/AC.10/11	STANAG's 4240, 4241, 4382, 4396, 4496	STANAG's 4240, 4241, 4382, 4396, 4496
7	Insensitive Munitions Assessment	AOP-39; STANAG 4439	AOP-39; STANAG 4439	AOP-39; STANAG 4439	AOP-39; STANAG 4439	AOP-39; STANAG 4439
8	Software Safety	AOP-52	ITOP 01-1-057; QAP-268; Joint Software Systems Safety Engineering Handbook	Def Stan 00-56	AOP-52	VG 95373, DIN EN 61508; ITOP 01-1-057

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APPENDIX I. ABBREVIATIONS/REFERENCES AND RELATED DOCUMENTS.
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	SHORT TITLE	NATO	US	UK	FR	GE
9	Fuze Safety Tests	STANAG 4157; ITOP 4-2-601; STANAG 4363; AOP-21; STANAG 4187	ITOP 4-2-601; MIL-STD-1316; MIL-STD-331	STANAG 4157; ITOP 4-2-601; STANAG 4363; AOP-21;	STANAG 4157; AOP-20; STANAG 4363; AOP-21; STANAG 4187	STANAG 4157; ITOP 4-2-601; STANAG 4363; AOP-21; STANAG 4187
10	Explosive Materials Qualification	STANAG 4170; AOP-7	STANAG 4170; AOP-7; NAVSEAINST 8020.5C	STANAG 4170; AOP-7	STANAG 4170; AOP-7; S-CAT 17500	STANAG 4170; AOP-7
11	Human Factors	STANAG 7201	MIL-STD-1472; TOP 01-1-015; TOP 01-2-610; MIL-HDBK-46855A	Def Stan 00-25; HSE Regulations	DGA/NO/FHG/913	VG 95115 ZDv 90/20 HdE; MIL-STD-1472
12	Environmental Testing	STANAG 4370; AECTPs 100, 200, 230, 240, 300, 400	MIL-STD-810	Def Stan 00-035	STANAG 4370; AECTPs 100, 200, 230, 240, 300, 400; GAM EG-13	STANAG 4370; AECTPs 100, 200, 230, 240, 300, 400; MIL-STD-810
12a	Global Climatic Data	STANAG 4370; AECTP 230 Leaflet 2311	MIL-HDBK-310; AR 70-38	Def Stan 00-035, Part 4	STANAG 4370	STANAG 4370
12b	Humid Heat	AECTP 300, Method 306	MIL-STD-810, Method 507	Def Stan 00-035, Part 3, Test CL6 Severity from Def Stan 00-035 Part 4 Ch2-01	AECTP 300, Method 306	AECTP 300, Method 306
12c	Low Temperature Storage	AECTP 300, Method 303	MIL-STD-810, Method 502	Def Stan 00-035, Part 3, Test CL5 Severity from Def Stan 00-035 Part 4 Ch2-01	AECTP 300, Method 303	AECTP 300, Method 303
12d	High Temperature Storage	AECTP 300, Method 302	MIL-STD-810, Method 501	Def Stan 00-035, Part 3, Test CL6 (for high humidity) & CL2 (for low humidity). Severity from Def Stan 00-035 Part 4 Ch2-01 if cyclic.	AECTP 300, Method 302	AECTP 300, Method 302
12e	High Temperature Cycle	AECTP 300, Method 302	MIL-STD-810, Method 501	Def Stan 00-035, Part 3, Test CL6 (for high humidity) & CL2 (for low humidity) Severity from Def Stan 00-035 Part 4 Ch2-01	AECTP 300, Method 302	AECTP 300, Method 302

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	SHORT TITLE	NATO	US	UK	FR	GE
12f	Solar Radiation	AECTP 300, Method 305	MIL-STD-810, Method 505	Def Stan 00-035, Part 3, Test CL3	AECTP 300, Method 305	AECTP 300, Method 305
12g	Thermal Shock	AECTP 300, Method 304	MIL-STD-810, Method 503	Def Stan 00-035, Part 3, Test CL14	AECTP 300, Method 304	AECTP 300, Method 304
12h	Temperature-Altitude-Humidity	AECTP 300, Method 317	MIL-STD-810, Method 520	Def Stan 00-035, Part 3, Test CL13	AECTP 300, Method 317	AECTP 300, Method 317
12i	Salt Fog	AECTP 300, Method 309	MIL-STD-810, Method 509	Def Stan 00-035, Part 3, Test CN2	AECTP 300, Method 309	AECTP 300, Method 309
12j	Sand and Dust	AECTP 300, Method 313	MIL-STD-810, Method 510	Def Stan 00-035, Part 3, Test CL25	AECTP 300, Method 313	AECTP 300, Method 313
12k	Immersion	AECTP 300, Method 307	MIL-STD-810, Method 512	Def Stan 00-035, Part 3,Test CL29	AECTP 300, Method 307	AECTP 300, Method 307
12l	Rain/Water Tightness	AECTP 300, Method 310	MIL-STD-810, Method 506	Def Stan 00-035, Part 3, Test CL27	AECTP 300, Method 310	AECTP 300, Method 310
12m	Icing	AECTP 300, Method 311	MIL-STD-810, Method 521	Def Stan 00-035, Part 3, Test CL10	AECTP 300, Method 311	AECTP 300, Method 311
12n	Mold Growth	AECTP 300, Method 308	MIL-STD-810, Method 508	Def Stan 00-035, Part 3, Test CN1	AECTP 300, Method 308	AECTP 300, Method 308
12o	Contamination by Fluids	AECTP 300, Method 314	MIL-STD-810, Method 504	Def Stan 00-035, Part 3, Test CN4	AECTP 300, Method 314	AECTP 300, Method 314
12p	Aircraft Cargo Decompression	AECTP 300, Method 312	MIL-STD-810, Method 500	Def Stan 00-035, Part 3, Test CL9	AECTP 300, Method 312	AECTP 300, Method 312
12q	Vibration Test	STANAG 4370; AECTP 400	MIL-STD-810, Method 514	Def Stan 00-035, Part 3, Test M1	STANAG 4370; AECTP 400	STANAG 4370; AECTP 400
12r	Vibration Test Schedule Development	STANAG 4370; AECTP 240, Leaflet 2410	MIL-STD-810, Methods 514, 527	Def Stan 00-035, Part 5	STANAG 4370; AECTP 240, Leaflet 2410	STANAG 4370; AECTP 240; ITOP 01-01-050
12s	Commercial (Common Carrier) Transportation Vibration	AECTP 400, Methods 401, 421	MIL-STD-810, Methods 514, 527	Def Stan 00-035, Part 3, Test M1, Annex A and M2	AECTP 400, Method 401	AECTP 400, Method 401
12t	Military Wheeled Vehicle Transportation Vibration	AECTP 400, Methods 401, 421	MIL-STD-810, Methods 514, 527	Def Stan 00-035, Part 3, Test M1, Annex A and M2	AECTP 400, Method 401	AECTP 400, Method 401
12u	Restrained Cargo Transport Shock	AECTP 400, Method 403	MIL-STD-810, Methods 516, 527	Def Stan 00-035, Part 3, Test M3	AECTP 400, Method 417	AECTP 400, Method 417
12v	Fixed Wing Aircraft Cargo Transportation Vibration	AECTP 400, Methods 401, 421	MIL-STD-810, Methods 514, 527	Def Stan 00-035, Part 3, Test M1, Annex A and M2	AECTP 400, Method 401	AECTP 400, Method 401
12w	Helicopter Cargo Transportation Vibration	AECTP 400, Methods 401, 421	MIL-STD-810, Methods 514, 527	Def Stan 00-035, Part 3, Test M1, Annex A and M2	AECTP 400, Method 401	AECTP 400, Method 401
12x	Under Water Explosion (UNDEX)	STANAG 4549; STANAG 4150; ANEP 43	MIL-DTL-901; ANEP 43	Def Stan 00-035, Part 3, Test M7 (or Test M3).		STANAG 4150

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	SHORT TITLE	NATO	US	UK	FR	GE
12y	Shipboard Vibration	AECTP 400, Methods 401, 421	MIL-STD-810, Method 528, 527; MIL-STD-167-1	Def Stan 00-035, Part 3, Test M1, Annex A and M2	AECTP 400, Method 401	AECTP 400, Method 401
12z	Fixed and Rotary Wing Captive Carriage Vibration	AECTP 400, Methods 401,420, 421; AECTP 240 Leaflet 2410	MIL-STD-810, Methods 514, 527	Def Stan 00-035, Part 5 (test spec development) Def Stan 00-035, Part 3, Test M1and M2 (tailored severities)	AECTP 400, Method 401; AECTP 240 Leaflet 2410	AECTP 400, Method 401; AECTP 240 Leaflet 2410
12aa	Gunfire Shock (Time Waveform Replication)	AECTP 400, Methods 405, 417, 421	MIL-STD-810, Methods 519, 525, and 527	Def Stan 00-035, Part 3, Test M19 (tailored severities)	AECTP 400, Methods 405, 417, and 421	AECTP 400, Methods 405, 417, and 421
12ab	Packaged Transit Drop	AECTP 400, Method 414	MIL-STD-810, Method 516	Def Stan 00-035, Part 3, Test M5	AECTP 400, Method 414	AECTP 400, Method 414
12ac	Catapult Launch and Arrested Landing		MIL-STD-810 Methods 513, 516, and 525			
13	Unpackaged Handling Drop	STANAG 4375	MIL-STD-810, Method 516	Def Stan 00-035, Part 3, Test M5	STANAG 4375	STANAG 4375
14	Packaged and Unpackaged Safety Drops	STANAG 4375	MIL-STD-810, Method 516	Def Stan 00-035, Part 3, Test M5	STANAG 4375	STANAG 4375
15	Logistic Drop Test (12 m drop)	STANAG 4375	MIL-STD-2105; ITOP 04-2-601	STANAG 4375	STANAG 4375	STANAG 4375; ITOP 04-2-601
16	Parachute Drop	ITOP 4-2-601; AECTP 400, Methods 403; AOP-20	ITOP 4-2-601; ITOP 07-2-509; TOP 04-2-509; MIL-STD-331	Def Stan 00-035, Part 3, Test M5; AP101A 1102-1		ITOP 4-2-601
17	Dynamic Firing	STANAG 4157; AOP-20; STANAG 4363; AOP-21	ITOP 04-2-806	STANAG 4157; AOP-20; STANAG 4363; AOP-21	STANAG 4157; AOP-20	STANAG 4157; AOP-20; STANAG 4363; AOP-21
18	Warhead Minimum Arming Distance	STANAG 4157; AOP-20; STANAG 4363, AOP-21	ITOP 04-2-806	STANAG 4157; AOP-20; STANAG 4363; AOP-21	STANAG 4157; AOP-20	STANAG 4157; AOP-20; STANAG 4363; AOP-21

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	SHORT TITLE	NATO	US	UK	FR	GE
19	Warhead Arena Test		ITOP 04-2-813; MEM NA 00-130ASR-2-1 (Army FM 101-51-3-CD (EM 0260))	ITOP 04-2-813	ITOP 04-2-813	ITOP 04-2-813; TL 1300-0011 Part 2, BWB WM VI 2 Hdb. ammunition sbewertung A 1981
20	Weapon Danger Area	STANAGs 2240, 2401; ARSP-1 Vol I and II; STANAG 2470; ARSP-2, Vol. 1	ITOP 05-2-505; AR 385-63	STANAGs 2240, 2401; ARSP-1 Vol I and II; STANAG 2470; ARSP-2, Vol. 1	TTA 208; STANAG 2921	STANAGs 2240, 2401; ARSP-1 Vol I and II STANAG 2470; ARSP-2, Vol. 1; A2-2090/0-0-1; ITOP 05-2-505
21	Health Hazards		TOP 06-2-507; TOP 10-2-508; OPNAVINST 5100.19E; OPNAVINST 5100.23G	HSE Regulations		
22	Toxic Gas/ Materials	STANAG 7201	MIL-STD-1472; TOP 2-02-614; ITOP 05-2-502	HSE Regulations		Erl. BMVg InSan I4-42-19-01; ITOP 05-2-502
23	Ionizing Radiation Hazards		TOP 03-2-711	HSE Regulations		
24	Electronic Equipment Hazards		MIL-HDBK-45	Def Stan 00-10 HSE Regulations		
25	Radiofrequency Health Hazards	STANAG 2345	TOP 03-2-616; OP3565 Vol. 1; DOD 6055.11	Joint Services Publication 392 Leaflet 35	ENV 501661; ENV 50061	DIN VDE-0848. T.1-4; DIN VDE-0848. T.2
26	Acoustic Noise	ISO 10843: 1997; AECTP-400, Method 402	MIL-STD-1474; ISO 10843: 1997; ISO 266: 1997	Def Stan 00-27 HSE Regulations	AT-83/27/28	ZDv 90/20 VM Blatt 1993

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APPENDIX I. ABBREVIATIONS/REFERENCES AND RELATED DOCUMENTS.
ANNEX 3. REFERENCES AND RELATED DOCUMENTS.

	SHORT TITLE	NATO	US	UK	FR	GE
27	Blast Overpressure	STANAG 4569 with references; Final Report RTO-HFM-089, -090, -148	ITOP 04-2-822; DOD 6055.9-STD		Consignes et instructions relatives à l'enregistrement et à l'exploitation des bruits d'armes et des bruits de détonation	Vorschriften und Richtlinien zur Registrierung und Auswertung von Waffen und Detonationsknallen and STANAG 4569 with references. Final Report RTO-HFM-089, -090, -148
28	Electromagnetic Environmental Effects (tests)	STANAG 4370; AECTP 500	MIL-STD-464; TOP 01-2-511; MIL-HDBK-240	Def Stan 59-411	NORMDEF 0301; GAM DRAM 02	STANAG 4370; AECTP 500;VG 95370; VG 95373; VG 95379
29	Electromagnetic Environmental Effects (environment description)	STANAG 4370; AECTP 250, Leaflet 258	MIL-STD-464; MIL-HDBK-240	Def Stan 59-411	NORMDEF 0301; GAM DRAM 01	STANAG 4370; AECTP 250, Leaflet 258; VG 95370; VG 95373; VG 95379
30	Electromagnetic Environmental Effects (HERO)	STANAG 4370; AECTP 508 Leaflet 3	MIL-STD-464; MIL-HDBK-240; JOTP-061; OP3565 Vol. 2	Def Stan 59-114; Def Stan 59-411	NORMDEF 0301; GAM DRAM 02	STANAG 4370; AECTP 508, Leaflet 3; VG 95378; VG 95379
31	Electrostatic Discharge (ESD) Environmental Test	STANAG 4370; AECTP 250, Leaflet 253; AECTP 508, Leaflet 2	MIL-STD-464; MIL-HDBK-240; JOTP-062	Def Stan 59-411	NORMDEF 0301; GAM DRAM 01; GAM DRAM 02	STANAG 4370; AECTP 250, Leaflet 253
32	Lightning Environmental Test	STANAG 4370; AECTP 508, Leaflet 4; AECTP 250, Leaflet 254	MIL-STD-464	Def Stan 59-411	NORMDEF 0301; GAM DRAM 01; GAM DRAM 02	STANAG 4370; AECTP 508, Leaflet 4; AECTP 250, Leaflet 254; VG 95379
33	Electromagnetic Interference	STANAG 4370; AECTP 501	MIL-STD-461; MIL-STD-464	Def Stan 59-411	NORMDEF 0301; STANAG 4370; AECTP 501	STANAG 4370; AECTP 501; VG 95370; VG 95373
34	Electromagnetic Compatibility	STANAG 4370; AECTP-250 and 500; IEC 61000 4-2	MIL-STD-461; MIL-STD-464; MIL-HDBK-237	Def Stan 59-411; IEC 61000 4-2	NORMDEF 0301	IEC 61000 4-2; VG 95370; VG 95373

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